

IMPLICATIONS OF SPATIAL POPULATION DYNAMICS FOR ABUNDANCE ESTIMATION AND CATCH
APPORTIONMENT OF ALASKA SABLEFISH (*ANOPLOPOMA FIMBRIA*)

By

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Abstract

Spatial dynamics in fish populations are challenging to identify and incorporate into population dynamics models due to the large amount of data required and the uncertainty associated with spatial processes. However, ignoring spatial dynamics can result in biased population estimates and negatively affect population sustainability or the ability to meet fishery management goals. My research explored implications of spatial dynamics for the assessment and management of Alaska sablefish (*Anoplopoma fimbria*). Sablefish are highly mobile, targeted in commercial and recreational fisheries, and documented to have spatial variation in population dynamics such as abundance and biomass. However, gaps remain in understanding the drivers of these spatial dynamics and their effects on fishery management. Chapter 1 focused on sablefish population dynamics in Alaska by developing a spatial stock assessment model with movement between modeled areas. Chapters 2 and 3 detail the development (Chapter 2) and application of a simulation and estimation model framework (Chapter 3) to examine alternative management scenarios and inform fisheries management.

The spatial stock assessment models developed in Chapter 1 showed that a combination of population processes (movement, recruitment) and fishing have resulted in spatial differences in sablefish spawning biomass and fishing mortality, and some management areas are likely below target biomass levels despite the population as a whole being at or near biomass targets. The implications of this apparent localized depletion are unknown, but could negatively impact recruitment success if depleted areas are crucial to population sustainability. Chapter 1 also highlights the complexity of spatial stock assessment models, which require large quantities of data and pose challenges for parameter estimation.

In Chapters 2 and 3, sablefish spatial population processes were simulated based on parameters estimated in the spatial stock assessment developed for Chapter 1 then assessed using a single-area, panmictic stock assessment model. Chapter 2 focuses on the simulation framework development and tests a suite of alternative population assumptions regarding movement and recruitment, which were found to be very influential in preliminary analyses. The analyses also revealed the challenges associated with a simulation framework that relies on

a relatively complex stock assessment model and cannot rely on human scientists to monitor and tweak assessment model estimation processes during the automated simulations.

In Chapter 3, five alternative methods for apportioning harvest opportunity among management areas were examined through the simulation to understand whether there were unexpected consequences of how catch opportunities were apportioned across space, given the assumed underlying population dynamics. We found that recruitment was poorly estimated by the single-area stock assessment model, and recruitment and movement rates between areas were drivers of population dynamics in each modeled region. We also found that no method for apportioning sablefish resulted in long-term population declines under the simulation and management assumptions, but recommended an apportionment method that used data from an annual survey so that catch opportunities are relational to observed spatial biomass. Each apportionment method resulted in different levels of catch for management areas and different year-to-year stability in catch, and these differences may be meaningful to stakeholders.

This research highlights the challenges and benefits of incorporating spatial population dynamics in assessment of a relatively data-rich species with high movement rates. For species with known processes that could create spatial dynamics but insufficient data for development of spatial stock assessment models, the development and implementation of monitoring and research plans to close data gaps could be a long-term strategy for improving understanding of the species' dynamics. However, monitoring can be logistically challenging and economically prohibitive in some situations, so simulation analyses such as those conducted here may help understand which data are most important to collect or inform the development of management procedures which are robust to the most plausible hypothetical population dynamics or future environmental conditions.

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Dedications

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General Introduction

Introduction

Fish population dynamics occur over space and time, but population dynamics models and fishery management are often reductionist in nature by necessity (Goethel et al. 2011, Berger et al. 2017, Cadrin 2020). Stock assessments frequently ignore spatial dynamics in lieu of a focus on temporal dynamics, particularly when spatially-stratified data are limited or have high uncertainty, or when political boundaries do not align with biological boundaries of populations (Goethel et al. 2011, Cadrin 2020). However, there are consequences to ignoring spatial complexity in populations when developing assessment models and management procedures; biased population estimates and mis-identifying population sources or sinks can occur and might keep systems from reaching management goals (i.e., forgone yield) or, in worst case scenarios, might seriously harm populations (Smedbol and Stephenson 2001, Fu and Fanning 2004, Lorenzen et al. 2010, Cope and Punt 2011, Goethel and Berger 2017). Thus, understanding the spatial dynamics of populations and properly identifying and accounting for spatial dynamics in stock assessment modeling and management can mitigate negative impacts.

Our ability to develop complex models and simulate temporally- and spatially-varying fish population dynamics is continually expanding as computing power grows, biological and environmental data are collected, and mechanistic drivers of population dynamics are identified (Begg and Waldman 1999, Begg et al. 1999, Goethel et al. 2011, Cadrin 2020). Simultaneously, the more we learn, the more we understand the scope of and challenges created by these dynamic interconnections among fish species, the ecosystem, and the environment around them (Link 2002, Christiansen and Pauly 2004, Schmidt et al. 2019). We have reached a point where complex models are computationally possible and where we understand some of the connections among ecosystems and populations, but where data can still limit our ability to represent dynamics across ever finer spatial scales. This, in turn, leads to intrinsic questions about the optimal level of complexity for our population and ecosystems models to strike an appropriate balance between meeting our modeling objectives and

addressing uncertainty introduced from reduced samples sizes as a result of analysis at a finer scale and model or parameter assumptions needed for finer scale models. Management Strategy Evaluation (MSE) is a process by which fishery managers and scientists can identify and evaluate fishery management objectives with respect to uncertainty in fisheries and ecosystem data, population dynamics models, and states of nature (Smith et al. 1999, Butterworth 2007, Punt et al. 2016). In my dissertation, I explored spatial complexity in Alaska sablefish population dynamics models and management using two techniques, estimation modeling of spatial population dynamics (Chapter 1) and simulation modeling that draws from techniques used in MSE (Chapters 2 and 3).

Every fish species presents unique challenges to assessing or estimating their population abundance and sustainable management of fisheries that target the species. Abundance on a population scale or local scale may be tied to dispersal patterns throughout the life history stages, prey-predator interactions, environmental drivers, anthropogenic effects such as fishing, and basic life history. Fishes that have low mobility (i.e., many reef fishes) may be susceptible to localized depletion due to high fishing mortality, but fishes with high mobility also create challenges for management. Highly migratory fish species may cross management boundaries and could face mortality that is not accounted for in stock assessment models (Morse et al. 2020). However, movement may also reallocate individuals in and out of areas with high fishing mortality and help buffer against population depletion (Boerder et al. 2019).

Fishes are a renewable resource and careful management can help ensure that they are available in perpetuity. In the U.S. exclusive economic zone (EEZ) extending 3–200 nautical miles off our oceanic coasts, the Magnuson-Stevens Fishery Conservation and Management Act (commonly referred to as the Magnuson-Stevens Act or MSA) details ten National Standards that guide fishery resource management (16 U.S.C. §§1801-1891d; National Marine Fisheries Service 2006). Specifically, five of the ten National Standards (NS) call for use of the best scientific information available (NS2), prevention of overfishing while achieving optimum yield (NS1), management of the appropriate stock units (NS3), achievement of equitable allocation (NS4), and mindfulness about the economic impact of management actions on fishing communities (NS8). It is within the context of these five Standards that the research and

modeling were conducted for this dissertation. Meeting the goals and guidance set forth by the National Standards is challenging and complicated due to data limitations and uncertainties, complex fish life histories and behaviors, conflicting goals for resource management by diverse resource users, and an ever-changing ecosystem.

My dissertation research focused on Alaska sablefish, a long-lived (max age 94–114 years; Kimura et al. 1998, Beamish and McFarlane 2000), economically valuable (\$123.8 million first wholesale value in 2017; Fissel et al. 2020), and widely distributed bottomfish in the Pacific Ocean (Sasaki 1985). From a purely scientific perspective, sablefish are fascinating to study due to the spatial ontogeny of their life history which places them in shallow water, frequently in nearshore bays, as juveniles, then gradually in deeper water as they mature. Alaska sablefish mark and recapture data show they can also move over great distances in relatively short timeframes, but that movement is not necessarily uniform across ages or individuals (Maloney and Heifetz 1984, Hanselman et al. 2015). Furthermore, their wide distribution across the Pacific basin, coupled with these movement rates, means they cross political boundaries where disparate harvest strategies could reduce stock-wide yields. Their high economic value adds to pressure from management bodies and stakeholders to maximize long-term sustainable yields. Alaska sablefish are managed by the North Pacific Fishery Management Council and are primarily harvested by a trawl fleet and a fixed gear fleet (pot and longline gears). Since 1995, the fixed gear fleet operates under a catch shares program, in which spatially-apportioned Individual Fishery Quota (IFQ) limit harvest amount and location. Total harvests have ranged between 10,231 and 38,406 metric tons during 1980–2020, with the peak occurring during a series of high catches in the late 1980s and early 1990s (Goethel et al. 2020).

In addition to spatial population dynamics, my research has explored the implications and sensitivity of both estimation models and management procedures to sablefish recruitment. The population trajectory of Alaska sablefish has historically been driven by episodic extreme large recruitment events (Goethel et al. 2020). We do not fully understand what contributes to sablefish recruitment success, the importance of particular spatial areas for spawning, whether spawning migrations are an important characteristic of sablefish reproduction, or the potential for skipped spawning.

The spatially complex life history of sablefish exposes them to a wide range of environmental conditions. Adult sablefish are believed to spawn offshore, near or beyond the continental slope, at depths greater than 300 m (Mason et al. 1983). However, eggs have been found between 200 and 800 m (Thompson 1941, Moser et al. 1994, Wing and Kamikawa 1995). Sablefish eggs develop at depth and then larvae further develop in surface waters offshore (McFarlane and Beamish 1992) before being advected to nearshore nursery areas. Eggs and larvae are subject to dispersal through entrainment in oceanic currents. Temporally and spatially variable sea surface temperatures and current speeds will thus affect larval transport and survival. Spatial and temporal overlap of larval sablefish with their prey may also be affected by water temperature that drives prey hatch timing and prey abundance (McFarlane and Beamish 1992) and may affect the overall contribution of a year class to recruitment (McFarlane and Beamish 1992, Schirripa and Colbert 2006, Shotwell et al. 2012). The post-settlement, juvenile life stage could also serve as a bottleneck in recruitment success when environmental conditions are not conducive to growth and survival (Krieger et al. 2020, Callahan et al. 2021). In summary, good survival of larval and juvenile sablefish may be due to a convergence of appropriate water temperatures, placement in currents that help bring them towards suitable habitat (Cowen et al. 2006), adequate prey availability (Krieger et al. 2020, Callahan et al. 2021), and predator avoidance. Perhaps not all of these factors contribute equally to successful recruitment. However, the relative infrequency of strong recruitment years (see Hanselman et al. 2018) suggests that successful recruitment is potentially due to the confluence of complex conditions that vary on small, medium (annual scale) and long (decadal) time scales. This makes recruitment dynamics challenging to model and contribute to uncertainty in stock assessment models.

Given the complex spatial population and recruitment dynamics exhibited by Alaska sablefish, the overarching requirements of the National Standards under MSA, and the modeling tools available, the goals of my dissertation were to: 1) improve understanding of the consequences of these uncertainties on Alaska sablefish management, and 2) provide further recommendations and insight regarding the management of Alaska sablefish through stock assessment modeling and simulation. Through the three chapters of my dissertation, I sought

to improve understanding of sablefish spatial population dynamics through the development of a spatial stock assessment model that incorporated movement between spatial regions (Chapter 1), improve understanding of data and population dynamics uncertainties and how they affect management decisions using simulations (Chapter 2), and use simulation and historical data analyses to provide sablefish management recommendations (Chapter 3). These models and simulation analyses were done through the lens of applied management, informed to the extent possible by observed data and existing management procedures instead of hypothetical data based on a 'sablefish-like' species. While this approach can be limiting, it was a conscious choice to maintain realism and provide direct application to Alaska sablefish management.

Specifically, in Chapter 1, I detailed the development of a spatial stock assessment model and examined the performance of alternative model structures of increasing complexity, with complexity measured as the number of spatial parameters estimated for selectivity and catchability. Through this suite of stock assessment models, I explored how movement (or assumptions about movement) affects sablefish population dynamics and the extent to which spatial complexity and movement affect population sustainability and depletion.

In Chapter 2, I developed a simulation framework that attempted to mimic Alaska sablefish population dynamics for six spatial areas across the Gulf of Alaska, Bering Sea, and Aleutian Islands, while balancing the limits of model feasibility and realism. I present the lessons learned about the panmictic, age-based stock assessment model used to assess the simulated spatial dynamics of the sablefish population. I also explored scenarios to understand drivers of population dynamics, focusing on recruitment and movement as 'axes of uncertainty.'

Finally, in Chapter 3, I applied what I learned in Chapters 1 and 2 about sablefish spatial dynamics and simulation techniques, and explored alternative approaches for apportionment of sablefish annual catch targets to management areas. This chapter was driven by real management questions and concern from sablefish stakeholders about how to partition the resource and a desire to understand if there were unexpected negative population

consequences to any catch apportionment alternatives. I developed paired but complimentary analyses to examine five catch apportionment alternatives: one technique was a complex simulation model with a feedback loop, the other was a simple retrospective analysis that used real observed dynamics but had no feedback loop between management action and the population over time. Both analyses used a suite of performance metrics that showed how apportionment alternatives compared. Together, these three chapters helped shape our understanding of Alaska sablefish spatial population dynamics. Moreover, I believe some of the findings and lessons are general to most applied spatial stock assessments.

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Chapter 1: Tradeoffs and uncertainties in developing a spatially explicit stock assessment for sablefish (*Anoplopoma fimbria*) in Alaska¹

1.1 Abstract

Spatial structure is an emergent property of many fish species, but it is often ignored in fisheries stock assessment models because spatial dynamics can be hard to model. Spatial structure arises in fish populations in multiple ways, including spawning migrations, ontogenetic movement, or phenotypic differences across environmental gradients. Because spatial population structure can affect the availability of fishes to harvesters, fishery managers may be interested in understanding fish abundance and distribution at finer spatial scales than available from a population-wide stock assessment. We developed a flexible, spatially-explicit three-region stock assessment model, which was applied to the U.S. federally managed portion of the North Pacific sablefish (*Anoplopoma fimbria*) population in Alaska. The model used externally estimated movement rates based on a thirty-year time series of mark-recapture data to produce spatially-explicit biomass estimates for three regions. The spatial model results were compared to two versions of a single-area panmictic model. While all three models produced similar estimates of population-wide biomass, the spatial model allowed the development of region-specific reference points, thereby revealing differences in regional stock status. Within the spatial model, the estimated biomass for the western region (containing the Bering Sea, Aleutian Islands, and the western Gulf of Alaska) was almost double the estimates for the central and eastern Gulf of Alaska regions. Because Alaska sablefish are managed under an Individual Fishing Quota program where quota shares are tied to management areas, fishery managers and quota shareholders have a strong interest in maintaining regional abundance to avoid economic and social hardships that may result if localized depletion occurs in a given management area. These results provided insight into the spatial distribution of the sablefish resource in Alaska to aid management advice, highlighting the major impediments to implementing spatial stock assessment models for complex, real-world populations even where reliable data are available.

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1.2 Introduction

Fishery managers are tasked with balancing several goals, including ensuring equitable economic opportunity, maintaining ecosystem health, minimizing the risk of stock collapse, and maximizing sustainable yield. It can be a challenge to achieve or define sustainable yield when fish species have complex spatial and demographic structure resulting from annual migrations for spawning, ontogenetic movement, physical dispersal of eggs or larvae, species-habitat relationships, and spatiotemporal variation in harvest (Gorman and Karr 1978, Dieckmann et al. 1999, Cowen et al. 2006, Bosley et al. 2019). Spatial population structure can increase the stability and resilience of a population when contingents exhibit divergent life histories and habitat use, which may help buffer periodic changes in environmental conditions and overharvest (Kerr et al. 2010). However, ignorance of species biocomplexity can lead to localized depletions and erosion of population structure diversity (Fu and Fanning 2004, Kerr et al. 2017).

As fisheries stock assessment methods evolve, data accumulate, and computing power increases, publication of spatial assessments is becoming more common (Goethel et al. 2011, Punt 2019). However, few applied spatial assessments are used as the basis of management advice, particularly for U.S. stocks (but see assessments of tuna stocks in the Western Pacific for examples; Langley et al. 2013, McKechnie et al. 2017). Spatial assessments are rarely applied to management for myriad reasons, including insufficient time series of data at fine spatial scales, as well as institutional inertia preventing the development and adoption of more complex spatial stock assessments (Wilén 2004, Berger et al. 2017, Punt 2019). Yet, properly accounting for spatial structure can determine whether management goals are met (Smedbol and Stephenson 2001, Lorenzen et al. 2010, Cope and Punt 2011, Goethel and Berger 2017). The possible consequences of mis-specification of stock or population structure include localized depletion, economically or biologically inefficient harvest allocation, and biased stock assessment results (Fu and Fanning 2004, Goethel and Berger 2017). For example, Goethel and Berger (2017) showed that assuming a population is panmictic when the population has spatial structure led to overharvest when exploitation rates were not uniformly distributed, and that the underlying population structure (e.g., a natal homing population or a metapopulation)

along with the level and direction of connectivity among population units strongly influenced sustainable harvest levels. A similar high risk of depletion was found to occur when a fish population was managed as separate closed populations, but there was underlying mixing between the sub-populations (Fu and Fanning 2004, Ying et al. 2011).

Several techniques have been developed to incorporate movement in stock assessments or address spatial population structure. One method uses ‘areas-as-fleets,’ which allows selectivity functional forms to serve as a proxy for spatial structure (Waterhouse et al. 2014, Punt 2019). However, spatially implicit areas-as-fleets models are often only moderately less biased than panmictic models (e.g., Hurtado-Ferro et al. 2014). Thus, spatially explicit assessment models are often warranted when spatial structure exists because they are better able to explicitly account for spatial population processes and movement (Berger et al. 2017, Punt 2019, Goethel et al. 2020). A primary limitation in the application of a spatial stock assessment is being able to accurately quantify movement among assessment regions (Goethel et al. 2020). Despite examples of applying reliable spatial models without tagging data within simulation studies (e.g., Hulson et al. 2011, 2013; McGilliard et al. 2015; Goethel et al. 2019, 2020), there remain relatively few demonstrations with real-world case studies.

Spatial tag-integrated models, which directly incorporate mark and recapture data within the objective function of the assessment, are the gold standard for spatial models because they permit uncertainty to be propagated through the estimation model, and tagging data can help inform assessment model parameters (e.g., movement and mortality; Goethel et al. 2011, Maunder and Punt 2013). However, few species have sufficient tag data for incorporating this level of spatial specificity. More commonly, there are limited tag data on movement from shorter-term, or small-scale tagging studies that may allow for estimation of movement among spatial regions (e.g., Hulson et al. 2011). Additionally, several caveats exist to using tagging data, which require careful consideration when combined with other data sets within an assessment model. A number of assumptions must be addressed including how to estimate spatiotemporal variation in tag reporting rate, tag induced mortality, tag mixing, and tag loss (Vincent et al. 2017, 2020; Goethel et al. 2019). Estimating these parameters can be extremely difficult and large bias may occur when tag ‘nuisance’ parameters are incorrectly

specified (e.g., reporting rate; Goethel et al. 2015; Vincent et al. 2017, 2020). Similarly, bias may arise when fitting tagging data in an age-based assessment model when the tagging protocols do not accurately measure release and recovery lengths or when age-length keys are inaccurate (e.g., due to aging sample size limitations at the desired spatial scale of the assessment model; Goethel et al. 2019). Tag-integrated models must also deal with data weighting issues (Punt et al. 2017) and potentially account for the lack of independence in tag samples (i.e., overdispersion; Hanselman et al. 2015a). Perhaps most importantly, the movement dynamics of individual tagged fish must, on average, represent those of the general population (Goethel et al. 2020), which includes sampling in such a way as to match the age or length structure of the population (Ziegler et al. 2013).

Although the benefits of including tagging data in spatial models generally outweigh the potential pitfalls, simplified spatial models that incorporate movement in the form of externally generated movement estimates may still provide useful information to managers (e.g., Quinn et al. 1990). Even when tagging data exist, initial explorations of spatial models may benefit from fixing movement and exploring sensitivity to connectivity over a range of plausible or bounding movement scenarios. By avoiding the added complexities associated with the direct incorporation of tagging data, spatial models with fixed movement can better focus on the implications of various plausible movement scenarios without the associated bias that may accompany estimation of tag nuisance parameters. Similarly, given the potential estimation difficulties and increased correlation that often occurs when movement is directly estimated (Cadrin et al. 2019), fixing movement rates and exploring a variety of connectivity sensitivity runs may prove beneficial to spatial model performance. However, care must be taken when interpreting the results of models that fix movement rates because they can sometimes perform worse than ignoring movement altogether (Goethel et al. 2015, 2020).

We explore the potential consequences for management of ignoring population structure for the U.S. federally managed portion of the North Pacific sablefish (*Anoplopoma fimbria*, hereafter ‘Alaska sablefish’) population in Alaska by comparing a three-area spatial assessment model incorporating movement among areas to a single-area panmictic assessment model. Sablefish are long-lived (maximum age 94–114 years; Kimura et al. 1998, Beamish and

McFarlane 2000) and widely distributed along most of the west coast of North America, from the Baja Peninsula in northern Mexico through the Aleutian Islands and the Bering Sea, and along the western borders of North Pacific as far south as Japan (Sasaki 1985). A recent genetic study has also shown that northeastern Pacific sablefish are not genetically distinct (Jasonowicz et al. 2017). In addition, Hanselman et al. (2015a) used tagging data to show that some movement does occur among all regions. Understanding the spatial abundance of Alaska sablefish is particularly important because they are highly valuable (U.S. \$94 million ex-vessel value for 2015 in Alaska; Fissel et al. 2016). Furthermore, sablefish are managed under a spatially-apportioned Individual Fishery Quota (IFQ) program, so properly characterizing spatial abundance has direct economic consequences for IFQ shareholders as they may only fish in areas where they own quota. Alaska sablefish are relatively data-rich, with estimates of movement rates between management regions from a long-term (1979–present) tagging program (Hanselman et al. 2015a), extensive fishery-independent survey spatial coverage, and fishery-dependent data from at-sea observers (Hanselman et al. 2015b). However, relatively little is known about the drivers of recruitment variation, spatial recruitment patterns, and the stock-recruitment relationship functional form.

We developed a flexible, spatially-explicit stock assessment model that is informed by externally estimated movement rates from tagging data. We fixed movement and then explored sensitivity to the movement rates without directly incorporating tagging data in the model because it allowed a focus on the implications of movement without the potential bias that might be added from the direct incorporation of tagging data, namely converting length composition to age composition for tagging data used in the assessment and estimating tag nuisance parameters. Our modeling approach represents a first step in exploring spatial models for potential use in sablefish management in Alaska to determine whether more sophisticated and complex tag-integrated models may be warranted. Additionally, the decision not to directly use tag data was one of dozens made in choosing spatial area breakdowns, examining data quality, and selecting data to inform the model. We applied this model to Alaska sablefish and 1) compared the spatial model estimates with two versions of a single-area panmictic model, 2) implemented a suite of spatial model parametrizations to address the sensitivity of biomass

estimates and biological reference points to assumed movement and model complexity, and 3) examined the source-sink dynamics associated with the assumed movement rates and estimated population dynamics.

1.3 Data and methods

Two model structures were developed and compared to examine the potential differences in estimates of Alaska sablefish biomass and interpretation of population trends: 1) a ‘Spatial’ age-structured stock assessment model which has three spatial areas, and data are provided to the model for each area, parameters may be estimated for each area, and movement occurs between areas, and 2) a panmictic, non-spatial (area-aggregated) age-structured assessment (hereafter ‘Non-spatial’) model. These two models were compared to a third ‘Management’ stock assessment model (Hanselman et al. 2015b), which was non-spatial (i.e., matches the spatial population structure of the Non-spatial model) and is currently used for the assessment and management of Alaska sablefish. Many of the basic assumptions and model structural decisions used in the Spatial model development match assumptions used in the Management model, except those regarding spatial structure. The Non-spatial model attempts to mimic the Management model, but within the context of a model code framework that matches the Spatial model (i.e., it imitates the Management model, but within the same structural and estimation framework as the Spatial model).

1.3.1 Spatial model data

For the Spatial model, data collected on the scale of management regions were aggregated into three spatial regions, r , which contain single or grouped management regions as defined by the North Pacific Fishery Management Council (NPFMC, NPFMC 2017; Figure 1.1). The western region included the U.S. exclusive economic zone waters of the Bering Sea, Aleutian Islands, and western Gulf of Alaska fishery management regions (denoted ‘West,’ Figure 1.1). The central region consisted of the central Gulf of Alaska (‘Central’) management region. The eastern Gulf of Alaska (‘East’) region combined the NPFMC regions West Yakutat and East Yakutat/Southeast Outside. The Bering Sea, Aleutian Islands, and Western Gulf of Alaska regions were combined to ensure sufficient sample sizes for age and length compositions because the Bering Sea and the Aleutian Islands are only surveyed every other

year, and fishery-dependent data are limited for these regions. The eastern management sub-regions were combined into a single East region to match movement rate estimates from available movement analyses (Hanselman et al. 2015a) and because sample sizes in the eastern sub-regions were small. After aggregating into the three areas, each region's sample sizes for age and length composition data were similar.

Sablefish are caught predominately in a fixed gear fishery that includes longline and pot gears and is fished by IFQ shareholders (91% of the average total catch for 2010–2015 was from fixed gear). Trawl catches make up the remainder of the total catch and are not managed under the IFQ program. Approximately 23%, 37%, and 40% of fixed gear catches were from the West, Central, and East regions, respectively, since the implementation of the IFQ program in 1995. Over this same period, 69% of the trawl fishery catch was from the Central region.

The Spatial model was informed by fishery-dependent and two fishery-independent indices of abundance (Table 1.1). The U.S.-Japan cooperative longline survey (US-JP; 1979–1994) and the U.S. longline survey (1990–2015) collected data on sablefish catch, effort, and length and subsampled catch for age, weight, sex, and maturity (Hanselman et al. 2015b). For each survey, relative population abundance (in numbers) for NPFMC management regions was calculated using the longline survey catch and effort data for a relative abundance index (Hanselman et al. 2015b). Fishery-dependent catch-per-unit-effort (CPUE) indices of relative abundance (in weight) were available from the U.S. fixed gear fishery for 1990–2014 and the foreign fixed gear fishery for 1977–1981.

The open-source software R (R Core Team 2015) was used for data compilation and formatting.

1.3.2 Spatial model structure and dynamics

The Spatial model is sex-specific, spatially-structured and age-structured, was developed in AD Model Builder software (Fournier et al. 2012), and fit to Alaska sablefish data (Table 1.1). Age groups $a_1 - a_n$ (representing ages 2–31) were modeled, where n is the number of age groups in the model. A plus group, a_+ , accumulated fish age a_n and older.

Sablefish recruitment is not thought to be well modeled through stock-recruitment functions because some of the largest recruitment events have come from low spawning biomass (Shotwell et al. 2014). Therefore, the Spatial model estimated a single mean recruitment and spatial annual deviations from the estimated mean with no assumed stock-recruitment relationship. For a_1 , estimated recruitment is a function of the population-wide mean recruitment μ_{Rec} and spatial annual deviations $\tau_{y,r}$ from the mean, where subscript y represents year and r region (Eqs. 1, 2). A likelihood penalty constrained recruitment deviations to be normally distributed with mean 0 and standard deviation σ_{Rec} . The model was configured for a 1:1 sex ratio at recruitment age a_1 using a 0.5 multiplier, and with recruitment occurring at age 2. Abundance before movement (N_{BEF}) for ages a_2 to a_{n-1} for the first model year was the product of recruitment and an exponential mortality model of survival to the oldest age, where M was natural mortality and S is selectivity from the longline fleet and $\bar{F}$ is the average fishing mortality. Plus group abundance (a_+) in year 1 was modeled as an infinite series solution for the number of recruits surviving (Eq. 1).

$$(1) N_{BEF,y=1,r,a,s} = \begin{cases} 0.5 * e^{(\mu_{Rec} + \tau_{1,r})} & a = a_1 \\ 0.5 * e^{(\mu_{Rec} + \tau_{(y+1-a),r})} e^{-(a-a_1)(M+S_{a,s}\bar{F})} & a_1 < a < a_+ \\ 0.5 * e^{\mu_{Rec}} e^{-(a-a_1)(M+S_{a,s}\bar{F})} (1 - e^{-(M+S_{a,s}\bar{F})})^{-1} & a = a_+ \end{cases}$$

For subsequent years, abundance before movement of age a and sex s , in each region, was the product of abundance after movement (N_{AFT}) in the previous year and age and survival from the previous year and age, by region (Eq. 2).

$$(2) N_{BEF,y,r,a,s} = \begin{cases} 0.5 * e^{(\mu_{Rec} + \tau_{y,r})} & a = a_1 \\ N_{AFT,y-1,r,a-1,s} e^{-Z_{y-1,r,a-1,s}} & a_1 < a < a_+ \\ N_{AFT,y-1,r,a-1,s} e^{-Z_{y-1,r,a-1,s}} + N_{AFT,y-1,r,a,s} e^{-Z_{y-1,r,a,s}} & a = a_+ \end{cases}$$

Year-, region-, age-, and sex-specific total instantaneous mortality for each region, $Z_{y,r,a,s}$, was the sum of natural mortality (M) and annual fishing mortality (F) for each year, region, and gear g ,

$$(3) Z_{y,r,a,s} = M + \sum_g (F_{y,r,g} * S_{r,a,g,s}).$$

$F_{y,r,g}$ is the product of mean fishing mortality $\mu_{f,r,g}$ and an annual fishing mortality deviation $\phi_{y,r,g}$, and annual fishing mortality accounts for region-, age-, gear-, and sex-specific selectivity $S_{r,a,g,s}$. A penalty was imposed upon fishing mortality deviations, such that large fluctuations in fishing mortality summed over regions, gears, and model years increased the total log likelihood (see Appendix 1 for likelihood equations).

After natural and fishing mortality in each year, fish of all ages were assumed to move in a single pulse, $N_{AFT,y,r,a,s} = (N_{BEF,y=y,r,a=a,s=s} * \Phi_{i,j}^T)$, where Φ is an externally estimated movement matrix (see section 1.3.1, Table 1.2). Elements of Φ are box-transfer transition probabilities among regions, where rows (i) describe the starting region and columns (j) represent the regions to which movement occurs, where each row sums to 1. For each year, age, and sex, the vector of abundance in numbers by region (before movement) is multiplied by the transposed movement (transition probability) matrix Φ^T , which assumes that movement is a Markov process.

Total biomass, B , was the product of abundance and weight at age summed over ages and sexes, $B_{y,r} = \sum_a \sum_s N_{AFT,y,r,a,s} w_{a,s}$. Female spawning biomass (SB) was calculated similarly to total biomass but was for females only and includes the proportion mature at age, m_a , where $SB_{y,r,s=f} = \sum_a N_{AFT,y,r,a,s=f} w_{a,s=f} m_a$.

Spatial, sex- and gear-specific selectivity parameters are useful when sexually dimorphic growth is present because the relative age-specific probability of capturing individuals differs between gear types and also when there may be spatial differences in fishery or survey selectivity. For the Spatial model, a simple two-parameter asymptotic selectivity form was used for the fixed-gear fishery to estimate the age at 50% maturity parameter, $a_{50\%,r,g,s}$, and slope, $\delta_{r,g,s}$ for each sex and gear, $S_{r,a,g,s} = \left(1 + e^{(-\delta_{r,g,s}(a-a_{50\%,r,g,s}))}\right)^{-1}$.

A dome-shaped, 2-parameter gamma function (Punt et al. 1996) was used for the trawl fishery for flexibility in selectivity at older ages, $S_{a,g,s} = \left(\frac{a}{a_{max}}\right)^{a_{max}/p} e^{(a_{max}-a)/p}$, where $p = 0.5 \left[\sqrt{a_{max}^2 + 4\delta_{g,s}^2} - a_{max} \right]$.

Selectivity in the U.S. fixed gear fishery was broken into two time blocks, pre-IFQ (1977–1994) and post-IFQ (1995–2015), each with separate logistic selectivity parameters. The pre-IFQ fishery was modeled using a single set of selectivity parameters shared across regions; the post-IFQ fishery selectivity parameters were spatial (up to three $a_{50\%,r}$ and three δ_r parameters estimated in the base Spatial model were possible). The U.S. fixed-gear fishery selectivity was fitted separately before and after the implementation of the IFQ program because there were changes to harvester behavior as a result of the IFQ program, which ended the derby fishery and reduced the number of vessels fishing (Sigler and Lunsford 2001, Brinson and Thunberg 2013, Warpinski et al. 2013). In addition, the fishing season transitioned from as short as five days to a prolonged season from March – November, allowing harvesters to target older and more valuable fish after IFQ implementation. Trawl fishery selectivity was assumed to be dome-shaped because the fishery operates predominantly in shallower waters than the fixed gear fishery, where older fish are less common and, thus, less available to the gear.

Non-spatial catchabilities (invariant across regions) were estimated for the U.S. longline survey, U.S.-JP survey, and the foreign fishery CPUE. Similar to selectivity, the U.S. fixed gear catchability was estimated in two time blocks, pre-IFQ and post-IFQ, to allow for changes in catchability with the new regulations, and a catchability parameter was estimated for each region and time block. The U.S. fixed gear abundance index was poorly fit in early model versions; thus, spatial catchability was incorporated to improve index fit.

Spatial model parameters were estimated using penalized maximum likelihood methods with penalties added for parameters with deviations from priors (see Appendix 1 for values). The objective function consisted of the individual likelihoods for the model components of catch, abundance indices, age compositions, and length compositions. Predicted length and age compositions use an age-length transition matrix and aging error matrix, respectively.

Equations for predicted values for indices of abundance and composition data, penalty components for catchability deviations, and recruitment deviations are detailed in Appendix 1.

The structure of the 'base' Spatial model was chosen based on a combination of *a priori* assumptions for which model components are likely to vary spatially and a desire for model parsimony and stability when estimating spatial selectivity parameters. We assumed that the U.S. longline survey selectivity and catchability should not vary spatially except in sensitivity explorations because a single vessel with fixed protocols for gear and bait standards, gear setting, and depths fished was used annually for surveying all regions. Model stability was a factor in fixing some components of spatial selectivity parameters; for example, early exploration revealed that models did not converge or performed poorly (e.g., high gradient) when estimating spatially-explicit selectivity for the U.S.-JP cooperative longline survey.

Several model input values and parameters were estimated outside the stock assessment model. Sablefish natural mortality was assumed to be 0.1 for all ages, and invariant across temporal and spatial scales (Johnson and Quinn 1988). Log recruitment variability σ_R was fixed at a relatively unrestrictive 1.2 based on historical precedent and to match the value assumed in the Management model.

Female maturity at age data (Hanselman et al. 2015b) were input as invariant among regions. Weight-at-age and length-at-age were sex-specific and assumed to be invariant over years and regions. These assumptions imply that Alaska sablefish are a single, well-mixed genetic population with no phenotypic variation in growth rates across regions, which is consistent with the methods used in Hanselman et al. (2015b). Weight-at-age was specified at the same values used in the Management model (Echave et al. 2012, Hanselman et al. 2015a).

In the base Spatial model, movement rates were not age- or length-structured and do not vary over time. Movement rates between spatial regions were estimated outside the assessment model with mark and recapture data from a long-term tagging study (1979–2009), following methods described in Hanselman et al. (2015a), except that movement rates were estimated for fish of all lengths combined. In the base movement matrix (Table 1.2), ~33-63% of fish move from a region annually. Retention rates, or the percentage of fish that do not leave a

region, were highest in the West (67.5%) and East (63.9%) and lowest in the Central region (37.1%). Because accuracy in movement rates was central to the spatial model validity and reliability, several sensitivity analyses using alternate movement matrices were examined; see the rationale in section 2.5.

1.3.3 Non-spatial model and management model structure and data

Both the Non-spatial and Management models are single area models and assume a panmictic population. The Non-spatial model used the same data and population dynamics equations as the Spatial model, but in spatially aggregated form (r subscript removed from equations), which combines data from the West, Central, and East regions. Length and age composition data from each region were weighted by catch in the respective region when aggregating for the Non-spatial and Management models. Catch and relative population in numbers or biomass are additive and were summed.

The Management model is the 2015 operational stock assessment model used for setting harvest limits for NPFMC Alaska sablefish and was the structural basis for the development of the Spatial and Non-spatial models (Hanselman et al. 2015b). This model begins in 1960 instead of 1977. The extended time series were not possible for the Spatial and Non-spatial models because the pre-1977 abundance index, length- and age-composition data did not contain sufficient spatial information to break the data into management regions.

While much of the data in the Spatial model were a spatially disaggregated version of the Management model data, there are a few minor differences (Table 1.1). Unlike the Management model, the Spatial and Non-spatial models do not include age composition data from the U.S.-JP cooperative longline survey because once spatially split, sample sizes were very small, and the spatial extent of samples was limited to small portions of the spatial regions, so they were not thought to be representative of individual regions. The Non-spatial model was intended to be a 'bridge' between the Spatial and Management models to elucidate the effect of a shortened time series and exclusion of the U.S.-JP cooperative survey and trawl survey data.

The Non-spatial and Management models estimated biomass, fishing mortality, and other output parameters for the entire population, whereas the Spatial model estimated these values for each region. For a direct comparison of the output between these three models, estimated biomass, recruitment, and other additive parameters were summed across Spatial model regions for each year.

1.3.4 Biological and management reference points

Spatial fishing mortality and fishing mortality-based biological reference points are key quantities of interest for management. A spatial spawner per recruit (*SPR*) model (Quinn and Deriso 1999) was used to estimate the biological reference points $F_{40,r}$ and $B_{40,r}$ for each region. $F_{40,r}$ was the exploitation rate in each region r which reduced the female potential spawning biomass per recruit, $SBR_{F,r}$, to 40% of the unfished level, $SBR_{U,r}$.

Spawning biomass for the *SPR* model was the sum over all ages of the product of maturity at age (m_a), weight at age (w_a), and the abundance-at-age of fish that survived until spawning ($e^{(-sf_r M_r F_{x\%r} S_{r,a,g})}$), where sf_r is the fractional timing of spawning during the year used to discount SSB for the assumed spawning date, and S is selectivity (defined above),

$$(4) SBR_{F_{x\%},r} = \sum_a \left(\left(N_{SPR_{F_{x\%}},r,a} \right) w_a m_a e^{(-sf_r M_r F_{x\%r} S_{r,a,g})} \right).$$

Abundance, $N_{SPR_{F_{x\%}},r,a}$, for the first age in the *SPR* model is recruitment (a_1) and was apportioned into regions based on the proportion of mean female age-2 recruitment in region r to the sum of mean 1977–2011 recruitment for all regions, assuming 1:1 sex ratio at recruitment (Eq. 4). This preserved the relative recruitment among regions. Subsequent ages had reduced spawning potential as natural and fishing mortality reduced abundance of fish, and subsequent movement occurred in a pulse prior to advancing to the next age to distribute spawning potential among regions after mortality. As in the main equations, movement, Φ , is an externally estimated movement matrix (see section 1.2.1, Table 1.2). For each year, age, and sex, the vector of abundance in numbers by region (before movement) is multiplied by the stationary distribution of the movement (transition probability) matrix Φ^T , transposed:

$$(5) N_{SPR_{F_{x\%},r},a} = \begin{cases} 0.5 \bar{N}_{AFT,r,a_1} / \sum_r \bar{N}_{AFT,r,a_1} & a = a_1 \\ \Phi^T * N_{SPR_{F_{x\%},r},a-1} e^{-(M_r + F_{x\%,r} S_{r,a-1,g})} & a_1 < a < a_+ \\ \Phi^T * (N_{SPR_{F_{x\%},r},a-1} e^{-(M_r + F_{x\%,r} S_{r,a-1,g})} + N_{SPR_{F_{x\%},r},a} e^{-(M_r + F_{x\%,r} S_{r,a,g})}) & a = a_+ \end{cases}$$

A $B_{40,r}$ reference point for each region was the product of $SBR_{F_{40},r}$ and the overall mean predicted recruitment, $(1/n) * \sum_r \hat{N}_{r,a_1}$, where n is the number of spatial regions r . The mean predicted recruitment parameter was used because there was no assumed stock-recruitment relationship and spawners from any region could contribute to recruitment in any region. $B_{40,r}$ was compared to spawning biomass in each region, $SB_{y,r}$, in the terminal year to assess stock status. The stock-wide B_{40} was the sum of regional estimates of $B_{40,r}$. Terminal year F , F_{2015} , from the assessment model was compared to the F_{40} reference point.

We also examined the spatial source-sink dynamics of the population spawning potential by calculating the contributions of spawning potential for region j to the other regions k ($k \neq j$) using the SPR calculations and no fishing mortality. For each region in turn, the number of a_1 recruits in region j is initiated with one recruit and the regions k with zero recruits. Movement and calculation of spawning potential for subsequent ages were the same as in Eqs. 4–5 above. The sum of spawning potential that moved from region j to regions k (loss to other regions) was compared to the sum of spawning potential that moved to region j from regions k (gain from other regions) for each region. A region was considered a source if the fraction of spawning potential lost to other regions exceeded the gain in spawning potential from other regions.

1.3.5 Model comparisons and sensitivity analyses

To examine how spatial parameterization of selectivity and catchability affected model output, we used a stepwise approach to move from the single area Non-spatial model (model 2 in Figure 1.2; model 1 is the Management model) with data aggregated into a single modeled region to increasingly spatially complex models that used disaggregated data split into three regions and region-specific parameters (models 3–17; Table 1.3, Figure 1.2). The simplest model step (model 3 in Figure 1.2) used disaggregated data (i.e., split into West, Central, and East

regions), did not have movement among regions, and non-spatial catchability and selectivity parameters were fixed at the values estimated in the Non-spatial model. This model is essentially conducting three separate regional stock assessments with a few shared parameters. Model 4 adds the base movement rates but leaves parameters fixed at the Non-spatial model values, and model 5 is identical to model 4 but estimates selectivity and catchability parameters that are shared across regions.

Models 6-8 have base movement rates, but estimate selectivity and catchability for either the pre-IFQ fishery, the post-IFQ fishery, or the longline survey (Table 1.3, Figure 1.2). Model 9 is the base 'Spatial' model, which estimated post-IFQ selectivity and pre-IFQ and post-IFQ catchability. The most spatially complex models, models 10–12, estimated different combinations of spatial parameters for the survey, pre-IFQ, and post-IFQ selectivity and catchability (Table 1.3, Figure 1.2). Not all parameters were estimable in every sensitivity step; for some models, poorly estimated parameters were fixed to Non-spatial model values or shared among model areas to obtain a positive definite Hessian matrix and model convergence.

Movement rates between regions were a key area of interest and uncertainty, so multiple sensitivity models were run to compare the base Spatial model (model 9) with age- or length-specific movement rates (Table 1.2, Table 1.3, Figure 1.2). Sensitivity models with alternate but plausible movement rate patterns included three age-based movement models in which fish movement rates varied among age groups (based on Hanselman et al. 2015a) instead of common movement rates for all ages (models 15–17; models 13–14 described below). For the first set of movement sensitivity analyses, externally estimated length-based movement rates were calculated for small (<57 cm), medium (57–66 cm), and large (>66 cm) sablefish (Hanselman et al. 2015a). Length-based movement rates were then converted to age-based rates for use in the spatial model using age-length relationships (Hanselman et al. 2015b; Table 1.2). Because sablefish have sexually dimorphic growth, but the movement model was not sex-specific, three alternatives for age-length relationships were used to link the size groups to age groups; female length at age (Option A), male length at age (Option B), and the average of female and male length at age (Option C). For these sensitivity analyses, we examined the

population as a whole or individual region relative to the standard NPFMC target of B_{40} (i.e., SB_{2015}/B_{40}).

Additional sensitivity analyses explored two hypothetical scenarios involving movement (Table 1.3, Figure 1.2). The base sablefish Spatial model was run with no retention of fish (model 14; movement matrix Φ main diagonal is zero; all fish leave their current region each annual time step) and full retention (model 13; all fish remain in their current region each annual time step; this is also like model 3 but with spatial parameters estimated). In these movement scenarios, off-diagonal movement was non-directional; fish not retained in a region moved in equal proportions to the remaining two regions. As retention in a region decreases, the overall movement rate of sablefish between regions increases.

Multiple sensitivity analyses and model explorations were conducted for several other areas of uncertainty while developing the models presented here, including an examination of alternate values and parameterizations for natural mortality, recruitment, and selectivity. However, the results of these analyses are not presented here for brevity. Uncertainty estimates for the Spatial and Non-spatial models were calculated using Markov Chain Monte Carlo (MCMC) simulation (10 million simulations, thinned to 5000 draws) for the base Spatial model (model 9). MCMC simulation was also conducted for models 2 (Non-Spatial model), 6, and 17, for 1 million simulations, thinned to 5000 draws. We also examined MCMC uncertainty in predicted spawning biomass in 2015 (SB_{2015}), predicted recruitment for spatial regions for the 2000 and 2008 large year classes, and for B_{40} for each region.

1.4 Results

1.4.1 Spatial, non-spatial, and management model results and comparisons

The estimated total (i.e., population-wide) biomass and spawning biomass from the sablefish Spatial model had similar trends and scale as the single-area Non-spatial and Management models (Figure 1.3A). Estimated terminal year total biomass for regions summed in the Spatial model was 219.9 kt, which was 12.3% more than the terminal year total biomass estimated by the Non-spatial, 195.9 kt, and 9.1% more than the Management model, 201.6 kt.

All models show a general decline in total biomass over the last ten years modeled, but the Spatial model had an upward trend in spawning biomass in the most recent year, whereas the Non-spatial and Management models declined or remained level (Figure 1.3B).

There were large differences in estimated sablefish total biomass between spatial regions, and biomass values were well estimated for all regions (2–5% coefficient of variation [CV] for 1977–2015). The Western region was estimated to contain 116.5 kt, or 53% of the total biomass of sablefish in 2015, whereas the Central and Eastern regions contained 59.4 kt (27%), and 44.0 kt (20%), respectively (Figure 1.3C).

Similar to total biomass, estimated terminal year spawning biomass (SB_{2015}) for regions summed was the highest in the Spatial model, 96.2 kt, compared to 84.0 kt in the Non-spatial and 90.0 kt in the Management model (Figure 1.3B). Regional Spatial model SB_{2015} was 50.0 kt (5% CV), 26.4 kt (4% CV), and 19.8 kt (5% CV) for West, Central, and East regions, respectively (Figure 1.3D).

The Spatial model fit the four survey and fishery indices reasonably well, except for the fit to the Western region for the U.S. Longline Survey and U.S. Fixed Gear Fishery (Figure 1.4; see the Appendix for other abundance indices). The Western region indices were much more variable and have poorer fits to observed values than the Central and Eastern region indices. The estimated U.S. Fixed Gear Fishery values were considerably higher than the observed values in recent years.

Fully selected fishing mortality estimates were similar between the Non-spatial and Management models (Figure 1.5). Fishing mortality for the Spatial model varied between regions and over time with generally similar trends to the Non-spatial and Management models. In the Spatial model, the F rates were the highest and most similar in the Central and Eastern regions, with a mean value of 0.11 and 0.12, respectively, for post-IFQ years 1995–2015. The Spatial model Western region's mean F rate was 0.04 for the same post-IFQ period. The drop in fishing mortality at the onset of the IFQ program in 1995 was apparent in both the Non-spatial and Management models, as well as East and Central Spatial model regions. Fishing mortality in the Western region of the Spatial model began declining in the late 1980s.

The 1977–2015 average estimated recruitment for the Spatial (summed over regions), Non-spatial, and Management models was 15.4, 15.4, and 21.5 million fish, respectively. In the Spatial model, log recruitment deviations were highly variable annually and between regions (Figure 1.6, top panel). Predicted recruitment of age-2 fish was lower in the East than the West or Central regions (Figure 1.6, middle panel). For years 1977–2015, the mean predicted recruitment for West, Central, and East was 6.6, 4.7, and 4.1 million sablefish, respectively. When summed across Spatial model regions, predicted recruitment patterns were similar between the Spatial and Non-spatial models (Figure 1.6, bottom panel), but uncertainty in estimates of predicted recruitment estimates were generally higher in the Spatial model.

MCMC simulations showed the posterior distribution for B_{40} , spawning biomass in 2015 (SB_{2015}), and predicted recruitment for two large year classes (Figure 1.7). B_{40} point estimates from the likelihood were within the range of the MCMC posterior distribution, but there was substantial overlap between the West and Central posterior. Predicted recruitment for two Spatial model strong year classes (2000 and 2008) had multi-modal posterior distributions, which overlapped the point estimates (Figure 1.7, bottom panels). The MCMC posterior for predicted recruitment in the West region of the Spatial model was wide and highly uncertain.

1.4.2 Biological reference points

Based on the Spatial model, the spawning biomass for all regions combined in 2015 (96.2 tons) was at 92% of the 104.3 t target B_{40} spawning biomass reference point used in the Alaska region, which was more optimistic than the estimates from the Non-spatial (82%) and Management models (88%). However, for individual regions of the Spatial model there were substantial spatial differences between the regional target B_{40} and the 2015 spawning biomass, with the West region at 155% of the 32.3 t target, the Central region at 87% of the 30.2 t target, and the East region estimated to be at 48% of the target of 41.7 t (Figure 1.3D). The F_{2015} for Western (0.02) and Central (0.10) regions were lower than their respective F_{40} reference points (0.10 and 0.13), whereas the Eastern region F_{2015} of 0.13 was higher than F_{40} (0.07).

Source-sink calculations indicate the West and Central regions were sources of spawning potential with ratios of gain from other regions, to loss to other regions, of 0.90 and 0.82, respectively, and the East was a sink with a gain:loss ratio of 1.33.

1.4.3 Sensitivity analyses: spatial selectivity and catchability parameters

The alternative spatial parameterizations of catchability and selectivity (Table 1.3, Figure 1.2 models 3–8,10–12) had a greater impact on the scale of spawning biomass estimates, compared with the proportional distribution of SB among regions or the depletion ratio (SB_{2015}/B_{40} ; Table 1.4). For example, the summed SB_{2015} across regions ranged between 56.0–118.5 kt, and the proportion of SB_{2015} by region were 48–57%, 23–28%, and 20–24% for West, Central, and East regions, respectively. SB_{2015} was precisely estimated for all stepwise models, with CVs of 3–5% for all models and areas. Predicted recruitment was estimated with greater certainty (defined as a lower mean CV for years 1977–2015) for the Non-spatial (mean CV 36%) and Management (mean CV 31%) models, compared with the Spatial or stepwise sensitivity models which had mean CVs of 51–83%. Predicted mean recruitment was greatest in the West (range 5.0–7.5 million age-2 fish across models) and lowest in the East (range 3.2–4.4 million age-two fish) for all stepwise models. In general, the predicted sum of recruitment across spatial areas had similar trends and temporal patterns for all stepwise models, though some deviations were observed, particularly in earlier years (Figure 1.8).

The addition of a spatial parameterization for catchability and selectivity did not always result in a functional, converged model (maximum gradient value < 0.001 and positive definite Hessian) without fixing some of the additional selectivity parameters. For example, model 10, which adds spatial pre-IFQ selectivity and catchability to the base Spatial model parameterization did not converge when all $a_{50\%,r}$ and δ_r selectivity parameters were freely estimated, but converged when the spatial δ_r parameters were fixed at the Non-spatial model values. In addition, some models which appeared to have converged (met our maximum gradient threshold and produced a positive definite Hessian matrix) produced biologically-implausible selectivity estimates (i.e., no fish selected at any age; Table 1.3 notes the models with biologically implausible selectivity) and high parameter correlation. Models attempting to estimate spatial pre-IFQ $a_{50\%,r}$ and δ_r selectivity parameters for multiple regions generally

failed to produce plausible parameter estimates. This highlights the potential cost in terms of estimation performance that is associated with increasing spatial complexity.

1.4.4 Sensitivity analyses: alternative movement rate assumptions

Alternative movement rates (models 13–17; Table 1.3, Figure 1.2) had a greater effect on the scale of SB and depletion estimates than on the proportional distribution of SB between regions (Table 1.4), and SB_{2015} was well estimated for all movement sensitivity models (CVs 4–6%). Alternative movement rates all resulted in the greatest biomass and least depletion in the West region. However, whether the Central or East region had the smallest proportion of biomass depended on which movement rates were used even though depletion was greatest in the East across all models that tested sensitivity to assumed movement rates (i.e., models 13–17). The 100% retention model (no movement, model 13), which is functionally the same as three single-area models with some shared parameters, produced spatial SB_{2015} estimates that were 20–33% higher than the Spatial model estimates for each region, and 28% greater SB_{2015} for regions summed compared to the Spatial model. Models 15–17 with age-based movement rates all had lower estimated total SB across regions and estimated more depletion (lower SB_{2015}/B_{40} for regions summed) when compared with the Spatial and Management models (Table 1.4). The sensitivity models with 0% and 100% retention served as bracketing ranges for sessility to extreme movement; the Spatial, Non-spatial, Management, and most other sensitivity models had SB_{2015} and depletion rate estimates that fell within the ranges of SB_{2015} and depletion estimated under the two extreme movement assumptions.

Predicted recruitment was not well estimated for movement sensitivity models, with mean CVs ranging from 50 to 93% across sensitivity models for years 1977–2015. Mean predicted recruitment was greatest in the West (range 6.8–7.7 million age-2 fish across models, Table 1.4) and lowest in the East (range 2.3–4.7 million age-2 fish) for all movement sensitivity models.

1.5 Discussion

Spatial stock assessment models are not new, but they are still infrequently used in an operational context for management advice (Berger et al. 2017, Punt 2019). Alaska sablefish were well suited for a spatial stock assessment model because extensive tagging has demonstrated that sablefish are capable of frequent, large-scale movements, survey data suggests heterogeneous distribution throughout their range in Alaska, and spatial patterns in exploitation exhibit diverse spatio-temporal harvest strategies. Furthermore, there was interest in spatial assessment research from sablefish fishery managers and constituents, who sought spatial resolution of reference points to ensure sustainable and equitable fishing opportunities. The models presented here represent the first applied spatial stock assessment models for Alaska sablefish and are a valuable step in understanding sablefish spatial dynamics.

The total and spawning biomass from the Spatial, Non-spatial, and Management models had similar scale and trends for most of the modeled years. This similarity, on the surface, gives confidence that the Spatial base model parameterization and movement rates are consistent with the assumptions of the single-area models. For example, some simulation work has demonstrated that panmictic models often provide unbiased estimates of total biomass, even when complex spatial processes occur (Li et al. 2015, 2018; Bosley et al. 2022). Thus, matching estimates of population-wide parameters may indicate that spatial and non-spatial models provide reliable parameters for the entire region. However, other spatial modeling work has shown that panmictic single-area models (i.e., the Non-spatial and Management models for sablefish) that ignore or incorrectly specify underlying spatial population structure can result in inaccurate estimates of productivity, which may result in overharvest and localized depletion (Fu and Fanning 2004, Ying et al. 2011, Goethel and Berger 2017). The utility of a spatial model is in the information on regional abundance and reference points for the management of sablefish that single-area models do not provide, which may help avoid overfishing on portions of the population (Ying et al. 2011).

Our Spatial model estimated large differences in abundance between the three modeled areas and highlighted source-sink dynamics between areas. The estimated differences in sablefish abundance among regions likely resulted from a combination of sablefish

movement rates and spatial differences in exploitation, selectivity, and recruitment to the fishery. Alaska sablefish biomass was estimated to be high in the West region because model-estimated recruitment of age-2 fish was generally high in this region (Figure 1.6), which was supported by the age composition data showing that young fish were most abundant in Western areas. Fishing mortality was also estimated to be low in the West compared with other Spatial model regions (Figure 1.5). The full amount of apportioned allowable catch from the West region is rarely taken by the fishery due to low catch rates and long distances required to travel from ports to fishing grounds. Depredation by killer whales (*Orcinus orca*) is also pervasive in these areas, further reducing the economic viability of fishing opportunities (Peterson et al. 2014, Peterson and Hanselman 2017). Movement rates also contributed to the model estimates of large biomass estimated in the West because movement from the West to other regions is lower than from Central or East regions (Table 1.2). This lack of movement out of the West is also an artifact of the sheer size of that combined region, which spans thousands of linear kilometers.

1.5.1 Model parameterization

What is the ‘right’ amount of spatial parameterization and spatial complexity in an assessment model? Punt (2019) and Cadrin (2020) provide best practices and suggest common definitions for terms, noting that while some challenges are common across spatial assessments, there are still unique issues that need to be addressed based on individual species’ fisheries and population dynamics (Cadrin 2020). The number of spatial regions to include in the Alaska sablefish Spatial model was a complex decision involving tradeoffs for data quantity and quality and the amount of biological realism. In building the Spatial model, we examined the underlying biological structure, age and length composition sample sizes, and abundance indices for three, four, and six potential spatial regions and settled on three as a suitable first step. A four- or possibly five-region spatial model may be possible by further dividing the West region into additional sub-regions. This could potentially alleviate the high uncertainty in some parameter estimates, such as selectivity in the West region, by allowing for spatial differences in selectivity and abundance index trends, but could also further compound convergence issues and parameter correlation observed for models with additional spatial

selectivity and catchability parameters. Future work should also consider potential spatial differences in biological parameters (growth, weight, maturity) that may arise from environmentally induced phenotypic differences among regions.

The Spatial model parameterization was chosen to reflect the most reasonable amount of spatial differentiation given limitations in data. Older data sources, like the U.S.-Japan cooperative longline survey age data and catch data prior to good observer coverage, have insufficient information available to divide the data into spatial areas without using strong assumptions or fixed ratios. We decided *a priori* not to model selectivity spatially for the longline survey because the same vessel and gear is used within a given year. However, there may be temporal variations in how the vessel fishes as it covers the large survey area due to differences in benthic habitat, currents, and changing captain and crew. In exploring models that estimate spatial catchability and selectivity, dozens of model configurations were tested with varying bounds and combinations of spatial $a_{50\%,r}$ and δ_r selectivity parameters with limited success. Conflicting data, correlated parameters, uninformative data, or some combination of both hampered our ability to develop complex spatial models and may present a key challenge for other researchers balancing the costs of complexity in spatial assessment models.

Sensitivity analyses indicated that model results were sensitive to the number and type of spatial parameters estimated and the movement rates assumed. Stepwise changes in the number and type of spatial catchability and selectivity parameters estimated revealed two issues. First, spatially-varying selectivity for the pre-IFQ longline fishery could not be reliably estimated from the available data. Second, models with various combinations of estimated spatial selectivity and catchability for the longline survey, and pre- and post-IFQ longline fishery may have converged and resulted in a positive definite Hessian. However, the resulting selectivity forms were not biologically realistic or well estimated (i.e., zero selectivity until age 20 and selectivity parameter CVs > 10,000%), indicating that the model may have found a solution, but perhaps not a global minimum and not a viable model. This highlights the importance of rigorous application of model convergence verification steps, including verifying parameters are not on bounds and thoroughly examining the biological plausibility of estimated

parameters in model output, as lack of fit to data can indicate model mis-specification for key processes (Lee et al. 2017). Penalty functions for spatial models may need to be used to keep parameters biologically realistic (e.g., fishing mortality, selectivity, recruitment, and movement).

A key challenge in the development of spatial stock assessment models is the need to balance the model's ability to estimate spatial parameters (i.e., have enough data to estimate parameters), having reasonable justification for which parameters should be spatial, and avoiding confounding parameters as the number of spatial parameters increases (Berger et al. 2017, Lee et al. 2017, Cadrin et al. 2019). The models developed here frequently had high parameter correlation, particularly for catchability and selectivity. This suggests that the data may be insufficient to discriminate among parameters, or the spatial scales for data and modeled parameters may not be well-aligned. Spatial biological parameters (e.g., weight at age, fecundity, growth) might be more appropriate in some situations where there is evidence of distinct differences among regions or directional movement coupled with life history characteristics (i.e., ontogenetic movement, known spawning migrations, or a metapopulation with natal homing). For sablefish, we opted for non-spatial biological parameters because of the relatively high movement rates among areas, and because it is unrealistic to have individuals take on new biological characteristics during transient movements through areas. Individual-based models (IBMs) that track individuals through time or composite population-IBM models (Goethel et al. 2011; Archambault et al. 2016) may be better suited for populations with known spatial differences in biological parameters and may be able to discriminate how different areas contribute to recruitment at age-2.

1.5.2 Movement

This spatial assessment model used an externally estimated matrix of movement rates among regions. More complex options would be to directly incorporate tag and recapture data in the assessment model to estimate movement rates internally, or to estimate age-, sex-, or time-varying movement rates. The benefit of externally estimated movement rates is that the model is conceptually simple, has fewer parameters (particularly the difficult to estimate tagging nuisance parameters), avoids potential bias that may arise from tagging model

assumptions (e.g., mis-specifying tag parameters; Goethel et al. 2019), reduces the potential for movement and recruitment parameter correlation (Cadrin et al. 2019), runs relatively fast, and it is simple to test alternative movement patterns. Drawbacks to this approach are that uncertainty in movement rates is not propagated through assessment model estimates, thus uncertainty is under-represented, and tag data are not able to help inform fishing mortality or natural mortality (Maunder 2001). Both methods, internal or external estimation of movement, require many assumptions that may be quite influential on results.

Alternative movement rates and model parameterizations affected the scale of SB_{2015} , which affects management recommendations for annual catch limits. All movement scenarios except 100% retention (no movement, model 13) estimated lower SB_{2015} than the ‘base’ Spatial model. Conversely, the validity of spatial model outputs has been shown through simulation to be conditional on correctly characterizing movement and spatial heterogeneity in fishery removals and demographic parameters (Guan et al. 2013, Bosley et al. 2019, Goethel et al. 2020). When movement is incorrectly specified or parametrized, it can be more detrimental than ignoring movement entirely (Li et al. 2015, Goethel et al. 2015, Goethel et al. 2020), further highlighting the need to proceed with caution in applying the results of our spatial model to real-world management.

Throughout North America, sablefish are assessed as four separate populations for the U.S. west coast, Canadian west coast, Federal waters off Alaska, and Alaska State waters (Cox et al. 2011, Hanselman et al. 2015b, Johnson et al. 2016, Fenske et al. 2019). Each assessment assumes a closed population, though there is known movement among these areas based on tag return data. Violations of the closed population assumption can lead to bias in biomass estimates and related reference points (Fu and Fanning 2004, Goethel et al. 2015, Goethel and Berger 2017, Lee et al. 2017, Li et al. 2018, Bosley et al. 2019). The Spatial movement sensitivity model with 100% retention (i.e., no movement among areas) was functionally similar to conducting three single-area assessment models with some shared parameters. The biomass estimated in this no-movement sensitivity model was substantially higher, and depletion was lower, than the Spatial base model that includes movement among regions. Recent simulations have shown that a mismatch among management areas and underlying population distribution

across areas can lead to biased perceptions of biomass and productivity (Goethel et al. 2015, Goethel and Berger 2017). Thus, further suggesting that it is not only important to have accurate estimates of movement among spatial areas, but also to consider whether the management structure of sablefish is correctly matched to the underlying population structure. This highlights a potential danger of having stock assessment boundaries smaller than the true population extent. It could be interpreted as a wise management choice that Alaska sablefish is the only stock in the NPFMC managed across all of Alaska's federal waters. Conversely, this broad area designation may still be too fine-scale given known movement into other areas of the North Pacific.

Much like the decisions made in balancing the number of spatial parameters to estimate, the decision to model three spatial areas for only the federally-managed portion of Alaska and not include British Columbia, Alaska State waters, or the U.S. West Coast was made to balance model complexity and spatial data availability, particularly movement rate data. It should be considered an initial step in sablefish spatial modeling. Because sablefish are so widely distributed and genetic analyses have not been able to distinguish separate populations (Jasonowicz et al. 2017), there is an incentive to continue ongoing explorations of alternative sablefish models which account for observed movement across established management boundaries or a single coast-wide model which represents the entire population (Fenske et al. 2019).

Caution should be used when applying the Spatial stock assessment reference points to management of Alaska sablefish. The interpretation of source-sink dynamics is confounded by a key model structure configuration and data gap. Recruitment age in the assessment model is for age-2 fish, so there is a two-year lag between spawning and recruits entering the fishery, and spawning locations are unknown. Transport by predominant currents for sablefish eggs and larvae (Kendall and Matarese 1987, Shotwell et al. 2014, Gibson et al. 2019) and movement of juveniles into and out of nearshore and inshore nursery habitat (Ehresmann et al. 2018) may mean that where spawning occurs may not correspond to where it enters the fishery, decoupling the biological reference point from the region-specific data. In addition, little is known about whether mature sablefish undertake spawning migrations among regions within a

year, which may not be captured by the annual time step of the movement estimation model (Hanselman et al. 2015a). The movement study includes monthly movement rates, but an assessment model that operated on monthly time steps is impractical for sablefish. The observed age and length composition data suggest that larger and older fish are more prevalent in the East region compared to the West, so by exploiting heavily in the East region, there may be consequences to future sablefish recruitment if there is disproportionate harvest in the East of older, potentially more reproductively important fish. More research is needed to determine the locations of spawning and nursery habitats, and whether there are spatial-, size-, or age-based differences in sablefish fecundity or maturity.

1.5.3 Conclusions

The implications of spatial differences in biomass for fishery management depend on the structure of the fishery and management approach. For sablefish, properly characterizing and preserving spatial abundance has economic consequences for sablefish IFQ stakeholders because IFQ shares are tied to a management region. The development of spatially-explicit reference points in stock assessment is still in its infancy (Reuchlin-Hugenholtz et al. 2016). Because there is no assumed stock-recruitment relationship for Alaska sablefish, there is no implicit link between depleted spawning stock and recruitment consequences at fine spatial scales. Future work should focus on the identifying appropriate spatial reference points for Alaska sablefish. Currently, a six-area spatial operating model is being developed for Alaskan sablefish and may be useful for exploring spatially explicit reference points. Using simulations of a sablefish-like species, Bosley et al. (2019) demonstrated that total system yield could be maximized by fishing extremely hard in a sink region (e.g., East) while lightly exploiting a source region (e.g., West), but doing so could affect the distribution of social and economic benefits. Therefore, the consequences of fishing mortality targets that are disproportionate to available biomass in different regions requires further exploration.

For Alaska sablefish, the Spatial stock assessment model is intended as a research tool and the beginning of a deeper dive into coastwide biological and population dynamics, simulation-based exploration of apportionment, and the utility of spatial stock assessment for sablefish. At a minimum, the exploration of spatial models for a species identifies where

significant data gaps occur, such as the location of spawning and pre-recruits, and may guide future spatial coverage of biological collections to improve future modeling endeavors and biological understanding of spatial dynamics. However, given that the spatial models demonstrated such strong regional differences in biomass and harvest, our results indicate that further work is required to better account for spatial dynamics in applied assessment models for sablefish. We also recognize that complex models may be great in theory, but in practice, less complex models often behave better and may be adequate if they are able to capture the essential features of the system.

1.6 References

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1.7 Figures

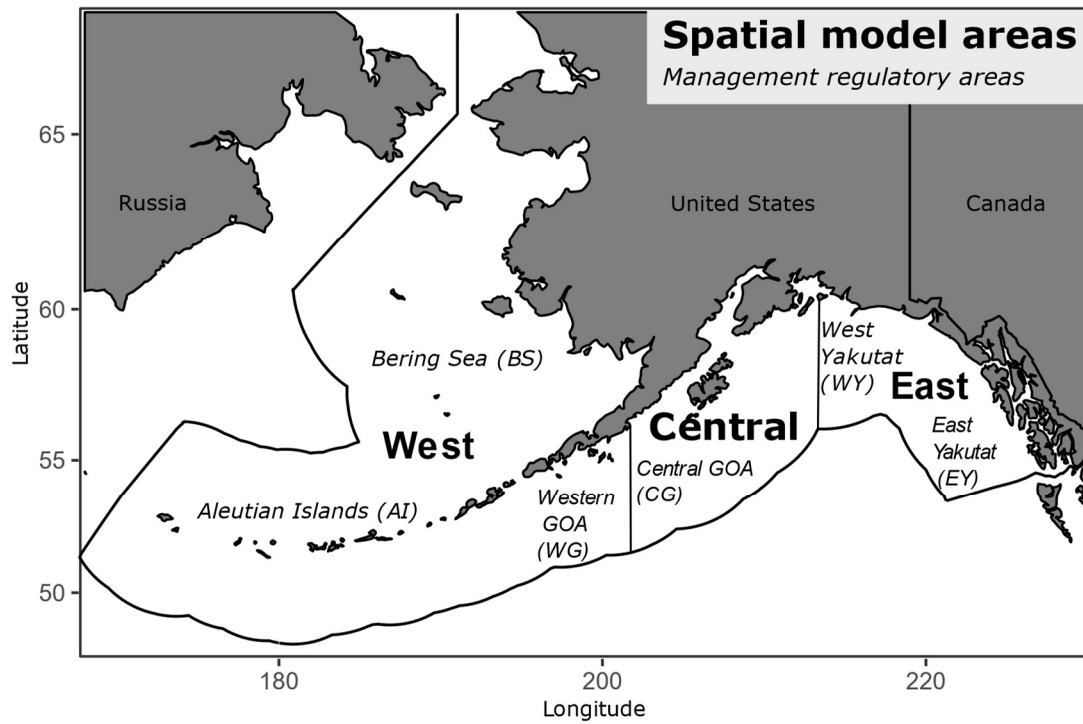


Figure 1.1. Spatial stock assessment model areas. The sablefish data in the Spatial model were aggregated into three Alaska regions (West, Central, East), each of which contained single or grouped management area as defined by the North Pacific Fishery Management Council; Bering Sea, Aleutian Islands, Western Gulf of Alaska (GOA), Central GOA, West Yakutat, and East Yakutat/Southeast Outside. Management area boundaries are approximate and not intended to be used for management purposes.

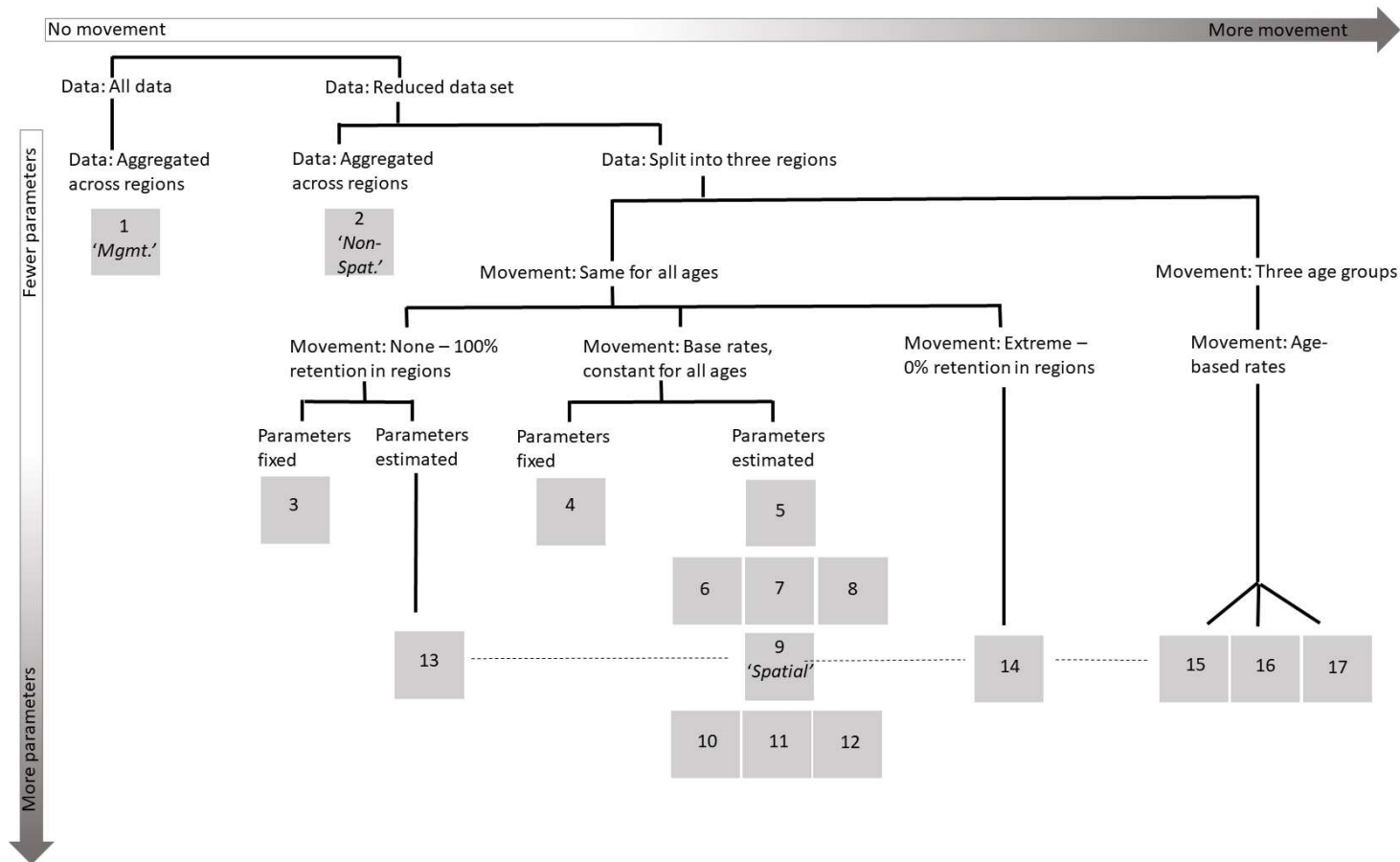


Figure 1.2. Relational diagram of Alaska sablefish spatial stock assessment models. Flow chart showing relationships between stock assessment models and sensitivity explorations. Model numbers correspond to named models in Table 2. There are two primary

gradients for model explorations: movement and number of estimated parameters. Dashed lines link models with the same number of spatial parameters and the same estimated spatial parameters, with different assumed movement rates among spatial areas.

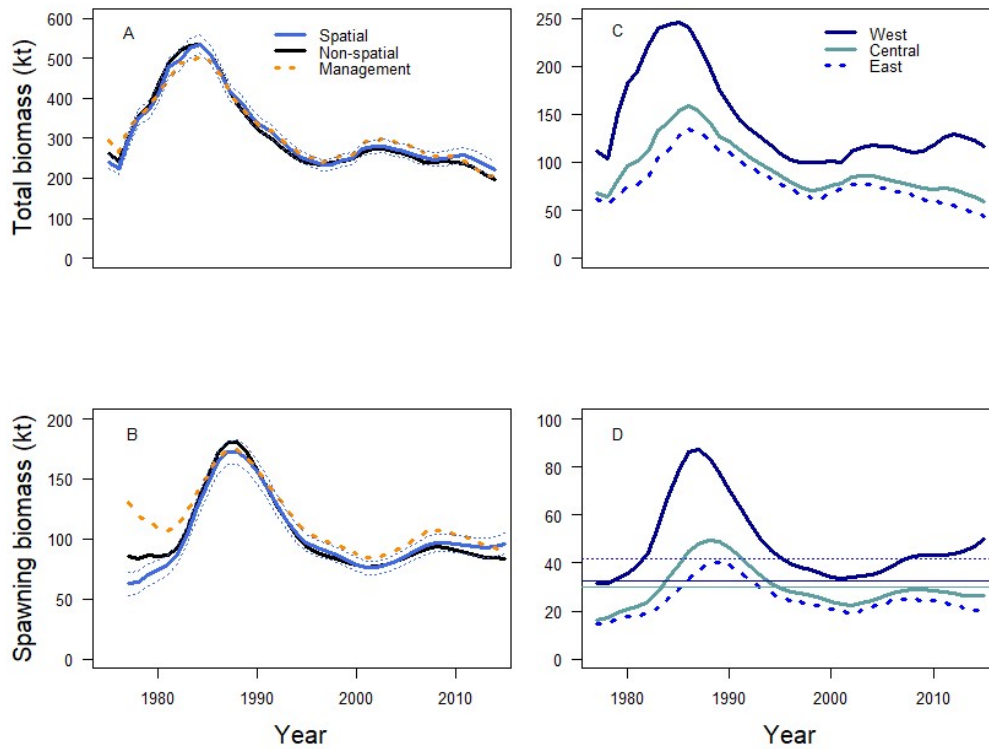


Figure 1.3. Total and spawning biomass comparison. Comparison of age-2+ total and spawning biomass for Alaska sablefish Spatial, Non-spatial, and Management models. A) Total sablefish biomass for the three spatial regions summed (solid blue; approximate 95% confidence intervals in dotted blue) compared to single Non-spatial model (black) and Management model (dashed orange) estimates. B) Spawning biomass for the Spatial model (blue, spatial areas summed; approximate 95% confidence intervals in dotted blue), single-area Non-spatial model (black) and Management model (dashed orange). C) Total biomass for each spatial region from the Spatial model. D) Spawning biomass for each spatial region from Spatial model and the corresponding spatial $B_{40,r}$ reference points. Note each panel has a different vertical axis scale.

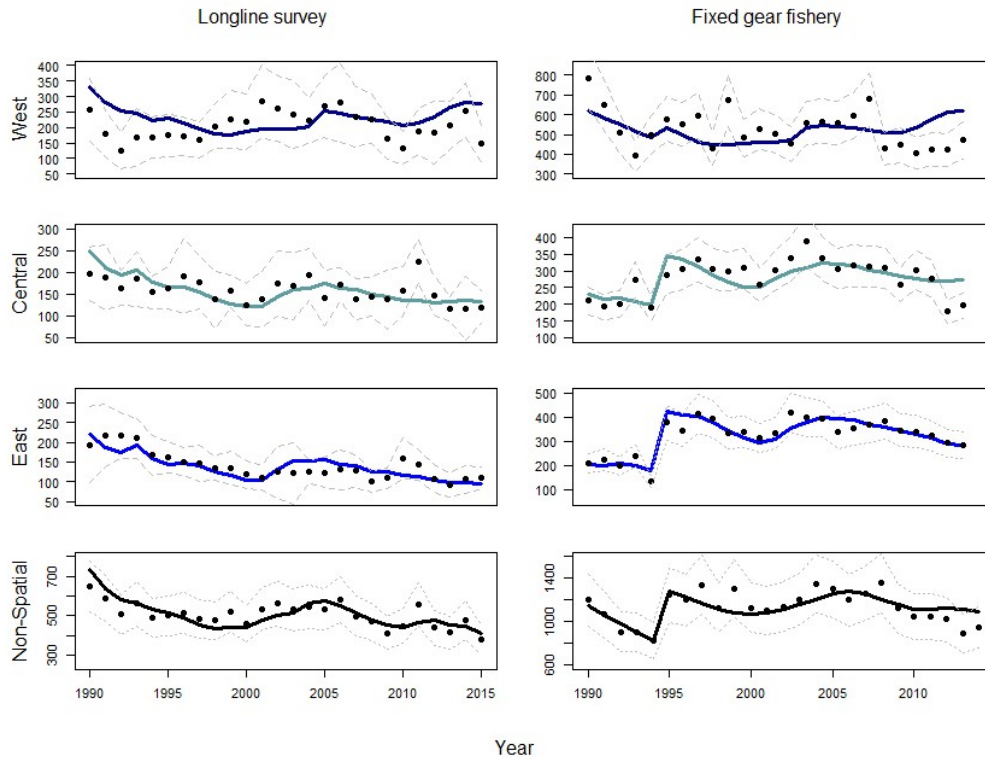


Figure 1.4. Fits to abundance indices. Observed (points) and predicted (line) Alaska sablefish abundance indices for the U.S. longline survey (left column) for 1990–2015 and the U.S. fixed gear fishery (right column) for 1990–2014. Dashed lines represent the approximate 95% confidence intervals of observed data. The first, second, and third rows are the West, Central, and East Spatial model regions, respectively. The bottom row is from the Non-spatial model. Note the different scale for each figure.

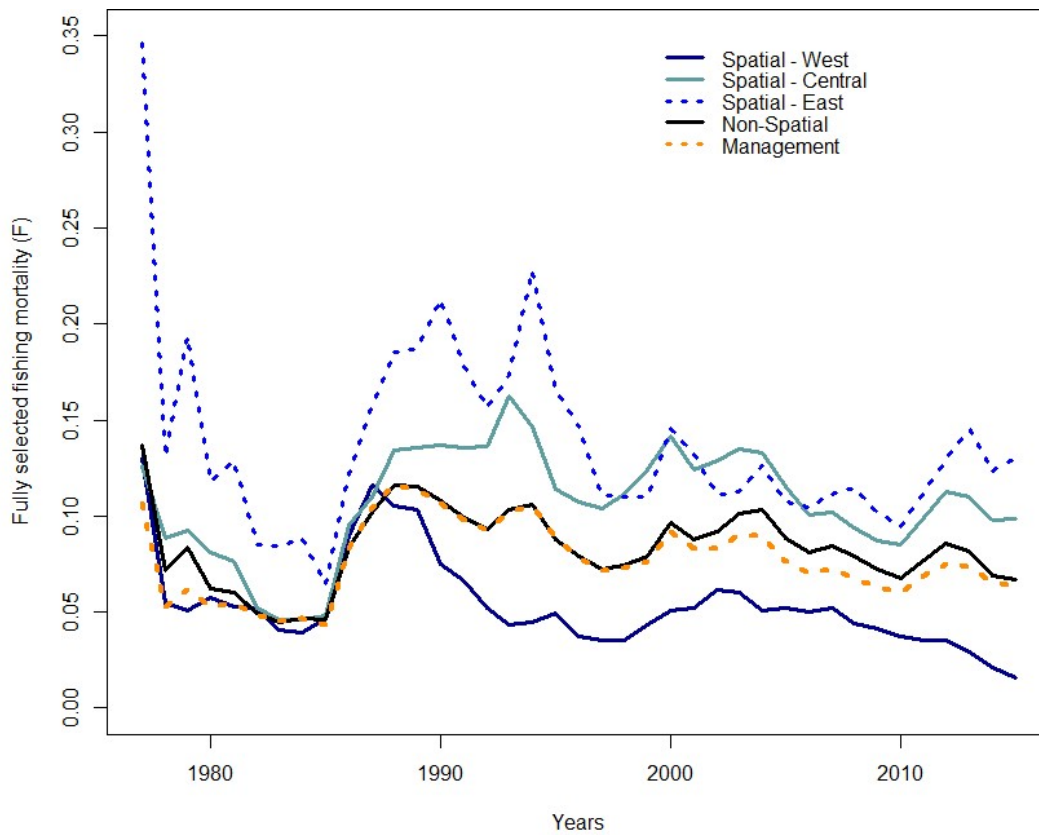


Figure 1.5. Fishing mortality comparison. Fully selected fishing mortality for each of the three modeled regions of the Alaska sablefish Spatial model and for the Alaska sablefish Non-spatial and Management models, 1977–2015.

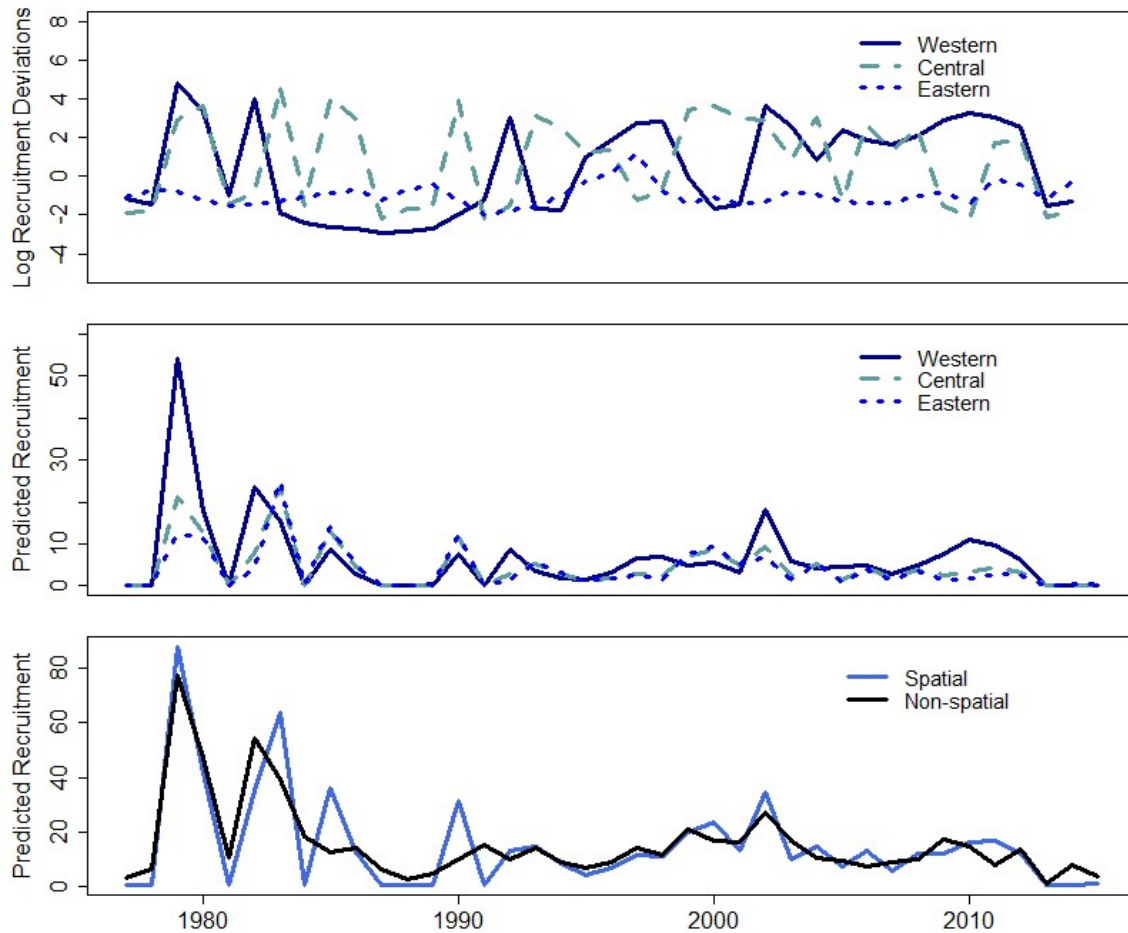


Figure 1.6. Recruitment comparison for Alaska sablefish models. Log recruitment deviations for 1977-2014 for each region in the Alaska sablefish Spatial model, top panel. Predicted recruitment (millions of fish) at age-2 for each spatial region, middle panel. Bottom panel is predicted recruitment (millions of fish) of age-2 sablefish for the Spatial (summed across spatial regions) and Non-spatial models, 1977-2015.

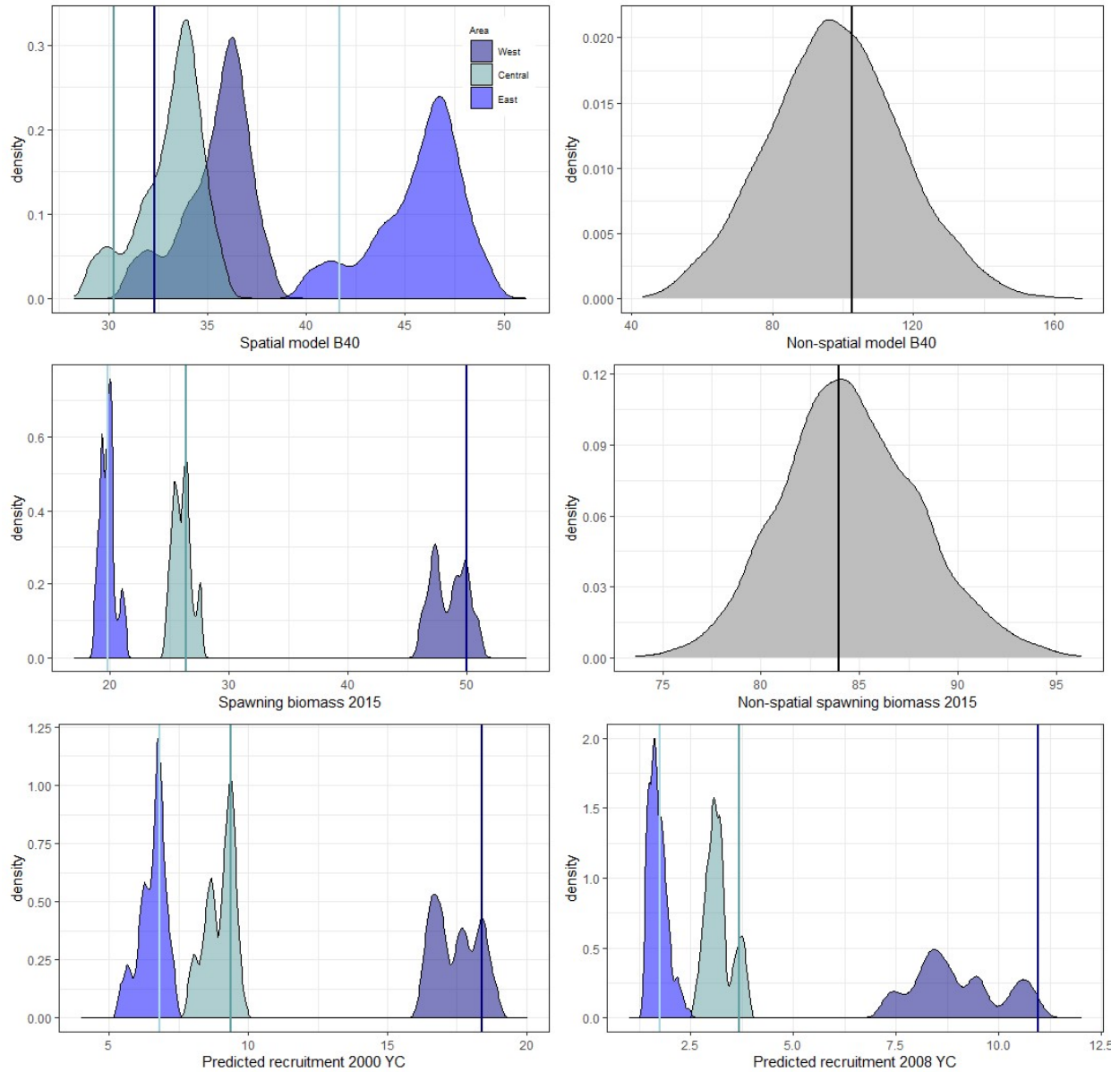


Figure 1.7. Markov chain Monte Carlo (MCMC) simulation results. MCMC simulations of the posterior distribution for selected Alaska sablefish Spatial and Non-spatial model parameters. The point estimates (vertical lines) for West, Central, and East regions are overlaid upon the posterior estimates of B_{40} , spawning biomass in 2015, and predicted recruitment for the large 2000 and 2008 year classes in the left side and bottom row panels. The top two right side panels show point estimates and MCMC posterior distribution for the Non-spatial model B_{40} and spawning biomass in 2015.

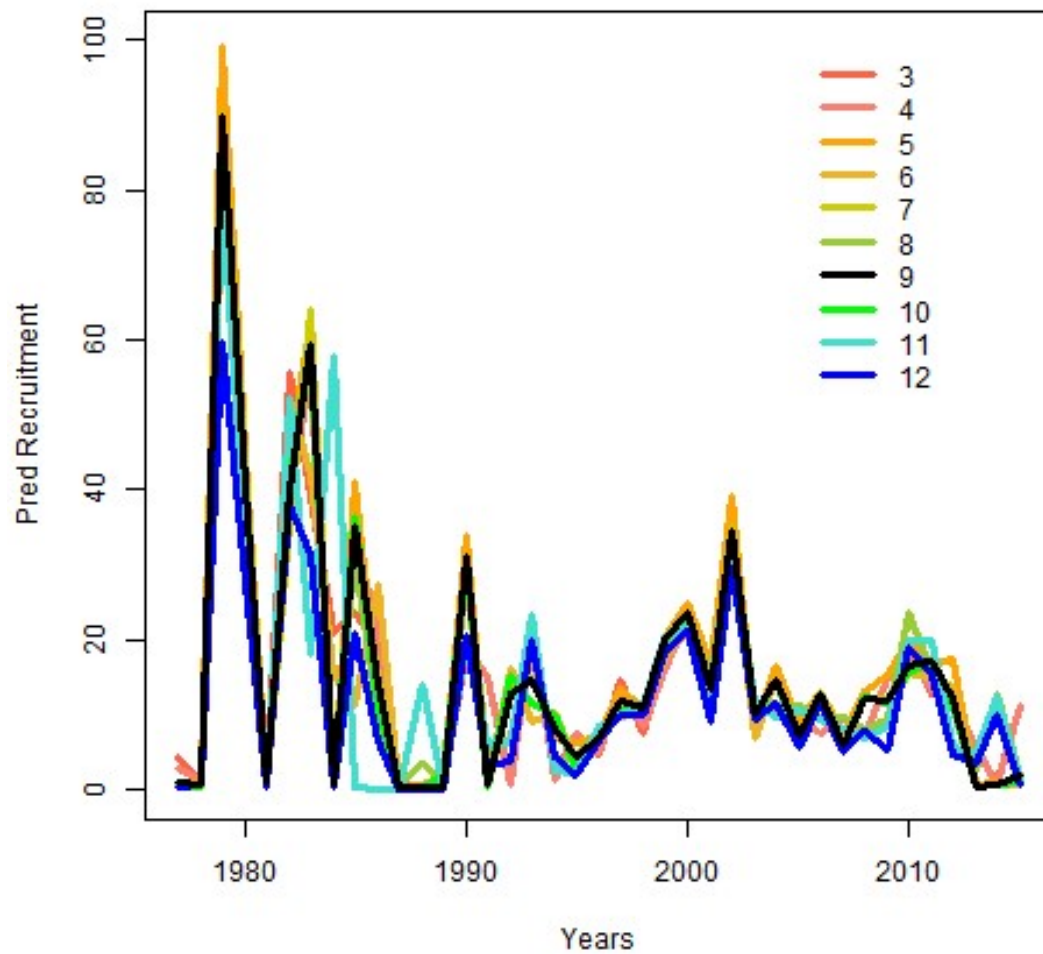


Figure 1.8. Comparison of predicted recruitment. Predicted (pred) recruitment (millions of fish) of age-2 Alaska sablefish (summed over three modeled spatial areas) for sensitivity models with increasing number of spatial selectivity parameters estimated.

1.8 Tables

Table 1.1. Sablefish stock assessment model data. Data available for the Alaska sablefish Spatial, Non-spatial, and Management assessment models. Data are available from the Alaska Fisheries Science Center and Alaska Fisheries Information Network.

Source	'Spatial' & 'Non-spatial' models	'Management' model
Fixed gear fisheries		
Catch	1977–2015	1960–2015
Foreign CPUE*	1977–1981	1964–1981
U.S. CPUE*	1990–2014	1990–2014
Length compositions	1990–2014	1990–2014
Age compositions	1999–2014	1999–2014
Trawl gear fisheries		
Catch	1977–2015	1960–2015
Length compositions	1990,1991,1999,2005–2014	1990,1991,1999,2005–2014
U.S.-Japanese longline survey		
Abundance index	1979–1994	1979–1994
Length compositions	1979–1994	1979–1994
Age compositions	not used	1981, 1983, 1985, 1987, 1989, 1991, 1993
U.S. longline survey		
Abundance index	1990–2015	1990–2015
Length compositions	1990–2015	1990–2015
Age compositions	1996–2014	1996–2014
U.S. Gulf of Alaska trawl survey		
Abundance index	not used	1984, 1987, 1990, 1993, 1996, 1999, 2003, 2005, 2007, 2009, 2011, 2013, 2015
Length compositions	not used	1984, 1987, 1990, 1993, 1996, 1999, 2003, 2005, 2007, 2009, 2011, 2013
Age compositions	not used	1984, 1987, 1996, 1999

* Denotes catch per unit effort (CPUE)

Table 1.2. Sablefish spatial stock assessment model movement rates. Alaska sablefish primary and alternative movement rates between West, Central, and East regions used in Spatial model and model sensitivity analyses.

a) Spatial model base movement matrix, invariant across ages and time where movement values from region x to region y are percentages. This movement transition matrix is used for models 4–12.

From:		To:		
		West	Central	East
	West	67.5	22.3	10.2
	Central	23.9	37.1	39.0
	East	7.9	28.2	63.9

Stationary distribution of the base movement used in spawning potential ratio (SPR) reference point calculations for models 4–12:

From:		To:		
		West	Central	East
	West	31.0	29.0	40.0
	Central	31.0	29.0	40.0
	East	31.0	29.0	40.0

b) Movement rates by size group for sensitivity analyses for small (<57 cm), medium (57-66 cm), and large (>66 cm) sablefish; values presented are percentages. These movement transition matrices are used for models 15–17.

From:		To:								
		Small			Medium			Large		
		West	Central	East	West	Central	East	West	Central	East
	West	69.4	18	12.7	61.6	24.1	14.2	50.1	24.8	25.1
	Central	29.4	32.9	37.7	27.4	33.9	38.8	22.3	31.1	46.6
	East	22.1	38.9	38.9	11.9	27.2	60.9	14.6	28.2	57.1

Table 1.2, cont.

Stationary distribution for size-based movement used in spawning potential ratio (SPR) reference point calculations for Alaska sablefish models 15–17:

From:		To:								
		Small			Medium			Large		
		West	Central	East	West	Central	East	West	Central	East
	West	45.7	27.7	26.6	32.3	28.1	39.6	26.0	28.2	45.8
	Central	45.7	27.7	26.6	32.3	28.1	39.6	26.0	28.2	45.8
	East	45.7	27.7	26.6	32.3	28.1	39.6	26.0	28.2	45.8

c) The length-based movement rates for three size groups in (see b) were converted to age groups for use in the Alaska sablefish Spatial model using female length at age (Option A), male length at age (B), or an average length at age across sexes (C). Movement rate size groups for each conversion option correspond to the following ages:

Size-Age Conversion	Size group		
	Small	Medium	Large
Female length at age (A)	2-3	4-6	7+
Male length at age (B)	2-4	5-13	14+
Average length at age (C)	2-4	5-7	8+

d) No movement, 100% retention in models 3 and 13.

From:		To:		
		West	Central	East
		West	Central	East
	West	1	0	0
	Central	0	1	0
	East	0	0	1

Stationary distribution movement used in spawning potential ratio (SPR) reference point calculations for models 3 and 13:

From:		To:		
		West	Central	East
		West	Central	East
	West	1	0	0
	Central	0	1	0
	East	0	0	1

Table 1.2, cont.

e) Extreme movement, 0% retention in model 14.

From:		To:		
		West	Central	East
	West	0	0.5	0.5
	Central	0.5	0	0.5
	East	0.5	0.5	0

Stationary distribution movement used in SPR reference point calculations for model 14:

From:		To:		
		West	Central	East
	West	33.3	33.3	33.3
	Central	33.3	33.3	33.3
	East	33.3	33.3	33.3

Table 1.3. Parameter and likelihood comparisons. The number of parameters and data likelihood values for explorations of Alaska sablefish stock assessment model sensitivity to parameterization (top) and movement (bottom) for each named model. A stepwise progression of models of increasing spatial complexity (number of spatial parameters estimated) were run, ranging from 165 to 487 estimated parameters. The Movement and Data columns indicate whether movement between spatial regions occurs and whether data were spatially aggregated (aggr.) across West, Central, and East regions. Spatial selectivity (S) and catchability (Q) parameters were estimated as indicated for the pre-IFQ fishery (Pre-IFQ), post-IFQ fishery (Post-IFQ), and the U.S. longline survey (Survey), or were fixed at the Non-spatial model values if 'No' indicated. Alternative movement models were based on the Spatial model with either a) alternative movement rates based on age-based movement (see Table 1.2) or b) hypothetical, symmetrical movement rates of 0% retention of fish in a region (all fish depart starting area each time step) and 100% retention (no movement). Models followed by an asterisk (*) note models that estimated biologically implausible selectivity.

61	Model name and number	Movement	Data	Spatial S	Spatial Q	Parameters	Data likelihood	Max. gradient component
	Stepwise model parameterization							
	<i>Management</i> (1)	No	Aggr.	No	No	222	1579	< 0.001
	<i>Non-spatial</i> (2)	No	Aggr.	No	No	165	2577	< 0.001
	NM.fixed-NSparams (3)	No	Split 3 areas	No	No	447	7920	< 0.001
	Mov.fixed-NSparam (4)	Yes	Split 3 areas	No	No	447	7741	< 0.001
	Mov.est-NSparam (5)	Yes	Split 3 areas	No	No	463	7388	< 0.001
	Mov.est-S-preIFQ (6)*	Yes	Split 3 areas	Pre-IFQ	Pre-IFQ	469	7210	< 0.001
	Mov.est-S-postIFQ (7)	Yes	Split 3 areas	Post-IFQ	Post-IFQ	469	7052	< 0.001

Table 1.3, cont.

Mov.est-S-surv (8)	Yes	Split 3 areas	Survey	Survey	475	6936	< 0.001
<i>Spatial</i> (9)	Yes	Split 3 areas	Post-IFQ	Pre- & Post-IFQ	471	7005	< 0.001
Mov.est-Spatial+preIFQ (10)*	Yes	Split 3 areas	Pre- & Post-IFQ	Pre- & Post-IFQ	475	6889	< 0.001
Mov.est-Spatial+survey (11)*	Yes	Split 3 areas	Post-IFQ, Survey	Pre- & Post-IFQ, Survey	483	6473	< 0.001
Mov.est-Spatial+preIFQ.surv (12)*	Yes	Split 3 areas	Pre- & Post-IFQ, Survey	Pre- & Post-IFQ, Survey	497	6048	0.029
Alternative movement							
100% Retention (no mvmt; 13)	No	Split 3 areas	Post-IFQ	Pre- & Post-IFQ	471	7151	< 0.001
0% Retention (all fish move; 14)	Yes	Split 3 areas	Post-IFQ	Pre- & Post-IFQ	471	6569	< 0.001
Age based, option A (15)	Yes	Split 3 areas	Post-IFQ	Pre- & Post-IFQ	471	6837	< 0.001
Age based, option B (16)	Yes	Split 3 areas	Post-IFQ	Pre- & Post-IFQ	471	6776	< 0.001
Age based, option C (17)	Yes	Split 3 areas	Post-IFQ	Pre- & Post-IFQ	471	6738	< 0.001

Table 1.4. Biomass, depletion and recruitment results. Model estimates for Alaska sablefish spawning biomass in 2015 (SB_{2015}), depletion (SB_{2015}/B_{40}), and mean predicted recruitment (millions of sablefish) for 1977-2015 for each modeled area (for models 3–17), and for areas summed.

Model name (Model number)	Proportion SB_{2015}				Depletion (SB_{2015}/B_{40})				Mean predicted recruitment			
	West	Central	East	Sum/All	West	Central	East	All	West	Central	East	Sum/All
Stepwise model parameterization												
<i>Management</i> (1)				90.0				0.92				15.6
<i>Non-spatial</i> (2)				84.0				0.82				15.4
NM.fixed-NSparams (3)	0.57	0.23	0.20	90.3	1.21	0.65	0.68	0.89	6.43	4.91	4.05	15.4
Mov.fixed-NSparam (4)	0.54	0.27	0.20	94.4	1.64	0.87	0.46	0.94	6.73	4.51	3.94	15.2
Mov.est-NSparam (5)	0.53	0.27	0.20	118.5	1.78	0.95	0.51	1.03	7.47	5.01	4.43	16.9
Mov.est-S-preIFQ (6)*	0.53	0.27	0.20	88.6	1.55	0.84	0.46	0.91	6.34	4.20	4.07	14.6
Mov.est-S-postIFQ (7)	0.52	0.27	0.21	97.8	1.57	0.87	0.48	0.93	6.73	4.65	4.06	15.4
Mov.est-S-surv (8)	0.54	0.26	0.20	95.5	1.61	0.85	0.46	0.93	6.83	4.54	4.08	15.4
<i>Spatial</i> (9)	0.52	0.27	0.21	96.2	1.55	0.87	0.48	0.92	6.65	4.66	4.06	15.4
Mov.est-Spatial+preIFQ (10)*	0.51	0.28	0.21	90.8	1.46	0.84	0.47	0.88	6.47	4.54	4.10	15.1
Mov.est-Spatial+survey (11)*	0.52	0.27	0.21	75.7	1.35	0.75	0.41	0.80	6.22	4.28	3.79	14.3

Table 1.4, cont.

Mov.est- Spatial+preIFQ.surv (12)*	0.48	0.28	0.24	56.0	1.13	0.71	0.43	0.73	4.98	3.52	3.15	11.6
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Alternative movement - *Spatial* model +

100% Retention (no movement) (13)	0.56	0.26	0.19	133.5	1.45	0.86	0.78	1.08	7.65	5.97	4.71	18.3
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0% Retention (all fish move) (14)	0.41	0.33	0.26	77.5	0.97	0.80	0.62	0.79	7.22	5.40	3.93	16.6
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Age based, option A (15)	0.47	0.26	0.27	65.2	1.46	0.74	0.49	0.82	6.78	2.62	2.30	11.7
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Age based, option B (16)	0.48	0.27	0.25	83.9	1.37	0.81	0.53	0.86	7.26	3.60	3.50	14.4
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Age based, option C (17)	0.47	0.26	0.27	71.2	1.44	0.76	0.50	0.83	7.06	2.94	2.68	12.7
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* model converged, Hessian was positive definite, but selectivity estimates not biologically probable

1.9 Appendix 1. Additional spatial model details

This appendix provides additional details on the equations (A.1.1), data (A.1.2), and model specifications (A.1.3) that were components of the Spatial stock assessment model. Numerous additional model variations were run to test the Spatial model sensitivity to parameterization, a selection of relevant results are included here for reference (A.1.4) along with additional results from the Spatial and Non-spatial models.

A.1.1. Additional equations for Spatial model

Predicted catches for each year y , region r , and gear g were calculated as the sum over ages and sexes (s) of the product of abundance at age a (after movement), mean weight at age, and full recruitment fishing mortality from the Baranov catch equation for yield (see Table A.1 for parameter definitions),

$$\hat{C}_{y,r,g} = \sum_a \sum_s w_{a,s} N_{AFT,y,r,a,s} \left(\frac{F_{y,r,a,g,s}}{Z_{y,r,a,g,s}} \right) (1 - e^{-Z_{y,r,a,g,s}}).$$

Predicted values for fishery biomass indices, $\hat{I}$, were calculated as the gear-specific and, where appropriate, region-specific catchability multiplied by survival, abundance, selectivity and mean weight at age, summed over ages and sexes,

$$\hat{I}_{y,r,g} = q_{r,g} \sum_a \sum_s (N_{AFT,y,r,a,s} S_{r,a,g,s} w_{a,s} e^{-0.5Z_{y,r,s}}).$$

Predicted survey abundance follows the same equation but does not include weight at age.

Predicted values for annual survey age composition by region were calculated as the sum over sexes of the product of predicted abundance, $N_{AFT,y,r,a,s}$, and sex and age-specific selectivity, $S_{a,s}$, for each year, region, and age multiplied by an aging error matrix, A ,

$$\hat{P}_{y,r,a} = \left(\frac{A \sum_s N_{AFT,y,r,a,s} S_{a,s}}{\sum_s \sum_a N_{AFT,y,r,a,s} S_{a,s}} \right).$$

Predicted values for fishery age compositions were calculated as the sum over sexes of proportional catch at age multiplied by the aging error matrix, A ,

$$\hat{P}_{y,r,a} = 0.5 \left(\frac{A \sum_s \hat{C}_{y,r,a,s}}{\sum_s \sum_a \hat{C}_{y,r,a,s}} \right)$$

Survey length composition predicted values were sex-specific and a sex-specific age-length transition matrix, K_s , was used to convert lengths to age in the model,

$$\hat{P}_{y,r,l,s} = \frac{N_{AFT,y,r,a,s} S_{a,s}}{\sum_a N_{AFT,y,r,a,s} S_{a,s}} K_s.$$

Fishery length compositions predicted values were sex-specific and a sex-specific age-length transition matrix, K_s , was used to match lengths to age in the model,

$$\hat{P}_{y,r,l,g,s} = \frac{\hat{C}_{y,r,l,g,s}}{\sum_l \hat{C}_{y,r,a,g,s}} K_s.$$

Model estimation

The Spatial model was fit by minimizing the negative log-likelihood of L_{Total} , which consisted of the individual log-likelihoods for catch L_C , indices of abundance L_I , age compositions L_{age} , and length compositions L_{length} , and included likelihood penalties for catchability deviations L_q , recruitment deviations L_τ , and fishing mortality, $L_{Total} = \sum_x L_x$.

The likelihood component for catch was the residual sum of squares between observed and predicted logarithms of catch, and assumed lognormally distributed error,

$$L_C = \sum_g \sum_y (\ln C_{g,y} - \ln \hat{C}_{g,y})^2.$$

Indices of relative abundance were assumed to have a lognormal error distribution and were fitted by minimizing the sum of squared differences between the log observed and predicted abundance or biomass,

$$L_I = \sum_r \sum_g \sum_y \frac{(\ln I_{y,r,g} - \ln \hat{I}_{y,r,g})^2}{(2\sigma_I^2)}.$$

Age and length compositions were assumed to have multinomial likelihood distribution with robust likelihoods (Crone and Sampson 1998),

$$L_{age} = \sum_r \sum_{n_{r,g}} -\psi_{y,r}^g \sum_a (P_{y,r,a} + v) \ln(\hat{P}_{y,r,a} + v), \text{ and}$$

$$L_{length} = \sum_r \sum_s \sum_{n_{r,g}} -\psi_{y,r,g} \sum_l (P_{y,r,l,g,s} + v) \ln(\hat{P}_{y,r,l,g,s} + v),$$

where $\psi_{y,r}^g$ was effective sample size for year and region, $n_{r,g}$ was the number of years of data for region and gear, and v was a constant equal to 0.001.

A prior for catchability was added to the total likelihood, which penalized deviations in estimated gear-specific catchabilities ($\hat{q}_{r,g}$), given an input expected value for catchability ($q_{\mu_{r,g}}$) and an assumed variance for the lognormal prior distribution for catchability parameters (σ_q^2).

$$L_q = \left(\ln \hat{q}_{r,g} - \ln q_{\mu_{r,g}} \right)^2 / 2\sigma_q^2.$$

A penalty, λ_r , with a value of 0.1 was imposed upon log recruitment deviations, where η was the number of years for which recruitment deviations were generated, σ_r was the standard deviation for recruitment deviations, and the total likelihood was penalized when recruitment deviations differed from zero,

$$L_\tau = \lambda_r \sum_y \sum_r \frac{(\tau_y)^2}{2(\sigma_{Rec})^2} + \eta \ln \sigma_{Rec}.$$

A penalty, λ_f with a value of 0.1 was imposed upon fishing mortality deviations, such that large fluctuations in fishing mortality summed over regions, gears, and model years increased the total log likelihood,

$$L_f = \lambda_f \sum_r \sum_g \sum_y (\phi_{y,r,g})^2,$$

where $\phi_{y,r,g}$ were annual deviations in regional fishing mortality, F (recall that $F_{y,r,g} = \mu_{f,r,g} * \exp(\phi_{y,r,g})$).

Finally, a penalty was included in the objective function to fit the spawning potential ratios,

$$L_{\hat{F}_{x,r}} = \sum_r 100 * \left(\frac{\sum_a SBR_{x,r}}{\sum_a SBR_{0,r}} - x \right)^2, \text{ where } x \text{ was } 0.40.$$

A.1.2. North Pacific sablefish data

Data are available from the NOAA Alaska Fisheries Science Center and Alaska Fisheries Information Network (AKFIN; www.akfin.org). Catch data for the West region included very small amounts of sablefish caught in Alaska state waters. Data from Alaska state-caught sablefish in the Central (Prince William Sound) and East (Clarence and Chatham Straits) regions were not included in the data used for the analyses described herein and are assessed separately by the state of Alaska. Catch reporting is assumed to be reliable in the North Pacific, so an attempt was made to fit catches closely using a high weight for the catch likelihood component. Catch data were reported in 1000s of tons (kt).

Spatial age composition data were available from the U.S. longline survey and the U.S. fixed gear fishery and are not sex-specific. Proportions of fish in single-age bins for ages 2–31+ were calculated using aged otoliths randomly sampled by fishery observers or survey staff. An aging error matrix generated using known-age sablefish was used when generating predicted age compositions (Heifetz et al. 1999).

Age compositions from the U.S. fixed gear fishery in the West region were weighted by the catch in each of the three component sub-regions (Bering Sea, Aleutian Islands, and western Gulf of Alaska) to reflect each sub-region's contribution. Length compositions from the U.S. fixed gear fishery in the West region were weighted by catch in each of the three component sub-regions as was done for age compositions. Due to fishery regulations, the majority of trawl fishery samples since 1995 have come from the Central region, so trawl fishery length composition data were not split into spatial regions.

Spatial, sex-specific length compositions from the fixed gear fishery, the U.S. longline survey, and the U.S.-JP longline survey were included in the assessment model. Trawl survey length compositions were sex specific but not spatial. All, lengths were binned into odd-numbered, 2-cm bins, including fish 41–99+ cm, with the plus group aggregating fish larger than 99 cm.

The Bering Sea and Aleutian Islands are surveyed in alternate years, the Gulf of Alaska sub-regions are sampled every year. For indices fit in numbers (U.S. longline survey and U.S.-JP

cooperative longline survey), the $w_{a,s}$ in Eq. A5 was omitted. The U.S. longline fishery index was calculated using data collected from trips targeting sablefish reported by a combination of on-board observers and vessel captains in both voluntary and mandatory logbooks.

Movement rates between spatial regions were estimated outside the assessment model as in Hanselman et al. (2015a), except that movement rates were estimated for fish of all lengths combined. Fish tagged or recaptured in Chatham Strait and Clarence Strait were excluded from the movement analyses because catches and indices from these areas were not included in the Spatial model.

A.1.3. Alaska sablefish Spatial model specifications

For the sablefish Spatial model, a total of 471 parameters were estimated (Table A.3).

Sablefish begin to recruit to the fishery and longline survey at age 2, and are assumed to have a 1:1 sex ratio at recruitment. Annual recruitment deviation estimation begins in 1945. Spawning is assumed to occur at the start of the year for *SPR* calculations.

Selectivity is often challenging to estimate. Sablefish trawl fishery selectivity was fitted using a dome-shaped, 2-parameter gamma function. The a_{max} parameter was difficult to estimate in early model versions, so the a_{max} for males was not estimated and was instead set equal to a_{max} for females.

In early versions of the Spatial model, logistic selectivity parameters were poorly estimated for males and females in the U.S.-JP cooperative longline survey and the sex-aggregated foreign fixed gear fishery. Consequently, we fixed the parameters at the values in Hanselman et al. (2015b). Non-spatial, sex-specific, asymptotic selectivity was estimated for the U.S. longline survey. The age at 50% maturity parameter, $a_{50\%}$, is estimated for each sex while gear- and sex-specific $\delta_{g,s}$ was fixed to values from Hanselman et al. (2015b) due to difficulty in estimation. The Spatial model did not include spatial selectivity for the pre-IFQ U.S. fixed gear fishery because selectivity parameters were poorly estimated in early model versions. The $a_{50\%}$ parameter was estimated for males and females, the δ_s parameters were fixed at Hanselman et al. (2015b) values. Finally, the U.S. fixed gear fishery post-IFQ selectivity $a_{50\%}$ parameters were

estimated for each region and sex, whereas the δ_s parameters were estimated for males and fixed at the Non-spatial model value for females.

Fishing mortality incorporates two time blocks for the fixed gear fishery to account for changes in selectivity due to the IFQ program. For 1994 and prior, the pre-IFQ selectivity was used, for 1995 onward, the post-IFQ selectivity was applied.

Because effective sample sizes for age and length compositions may be lower than actual sample sizes due to violation of independence, effective sample size for a given composition was chosen to average 20 across regions (as in Hanselman et al. 2015b), with the regional proportion of actual sample size as a scaling factor for each set of composition data. Iterative reweighting was done on the data weights until the standard deviation of normalized residuals (SDNR) values were approximately 1.0 (Francis 2011).

The values for catchability priors and CVs used in the Spatial model were based on values estimated by Hanselman et al. (2007), which were calculated from comparisons of mean trawl survey biomass estimates for the Gulf of Alaska and Aleutian Islands to longline survey or fishery estimates of relative biomass. The variance of the catchability was estimated using the delta method. For gears and time periods for which spatial catchability was estimated, the priors were the same among spatial regions under the presumption that, without additional information, catchability would be constant among regions.

The likelihood component for catch was comprised of a penalty (weighting) term λ and the residual sum of squares between observed and predicted logarithms of catch, and assumed lognormally distributed errors, with a CV of 0.02.

A.1.4. Additional results

Fits to U.S.-JP Cooperative Longline Survey and to Foreign Fixed Gear Fishery index for the Spatial model were generally good, though there were some patterns in residuals (Figure A.1).

Despite a relatively large weighting penalty in the objective function, the Spatial model does not fit catches closely, particularly for the fixed gear in the West and Central regions

(Figure A.2). The predicted catches were overestimated for the Western region fixed gear fishery in the late 1980s and early 1990s, prior to the implementation of the IFQ program and were underestimated in the Central region for the same period of years. For regions summed, the predicted and observed catches were closer (Figure A.2), with the over- and underestimates of catches canceling between regions.

The base Spatial model estimated a single mean recruitment for sablefish of -1.2 (log scale, CV 19%) and annual spatial deviations from the mean for 1949–2013. The abundance of age-2 predicted recruits for each region was highly variable, though the individual regions showed similar trends. The predicted recruitment values were the most variable and had the greatest mean for the West region, ranging from 0.05 to 54.4 million recruits (mean 6.6 ± 1.5 SE). The Central and East regions had predicted recruitment ranging from 0.08–23.7 million (5.5 ± 0.9 SE) and 0.10–25.1 million (5.1 ± 0.8 SE), respectively. The sum of predicted recruitment across regions had similar trends and means as the single Non-spatial model, with a range of 0.28–88.0 million (mean 15.3) for the Spatial and range of 0.87–77.8 million (mean 15.6) for the Non-spatial models.

Estimated selectivity parameters for the base Spatial model for the U.S. longline survey, the pre-IFQ U.S. fixed gear fishery, and the U.S. trawl fishery were well-estimated (CVs 1–5%) and generally very similar to the single-area Non-spatial model (Table A.4). The age at 50% selectivity ($a_{50\%}$) was estimated by sex and region for the post-IFQ U.S. fixed gear fishery. The estimated age at 50% selectivity was lower for the Western region than for the Central or Eastern regions, and CVs ranged from 3 to 20%. This pattern of selecting for smaller, younger fish in the West was consistent with age composition data, which showed a greater proportion of younger (and likely smaller) sablefish in the West (Figure A.3).

The Spatial model estimates for catchability were greater than the single Non-spatial model estimates for fleets or surveys with non-spatial catchability (Table A.1.1). Catchability was estimated spatially for the U.S. longline fishery pre- and post-IFQ implementation and the spatial values had a mean of 4.4 and 6.2, respectively, which was also greater than the single Non-spatial catchability estimates for those fleets.

A.1.5. References

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Appendix 1 Figures

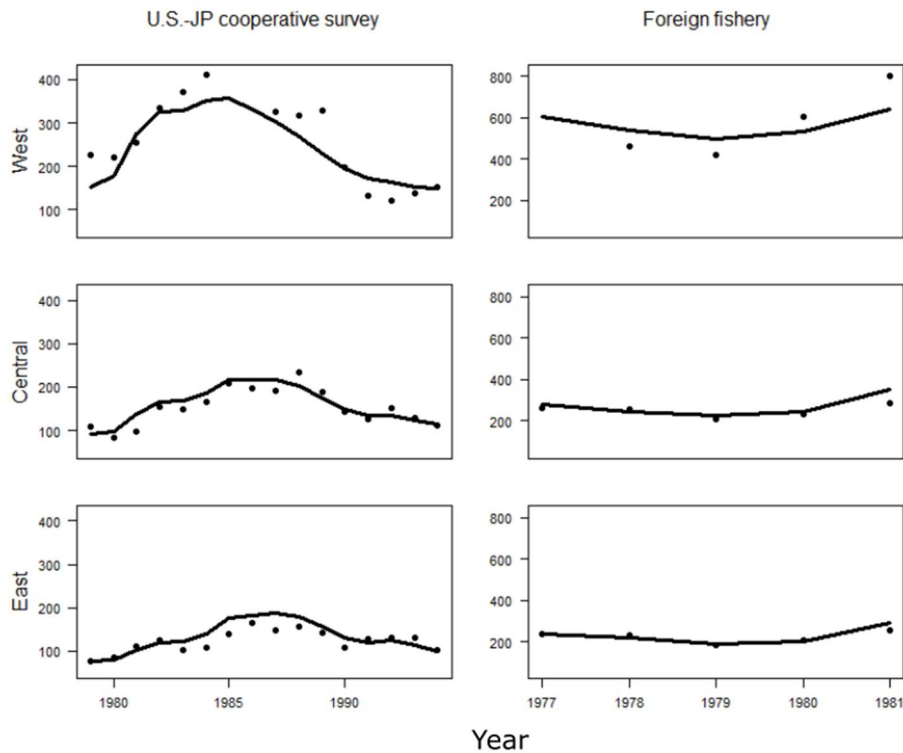


Figure A.1. Observed (points) and predicted (line) estimates of abundance indices for the U.S.-JP cooperative longline survey (left column) for 1977-1994 and the relative population weight for the foreign fixed gear fishery (right column) for 1977-1981. These indices were developed for Alaska sablefish in each of the three modeled spatial regions. Top, middle, and bottom rows are the West, Central, and East Spatial model regions, respectively. The sablefish data in the Spatial model were aggregated into three Alaska regions (West, Central, East), each of which contained single or grouped management area as defined by the North Pacific Fishery Management Council; Bering Sea, Aleutian Islands, Western Gulf of Alaska (GOA), Central GOA, West Yakutat, and East Yakutat/Southeast Outside. Note the different scale for left and right columns.

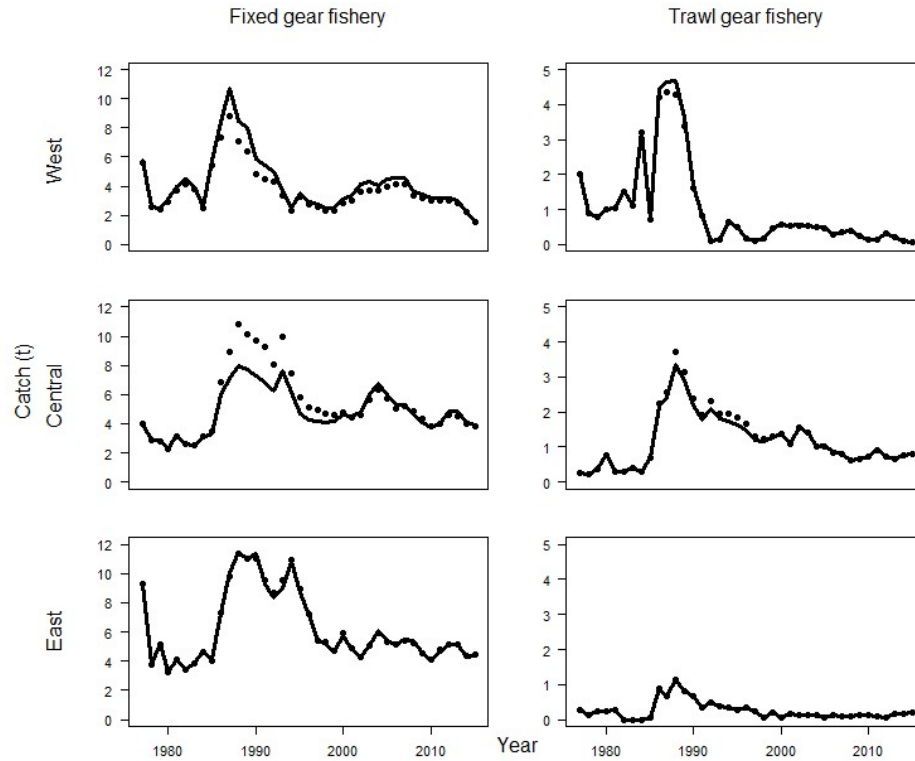


Figure A.2. Predicted and observed catch of Alaska sablefish by gear and region for the Spatial stock assessment model. The sablefish data in the Spatial model were aggregated into three Alaska regions (West, Central, East), each of which contained single or grouped management area as defined by the North Pacific Fishery Management Council; Bering Sea, Aleutian Islands, Western Gulf of Alaska (GOA), Central GOA, West Yakutat, and East Yakutat/Southeast Outside. Observed catches are black dots, predicted catches are black lines.

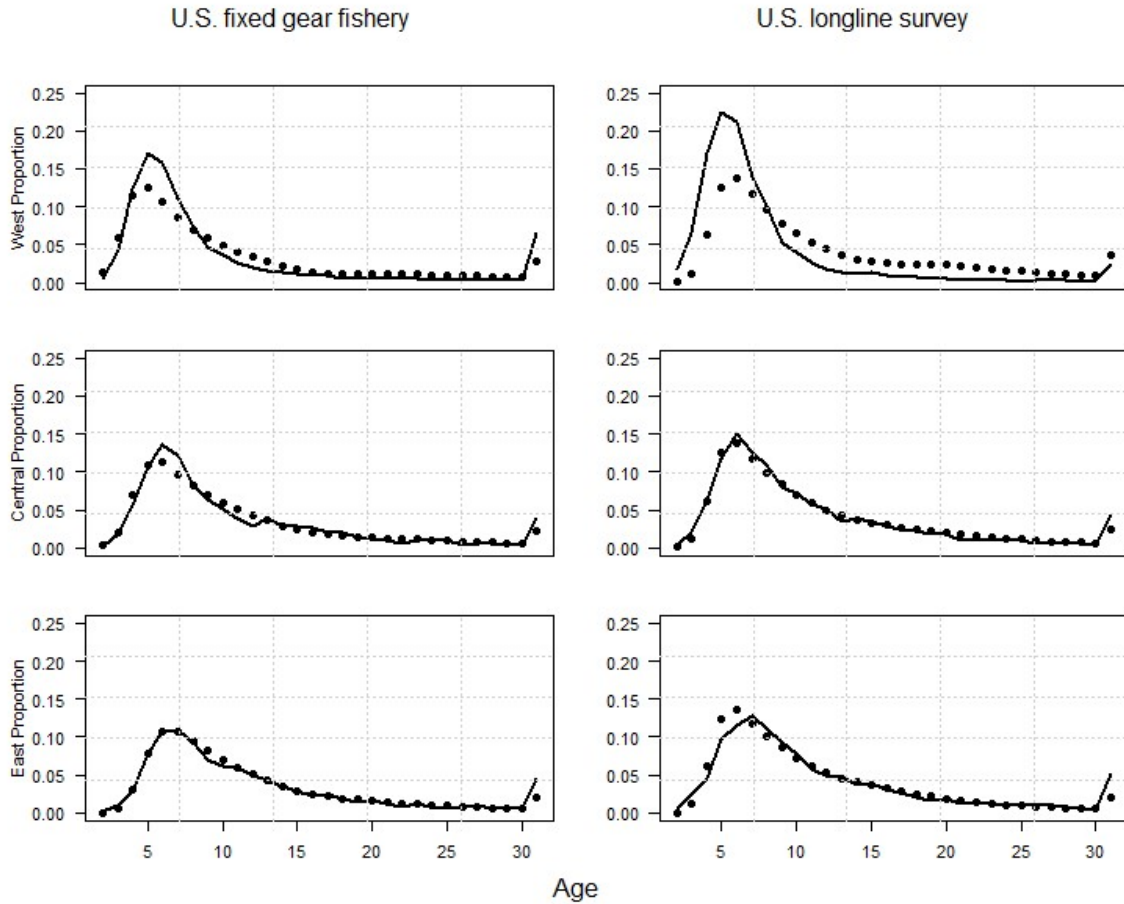


Figure A.3. Observed (dots) and predicted (lines) age compositions by region, aggregated across years for U.S. fixed gear fishery and U.S. longline survey for Alaska sablefish. The sablefish data in the Spatial model were aggregated into three Alaska regions (West, Central, East), each of which contained single or grouped management area as defined by the North Pacific Fishery Management Council; Bering Sea, Aleutian Islands, Western Gulf of Alaska (GOA), Central GOA, West Yakutat, and East Yakutat/Southeast Outside.

Appendix 1 Tables.

Table A.1. Parameters and subscript definitions for the Alaska sablefish Spatial model.

y	Year, $y = 1, 2, \dots$
a_1	Age at recruitment to the model
a_+	Plus-group age class
r	Region
g	Gear or fleet
s	Sex
l	Length
$w_{a,s}$	Mean weight at age a and sex s
m_a	Maturity at age a
μ_{Rec}	Mean log recruitment
$\tau_{y,r}$	Annual recruitment deviation [$\sim \ln(0, \sigma_{Rec})$] for year and region
σ_{Rec}	Recruitment standard deviation
$\mu_{f,r,g}$	Mean log fishing mortality for region and gear
$\phi_{y,r,g}$	Annual fishing mortality deviation
λ_r, λ_f	Penalty for recruitment deviations, fishing mortality deviations
M	Natural mortality
$N_{BEF,y,r,a,s}$	Abundance in number for year, region, age, and sex, before movement
$N_{AFT,y,r,a,s}$	Abundance in number for year, region, age, and sex, after movement
$\bar{F}$	Historic mean fishing mortality estimate for population initialization
$F_{y,r,g}$	Fishing mortality for year, region, and gear

$Z_{y,r,a,s}$	Total mortality for year, region, age, and sex
$S_{r,a,g,s}$	Selectivity for region, age, gear, and sex
$a_{50\%}$	Age at 50% selection for logistic selectivity
$\delta_{g,s}$	Slope/shape parameter for selectivity
a_{max}	Dome shaped gamma selectivity parameter
$q_{r,g}$	Catchability for region and gear/index
$q_{\mu_{r,g}}, \sigma_{r,g}$	Prior mean, standard deviation for catchability for gear/index and region
$I_{y,r}, \hat{I}_{y,r}$	Observed, predicted indices of abundance/biomass for year, region
$P_{y,r,a}, \hat{P}_{y,r,a}$	Observed, predicted age compositions for year, region, and age
$P_{y,r,l,g,s}, \hat{P}_{y,r,l,g,s}$	Observed, predicted length compositions for year, region, length, gear, sex
A	Age error matrix, dimensions a_+ by a_+
K_s	Age-length conversion matrix by sex, dimensions a_+ by number of length bins
L_x	Components of the likelihood function
$SB_{y,r}$	Female spawning biomass in year and region
$B_{y,r}$	Total biomass in year and region
$B_{SPR_x}, B_{SPR_x,r}$	Biomass which is $x\%$ of unfished biomass for the population or region r
$F_{x\%,r}$	Fishing mortality rate reference point that reduces spawning potential ration to $x\%$ of the unfished spawning potential in each region
$B_{SPR_x,r,a}$	Female spawning biomass for reference points $x\%$ of unfished spawning biomass for age a and region r
$\Phi_{i,j}^T$	Transfer coefficient for movement of sablefish between regions j and k , where $j \neq k$

Table A.2. Sablefish spatial stock assessment model catchability for the base model. For each index the prior value and log-scale model-estimated values are provided. CV indicates a coefficient of variation (CV) SD indicates the standard deviation.

Fleet	Prior mean	Prior CV	Log scale estimated value	Estimated value SD	Estimated value CV
U.S. longline survey	7.857	0.33	2.14	0.03	0.015
U.S.-JP cooperative longline survey	4.693	0.242	1.61	0.02	0.015
U.S. longline fishery pre-IFQ, West	4.967	0.328	1.71	0.06	0.032
U.S. longline fishery pre-IFQ, Central	4.967	0.328	1.19	0.06	0.049
U.S. longline fishery pre-IFQ, East	4.967	0.328	1.29	0.06	0.049
JP longline fishery	4.967	0.328	2.61	0.07	0.027
U.S. longline fishery post-IFQ, West	4.967	0.328	1.70	0.04	0.024
U.S. longline fishery post-IFQ, Central	4.967	0.328	1.62	0.04	0.024
U.S. longline fishery post-IFQ, East	4.967	0.328	2.04	0.04	0.021

Table A.3. Number and identification of parameters estimated inside the Alaska sablefish base Spatial stock assessment model.

Catchability	Number Parameters
U.S. longline survey q	1
U.S.-JP cooperative survey q	1
Foreign fixed gear fishery q	1
U.S. fixed gear fishery, pre_IFQ q_r	3
U.S. fixed gear fishery, post IFQ q_r	3
Log mean recruitment μ_R	1
$F_{35,r}, F_{40,r}, F_{50,r}$	9
Recruitment deviations $\tau_{y,r}$	198
Log mean fishing mortality $\mu_{f,r,g}$	6
Fishing mortality deviations $\phi_{y,r,g}$	234
Selectivity $S_{r,a,g,s}$	
U.S. longline survey $a_{50\%,s}$	2
U.S.-JP cooperative longline survey	0
Foreign fixed gear fishery	0
U.S. fixed gear fishery, pre-IFQ $a_{50\%,s}$	2
U.S. fixed gear fishery, post-IFQ $a_{50\%,r,s}$	6
U.S. fixed gear fishery, post-IFQ δ_s	1
U.S. trawl fishery a_{max}	1
U.S. trawl fishery $\delta_{g,s}$	2
Total Parameters Estimated	471

Table A.4. Age at 50% selectivity for the Alaska sablefish Spatial and Non-spatial models. The U.S. fixed gear fishery was split into two time periods, before the implementation of the sablefish individual fishing quota (pre-IFQ) program, and after implementation of the program (post-IFQ).

Spatial model					
	U.S. longline survey	U.S. fixed gear fishery, pre-IFQ	U.S. fixed gear fishery, post-IFQ		
			West	Central	East
Females	4.3	3.3	3.4	4.1	5.2
Males	4.2	10.8	3.3	4.2	4.7
Non-spatial model					
	U.S. longline survey	U.S. fixed gear fishery, pre-IFQ	U.S. fixed gear fishery, post-IFQ		
Females	4	3.2	4		
Males	3.4	8.9	4.5		

Chapter 2: Modeling spatial complexity in population dynamics: Application of a simulation-estimation model for Alaska sablefish (*Anoplopoma fimbria*)²

2.1 Abstract

Management strategy evaluation (MSE) is a useful tool for understanding risk associated with management procedures, arising from uncertainty in data, management implementation, or population dynamics. Key decisions are made during MSE development and implementation regarding the complexity of the operating model (OM) and the types of uncertainty and biological variability to incorporate. I used Alaska sablefish as a case study for exploring the level of complexity necessary within simulation and estimation tools to adequately represent population dynamics at scales relevant for management. I developed a simulation-estimation framework, which simulated population dynamics for Alaska sablefish across six management areas, then used a single-area, panmictic population estimation model to assess the framework's suitability for exploring management questions about apportionment of harvest. A suite of alternative assumptions about movement and recruitment dynamics were explored to validate the estimation model (EM) performance and understand EM performance during simulations, when human-led decision-making cannot easily be replicated. The framework produced sablefish spatial population dynamics that generally match our best understanding of the population under current and historical sablefish biomass and distribution. The EM was more robust to alternative assumptions about recruitment than about movement rates of the underlying population. Moreover, since the EM was tuned to the historical population, it lacked the flexibility to estimate population dynamics that substantially differed from historical conditions. When developing simulation tools analysts should consider the complexity of the OM and EM with the aim that a framework should be complex enough to represent the population dynamics at scales relevant for management, but simple enough to balance the lack of human-led decisions that are sometimes needed to optimize complex EMs when they are confronted with data outside of the historical conditions upon which they were developed.

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2.2 Introduction

Management strategy evaluation (MSE) is a valuable tool for engaging stakeholders and analyzing the strengths and weaknesses of current or proposed management actions relative to specified performance metrics (Smith et al. 1999, Butterworth 2007, Punt et al. 2016). MSEs have been conducted in name or in function since the 1980s, but computing power was limited for their full adoption (Quinn 2003). Computing power has increased substantially over subsequent decades, allowing full MSEs to be conducted for population assessment models that are also increasingly complex. The analytical models available for assessing population dynamics now range from surplus production models to age- or length-structured assessments, multi-area stock assessments, and occasionally multispecies stock assessments (Quinn and Deriso 1999, Maunder and Punt 2013, Ono et al. 2018). Concurrently, our understanding of population structure has also increased for many species through the continued accumulation of data and knowledge about fish movement and distribution and the development and refinement of genetic analysis tools (Begg and Waldman 1999, Begg et al. 1999, Cadrin 2020). Over time, other challenges have become apparent, like shifting distributions of fish stocks due to changing climate and oceanographic patterns (Thorson et al. 2016, O’Leary et al. 2020). The resulting uncertainty in spatial and temporal fishery and environmental dynamics has fostered the continued relevancy and expansion of MSE and related simulation tools to test the performance of management strategies and to quantify relative tradeoffs of potential to management strategies.

Spatial population dynamics, resulting from fishing practices, fish life history, habitat availability, or environmental conditions, translate into heterogeneity of stock population structure across space. However, single area, single species, single sex and age-structured stock estimation models are still among the most commonly used tools for assessing fish stock status and estimating management reference points (Goethel et al. 2011). As ecological monitoring time series grow longer and modeling capabilities become more common and accessible, opportunities arise to simulate population structure and management practices with more complex representations of fish population and fishery dynamics. However, key questions

remain about the level of complexity necessary within simulation and estimation tools to adequately represent population dynamics at scales relevant for management.

Alaska sablefish (*Anoplopoma fimbria*) is a high-value resource for fishers and fishing communities in Alaska and other regions of the North Pacific (Fissel et al. 2018). Sablefish are highly mobile and capable of moving large distances, but have also been tagged and recaptured in the same area (Maloney and Heifetz 1984, Hanselman et al. 2015). It is unknown whether this is due to individual variation in fish movement or other reasons. Relatively little is known about the spawning habits of Alaska sablefish, such as whether they undertake spawning migrations, the distribution of critical nursery habitats, and what drives temporal variation in recruitment. During 1977-2020, recruitment strength was highly variable, with some of the largest recruitments coming during periods of low spawning stock biomass (Goethel et al. 2020). This variability has created challenges for the management of Alaska sablefish because there is interest in balancing stability in year-to-year harvest levels in the face of spatially and temporally variable distributions of sablefish abundance and ensuring sustainable harvest that avoids localized depletion.

The portion of the Alaska sablefish stock (hereafter 'Alaska sablefish') in the federal Exclusive Economic Zone is managed and assessed annually on an Alaska-wide scale because movement rates among management areas are high and exploitation rates are relatively low (Hanselman et al. 2015, 2019). Alaska sablefish are managed by the North Pacific Fishery Management Council, which implemented a catch share program in 1995, with individual fishing quotas (IFQs) allocated for the fixed gear (pot and longline) fleet among six management areas (NPFMC 2018a, 2018b). Annual estimates of management reference points and corresponding acceptable biological catch (ABC) for the Alaska sablefish stock must be downscaled to provide regional management recommendations, specifically through apportionment of total ABC among the six management areas (Figure 2.1).

While methods for apportioning ABC to management areas have changed over time, and fishery stakeholders have proposed alternatives, there remains a lack of consensus regarding a preferred ABC apportionment method (Hanselman et al. 2014, 2019). A spatial

stock assessment model with three spatial areas and spatially-explicit parameter estimation was developed in Chapter 1, but parsing to more than three model areas was impeded by small sample sizes for age composition data, uncertainty in movement rates, poor area-specific data in the historical period, and poorly estimated selectivity. Given the continued urgency from fishery managers and stakeholders to evaluate the performance of alternative ABC apportionment methods with respect to stock dynamics, sustainability, and fishery performance, we developed the simulation-estimation framework described here to explore tradeoffs among candidate apportionment methods in meeting biological and stakeholder objectives. Our objectives were to 1) simulate spatial dynamics of the Alaska sablefish population using an operating model (OM) that incorporated our best understanding of the spatial population dynamics of Alaska sablefish, which result from movement, fishing dynamics, and recruitment, 2) evaluate the simulated population to verify it had sufficient functionality to be useful for simulations evaluating apportionment alternatives (Chapter 3), 3) identify universal challenges and tradeoffs that result from using a spatial OM to simulate population dynamics and a relatively complex age and sex-structured stock assessment estimation model (EM) that inherently assumes a panmictic population, and 4) provide guidance and considerations regarding the level of complexity that should be represented within an OM and EM.

In automated simulation analyses, no human-led decision-making can be incorporated, which would mimic real-world assessment and management. For example, when a model does not converge or perform adequately when confronting new data, an assessment author may fine-tune model parameterization or estimation phases. To address objectives three and four, we explored a range of scenarios for the OM to understand which simulated scenarios lead to EM convergence. Understanding that many things can lead to non-convergence, EM convergence rates helped understand potential limitations to the EM. In the process of setting up the simulations, as well as in response to the external reviewer comments from fishery stakeholders and management bodies, a large variety of configurations for OM process and observation error as well as alternative EM configurations were explored (singly and in combination), but not all are included here. The alternative OM configurations presented,

referred to as model scenarios, can be broken into three categories for comparison against the base model scenario and represent specific alternative states of nature regarding movement and recruitment dynamics; 1) OM movement rate assumptions, 2) OM recruitment assumptions, and 3) an OM-EM validation scenario.

2.3 Methods

We developed a simulation-estimation framework that contained two primary components; a six-area operating model (OM) for simulating age and sex-structured population dynamics developed in R (R Core Team 2018) with movement between areas to simulate mortality, growth, movement, and recruitment processes for an Alaska sablefish population, and a single-area, panmictic estimation model (EM) developed in AD Model Builder (Fournier et al. 2012). As this exercise was intended to develop a simulation tool capable of exploring tradeoffs among apportionment alternatives in the context of the current stock assessment process, the EM was similar in structure to the operational age-structured stock assessment model (hereafter ‘Management Model’; Hanselman et al. 2018). The six OM spatial areas correspond to the areas used by the North Pacific Fishery Management Council (NPFMC) for the management of Alaska sablefish; Bering Sea (BS), Aleutian Islands (AI), western Gulf of Alaska (WG), central Gulf of Alaska (CG), west Yakutat (WY), and east Yakutat/southeast outside (EY; Figure 2.1). The EM estimated parameters and management advice (e.g., stock status, and overfishing limits) for Alaska sablefish as a single, panmictic unit using data generated by the OM with observation error and subsequently aggregated across management areas. This aggregation of data across space reflects the practice in the current stock assessment process. The OM and EM operated on an annual time step (y), were age-structured, and had two sexes (s). Ages 2–31 were modeled in both the OM and EM, with age 31 serving as a plus group, accumulating fish aged 31 and older.

2.3.1 Operating model structure

The OM was spatially explicit so potential area-specific population dynamics and fleet or fish behavior (e.g., catchability, selectivity, or fish movement) could be simulated separately for the six sablefish management areas. The Base OM scenario was chosen after dozens of exploratory scenarios because it resulted in an OM population, which balanced a realistic level

of complexity in population dynamics across spatial areas, and provided informative data to the EM that was similar to data available for Alaska sablefish management, but also simplified movement and natural mortality.

The OM simulated sex, age, and spatially-structured population dynamics across two time periods, a deterministic conditioning period for years 1977–2018 that was the same across all OM scenarios, and the 25-year stochastic forward simulation period for years 2019–2043. Five hundred replicate iterations (i) of the simulation-estimation loop were run forward in time for 25 years for each alternative OM specification set (hereafter, “model scenario”). Replicate iterations differed in recruitment process error, fishery and survey age composition observation error, and fishery and survey abundance index observation error. To aid comparison between model alternative OM scenarios during forward projection, the time series of recruitment deviates was the same for every OM scenario. This consistency was accomplished by generating a single vector of annual recruitment deviates for each of the 500 iterations prior to forward simulation and using this same vector as an OM input when simulating all OM scenarios.

Numbers at age ($N_{y,s,a,m}$) for year (y), sex (s), age (a), and area (m) were tracked from the first age at recruitment (i.e., age 2) through a plus age (age 31+), as they moved among spatial areas (m) and were subject to both natural and fishing mortality. We initialized abundance in numbers at age before movement occurred, $N_{BEF,y=1976,s,a,m}$ for the simulated population in an initial year, $y=1976$, of the OM conditioning period. Abundance in this year (1976) was the product of an initial total population abundance value N_{init} that was a fixed input, and the proportions of abundance ρ by sex s at age a as estimated by the Management Model, and the mean proportions of abundance $\bar{p}_{m,init}$ in each management area m were estimated from the longline survey (1981–2017),

$$(1) \quad N_{BEF,y=1976,s,a,m} = N_{init} * \rho_{y=1976,s,a} * \bar{p}_{m,init}.$$

N_{init} in 1976 was assumed to be 93.4 million, which is the 1977 estimate of total abundance from the Management Model (Hanselman et al. 2019), and values for $\bar{p}_{m,init}$ were [0.13, 0.14, 0.11, 0.40, 0.10, 0.12], for the BS, AI, WG, CG, WY, EY, management areas respectively.

Biomass for the initialization year $B_{y=1976,s,a,m}$ and in all subsequent years was calculated from abundance, as the product of abundance and mean weight at age $w_{s,a}$ for each sex. Weight at age was assumed to be equal across spatial management areas and is equal to the fixed values in the Management Model (Hanselman et al. 2019) which were derived from longline survey samples. The longline survey has been conducted annually in the WG, CG, WY, and EY areas since 1987 and in alternating years for the AI and BS since 1996. The survey samples fixed stations of the upper continental slope and major gullies using baited longline gear, collects otoliths and lengths for age and length compositions, and catch data for an abundance index (Siwicke et al. 2022).

For 1977-2017, within the conditioning period, the magnitude of annual recruitment R_{cond} , or abundance of the first age in the OM, was fixed at the values estimated by the Management Model (Hanselman et al. 2019). Recruitment for 2018 was assumed to be approximately equal to the mean recruitment estimated by the Management Model over a recent and representative period 1995-2017 (16.5 million recruits). Sensitivity to this specification was also examined in alternative scenario R_alt2018 (Table 2.1).

In the forward projection period, simulated total recruitment R_{proj} was simulated with annual log-scale recruitment deviations drawn from a normal distribution (process error) with an arithmetic mean of 16.5 million recruits ($\overline{Rec}$), and a lognormal standard deviation of $\sigma_r = 0.8$ as $dev_y \sim Normal(0, \sigma_r^2)$ with no autocorrelation, and including the appropriate lognormal correction, $R_{proj,y} = e^{\overline{Rec}} * e^{(dev_y - (\sigma^2/2))}$. Total recruitment was the same across all model scenarios but differed for each year y and replicate simulation i . Note, the subscript i for replicate simulations is excluded from equations for simplicity. No autocorrelation in recruitment was specified in simulations, given the absence of correlated patterns in recruitments estimated by the Management Model.

Recruitment was divided into areas, m , in the OM to seed abundance by area for the first age of the OM. During the conditioning period, recruitment distributions to areas were based on the mean proportions $\mathbb{P}_m$ of age-2 fish from the longline survey (1981–2017) and were equal to [0.14, 0.07, 0.13, 0.43, 0.14, 0.09] for the BS, AI, WG, CG, WY, and EY areas,

respectively. These proportions were the same across simulation replicates and model scenarios. During the forward projection period, recruitment was distributed to areas based on a random draw from a multinomial distribution conditioned on the mean proportions P_m of age-2 fish from the longline survey (1981–2017). The multinomial sample size, controlling the variability in recruitment proportions across space, was set at 100, and multinomial draws for recruitment were standardized to sum to one for the sexes combined. For both the conditioning and forward projection periods, recruitment in each area was split equally between sexes, assuming a 50:50 female:male sex ratio at recruitment:

$$(2) \ N_{BEF,y,s,a=1,m} = 0.5 R_{cond,y} P_m \text{ and } N_{BEF,y,s,a=1,m} = 0.5 R_{proj,y} P_m.$$

No stock-recruitment relationship was assumed because some of the largest observed recruitments have come from low spawning biomass. Additionally, little is known about spawning behavior, spawning aggregations, or juvenile distributions other than the movement rates of tagged fish within and among management areas. Sensitivity to the recruitment specifications was examined in multiple alternate model scenarios (Table 2.1).

Abundance prior to movement in each area for subsequent ages was a function of fishing fleet-aggregated total instantaneous annual fishing mortality F_y and sex and age-specific natural mortality $M_{s,a}$,

$$(3) \ N_{BEF,y,s,a,m} = N_{BEF,y-1,s,a-1,m} e^{-(F_{y-1,s,a-1,m} + M_{s,a-1})}.$$

Natural mortality M was 0.1 for all ages and sexes in these simulations, but equations retain the subscript notation for natural mortality for completeness.

Abundance for the plus group ($a=A$, age 31+) was the sum of new entrants to the plus group and those already in the plus group that survived fishing and natural mortality,

$$(4) \ N_{BEF,y,s,a=A,m} = N_{BEF,y-1,s,a-1,m} e^{-(F_{y-1,s,a-1,m} + M_{s,a-1})} + N_{BEF,y-1,s,a=A,m} e^{-(F_{y-1,s,a=A,m} + M_{s,a=A})}.$$

Total annual age and sex-specific instantaneous fishing mortality F rate for each area was summed across fishing fleets g ,

$$(5) F_{y,s,a,m} = \sum_g (f_{y,m,g} * S_{s,a,m,g}),$$

and was the product of year-, area- and fleet-specific fishing mortality rates $f_{y,m,g}$ and selectivity by sex, age, area, and fleet $Sel_{s,a,m,g}$.

During the OM conditioning period (1976–2018), catch biomasses were fixed inputs equal to observed catches within Alaska management regions that inform the Management Model. The realized fleet-specific fishing mortality rate $f_{y,m,g}$ for each year's observed fleet- and area-specific catch during the conditioning period was found such that the OM predicted catch $\hat{C}_{y,m,g}$ was equal to the observed (input) catch $C_{y,m,g}$ in each year using the method of bisection, where

$$(6) \hat{C}_{y,m,g} = \sum_s \sum_a w_{s,a} \left(N_{BEF,y,s,a,m} * (F_{y,s,a,m,g} / (F_{y,s,a,m,g} + M_{s,a})) * (1 - e^{-(F_{y,s,a,m,g} + M_{s,a})}) \right)$$

and where fleet-specific fishing mortality rates by age were

$$(7) F_{y,s,a,m,g} = f_{y,m,g} * Sel_{s,a,m,g}.$$

Input or observed catch values $C_{y,m,g}$ were the same across replicates for the conditioning period. The estimated fishing mortality rate necessary to produce the observed catch level conditional on simulated abundance was then used to simulate abundance in the next year of an OM iteration (See Eqns. 3–4, above). For the forward projection period, the catch in each area was assumed to equal the ABC apportioned to each area in each year (i.e., 100% of apportioned ABC was harvested). The fishing mortality required to harvest ABC was calculated in the same way as the conditioning period by finding the instantaneous fishing mortality that would produce the apportioned ABC as a function of abundance and selectivity and subsequently used by the OM to simulate sablefish abundance in the next year ($y+1$).

Sex- and age-specific selectivity $Sel_{y,s,a,g}$ for each fleet or survey g were input to the OM. Input values for selectivity were fixed at values estimated by the sablefish spatially-explicit research estimation model (Chapter 1). Fixed gear and survey selectivity used a two-parameter asymptotic function; trawl selectivity used a two-parameter dome-shaped gamma function (Punt et al. 1996). Selectivity for each fleet or survey was the same across simulation replicate iterations and model scenarios.

Fixed gear fishery selectivity was assumed to be best represented by two time blocks, one with years 1977–1994 prior to implementation of IFQ (hereafter ‘pre-IFQ’) for fixed gear fleets (Eqn. 8), and the second representing post-IFQ years 1995–2043 (Eqn. 9). For the pre-IFQ years, selectivity was the same for all OM areas. For the post-IFQ fishery, three OM selectivity vectors were assumed that differed between spatial areas, informed by spatial stock assessment model estimates described in Chapter 1. For selectivity vector one, the OM selectivity was assumed to be the same for the BS, AI, and WG areas. OM selectivity vector two was specified for the CG area. A third distinct selectivity vector was assumed for the EY and WY areas. See Appendix 2 Table A.2 for additional details.

$$(8) \ Sel_{y < 1995, s, a, g}^{fixed\ fishery} = 1 / (1 + e^{-\delta_{y < 1995, s, g}(a - a_{50, y < 1995, s, g})}),$$

$$(9) \ Sel_{y \geq 1995, s, a, m, g}^{fixed\ fishery} = 1 / (1 + e^{-\delta_{y \geq 1995, s, g}(a - a_{50, y \geq 1995, s, m, g})}).$$

Dome-shaped trawl fishery selectivity was the same across areas,

$$(10) \ Sel_{s, a, g}^{trawl\ fishery} = \left(\frac{a}{a_{max}} \right)^{a_{max}/p} e^{(a_{max} - a)/p}, \text{ where } p = 0.5 \left[\sqrt{a_{max}^2 + 4\delta_{s, g}^2} - a_{max} \right].$$

Longline survey selectivity was the same across areas and years,

$$(11) \ Sel_{s, a, g}^{survey} = 1 / (1 + e^{-\delta_{s, g}(a - a_{50, s, g})}).$$

Note that in selectivity equations 8–11, while all selectivities fit within a general $Sel_{y, s, a, m, g}$ structure, in several instances, sex, age, and gear-specific selectivities were either constant across time or shared among spatial regions.

Catchability parameters q were fixed at values estimated in the sablefish spatial research model developed in Chapter 1 (Appendix 2 Table A.3). For the pre- and post-IFQ periods of the fixed gear fishery, catchability was fixed in the same area groupings as fixed gear selectivity. The longline survey catchability was the same across spatial management areas.

2.3.2 Movement

Movement of fish (in numbers) between areas occurred in a single pulse after fishing and natural mortality. $N_{AFT, y, s, a, m} = (N_{BEF, y=y, a=a, s, m} * \Phi_{k, j, a=a}^T)$, where Φ is an age-

dependent movement matrix (Table 2.3; see Appendix 2 for a complete set of age-based movement rates). Elements of Φ are transition probabilities among areas, where rows (k) describe the starting area and columns (j) represent the areas to which movement occurs, and each row sums to one. For each year, age, and sex, the vector of abundance in numbers by area was multiplied by the transpose of the movement (transition probability) matrix. Age-specific movement rates were estimated from tagging data for fish ages 1–16. Fish aged 17 or more were assumed to move at rates equal to those for age-16 fish. These estimates were derived using the model described by Hanselman et al. (2015) for estimating sablefish movement based upon mark-recapture data from a long-term (1979–2009) tagging program with more than 300,000 tag releases and 30,000 tag recoveries. Movement rates were time-invariant.

2.3.3 Observation process

The observed population abundance (relative population number RPN, from the longline survey) and biomass (relative population weight RPW, for the fixed-gear fishery) within the OM were sampled assuming random lognormal observation errors to obtain the relative abundance and biomass indices to which the EM was fit and thus informing management recommendations. For both indices, the simulated observation error standard deviation σ was assumed to be 0.3 for BS and AI areas and 0.15 for all other areas. These values were chosen because the BS and AI areas are sampled in alternating years, and observed coefficients of variation have historically been larger than for the WG, CG, WY, and EY areas (Hanselman et al. 2019). The longline survey RPN index is a function of abundance, catchability q for each area for fleet or survey g , and selectivity Sel , and summed across sex, age, and areas for the EM,

$$(12) \quad RPN_y = \sum_{s,a,m} \left[N_{AFT,y,s,a,m} q_{m,g} S_{s,a,m,g} * e^{(N(\mu=0,\sigma_m)-\sigma_m^2)/2)} \right], \text{ where } g \text{ here indicates the NOAA-NMFS Longline Survey.}$$

The longline survey alternates years in the Bering Sea and the Aleutian Islands. In a year without a survey in a given region, abundance in the unsurveyed area is approximated using the ratio of current year survey abundance for the Gulf of Alaska (GOA; summed area-specific abundance for WG, CG, WY, EY) to the previous year's GOA abundance, and multiplied by the last survey relative population number for the unsurveyed area:

$$(13) \ RPN_{y,m=BS \text{ or } AI} = (RPN_{y,m=GOA}/RPN_{y-1,m=GOA})RPN_{y-1,m=BS \text{ or } AI}.$$

The assumption that the expected annual change in BS or AI abundance is proportional to that in the GOA has been commonplace in the assessment since the assessment became an Alaska-wide model (Sigler 1999).

The fixed gear fishery catch per unit effort (CPUE) index (fishery RPW) was the same as for the survey RPN index (see Eqn. 12), but was derived from biomass B instead of abundance, included a lag in when data were provided to the EM, and used the estimated catchability from the fixed gear fleet. As a result, there was a one-year lag ($y-1$) in the availability of the fishery biomass index to the EM to represent the time required to collect and analyze fishery data.

Age compositions for each year and area for the longline survey and fixed gear fishery were sampled from the OM abundance with multinomial error, assuming an effective sample size of 200. Between 2000 and 2019, approximately 1200 otoliths are aged each year across all six areas. However, mean area-specific sample sizes range from a low of ~ 100 in the EY area to ~ 425 in the CG. Composition data for each year were summed over areas and sexes, then proportions at age were calculated. For the fishery age compositions, the aggregated age compositions were weighted by catch in numbers in each area for use in the EM as an area-weighted estimate of total age composition. This methodology was consistent with the current practice in the Management Model (Hanselman et al. 2019, Goethel et al. 2020).

2.3.4 Operating model – conditioning period

A critical component of developing any simulation-estimation framework is conditioning the operating model on observed data to ensure the OM used for simulating unobserved population dynamics accurately reflects the true but unobserved state of nature and key demographic processes (Punt et al. 2016, Sharma et al. 2020). We explored the performance of our spatial OM in simulating observed population dynamics during the operating model conditioning period (1977–2018) by comparing OM simulated outputs with the best estimates of observed quantities (e.g., fishery and survey indices) from the current stock assessment model used for Alaska sablefish management (Management Model). A well-conditioned and biologically-representative OM should produce a simulated population that closely matches the

historical estimates produced by the Management Model when provided with the catch in each of the six areas, total recruitment numbers, how that recruitment is apportioned to areas, and specified catchability and selectivity parameters during this period.

2.3.5 Estimation model

Given the objective of testing the current stock assessment process with simulated data, we intended to keep the EM as close as possible to the Management Model (Hanselman et al. 2019). However, a few structural differences were required. The Management Model is an age- and sex-structured estimation model parameterized in AD Model Builder (Fournier et al. 2012), which estimates annual recruitment and fishing mortality deviations by fitting to catch, catch age and length compositions, a fishery CPUE index, survey age and length compositions, and two survey abundance indices from the longline survey and the NOAA Alaska Fisheries Science Center bottom trawl survey. The Management Model fits to data and estimates the initial year of recruitment and demographic parameters beginning in 1960.

In contrast with the Management Model, our EM for the simulation-estimation analyses was fit to data and began to estimate abundance at age in 1977. To simplify the data simulation process and because the trawl survey is only conducted in the Gulf of Alaska areas (WG, CG, WY, and EY), the EM was not informed by length composition data or the trawl survey index of abundance. In addition, our EM incorporated new longline survey age composition data in the same year as each added year of the longline survey index, without the one-year processing lag that exists in the assessment model currently used for sablefish management (i.e., Management Model). This change was made to improve model convergence and EM recruitment estimation, which were compromised due to the lack of length composition data simulated in the OM and the lack of terminal year age composition data. We believe this change to the EM strikes an appropriate balance between the information that would ordinarily be provided by annual length composition in the year of collection and the one-year delay in otolith aging. Age composition data for the fixed gear fleet were included with a one-year lag, consistent with the Management Model.

The EM estimated fishing mortality, selectivity (not all selectivity parameters were estimated, some were fixed, see Appendix 2 Table A.2), and catchability for the two commercial fleets, fixed gear (pot, longline), and trawl gear. Two abundance or CPUE indices were fit by the EM, a U.S. longline survey (1990–2043) and a longline fishery CPUE index (1990–2042). The longline fishery CPUE index was broken into two time blocks with separate selectivity and catchability estimated, coinciding with the implementation of the IFQ program and resulting changes in fishing practices; differing selectivity for the fixed gear fleet was estimated for pre-IFQ years (1977–1994) and post-IFQ years (1995 onward). Further details about the EM are available in Hanselman et al. (2019).

The NPFMC has established harvest control rules based on a tier system (NPFMC 2018a, 2018b). The NPFMC tier 3 harvest control rule is applied to the biomass-based reference point to produce an ABC limit for each year, and the realized total allowable catch was assumed to equal the ABC. Under tier 3, ABC is calculated with an F_{ABC} equal to F_{40} , which is the fishing mortality rate that would lead to B_{40} . B_{40} is the biomass where spawners produced per recruit is at 40% of unfished biomass. If the spawning biomass/ B_{40} ratio falls below 1.0 (Eqn. 14), the harvest control rule reduces fishing mortality F below the projected F_{40} ,

$$(14) \quad F_{ABC,y+1} = F_{40} \left(\frac{SSB_{y+1}}{B_{40}} - 0.05 \right) / (1 - 0.05).$$

ABC for the next year was projected from the EM as the sum of projected catch in weight (w), by sex s and age a ,

$$(15) \quad ABC_{y+1} = \sum_s ProjCatch_{y+1,s,a} w_{s,a}.$$

And projected catch ($ProjCatch$) for the next year was a function of abundance N , survival, and selectivity Sel ,

$$(16) \quad ProjCatch_{y+1,s,a} = N_{y,s,a} Survival_{y,s,a} \left(Sel_a F_{ABC,y+1} / (Sel_{s,a} F_{ABC,y+1} + M) \right) * (1 - \exp(-Sel_{s,a} F_{ABC,y+1} - M)),$$

where M was natural mortality and was fixed at 0.1 in the EM, consistent with current practices.

For these analyses, ABC was apportioned annually to management areas using a 5-yr exponentially weighted moving average of fishery and survey abundance indices ($I_{fish,y,m}$ and $I_{surv,y,m}$, respectively) for each year y and area m . The survey index had double the weight ($w_{surv}=2$) of the fishery index ($w_{fish}=1$). This was the method accepted by the NPFMC for apportioning sablefish ABC for 2000-2013.

$$(17) \quad \psi_{y,m} = ABC_y * 1/(w_{surv} + w_{fish}) * \left[\left(w_{surv} * \sum_{k=1}^5 \vec{\rho}_k * \left(I_{surv,y=y-k+1,m} / \sum_m I_{surv,y=y-k+1,m} \right) \right) + \left(w_{fish} \sum_{k=1}^5 \vec{\rho}_k * \left(I_{fish,y=y-k,m} / \sum_m I_{fish,y=y-k,m} \right) \right) \right].$$

The exponential weighting factor ($\vec{\rho}_k$) for this apportionment type was a vector equal to [0.5, 0.25, 0.125, 0.0625, 0.0625] for years $y, y-1, y-2, y-3, y-4$ for the survey index and years $y-1, y-2, y-3, y-4, y-5$ for the fishery CPUE index.

Each forward projecting year, the apportioned ABC was split between fixed and trawl gears based on realized historical allocations (2008–2017) within each area: the BS fixed gear allocation was at 50% of ABC apportioned to this area, and trawl gear was 50%, AI 75% to fixed gears and 25% to trawl, WG and CG were both 80% to fixed gear and 20% to trawl, WY 95% to fixed gear and 5% to trawl, and EY 99.9% to fixed gear and 0.001% to trawl.

Early analyses encountered EM convergence issues, with convergence defined as an EM with a maximum gradient component (MGC) less than 0.01. During simulations, if the EM failed to converge (i.e., meet the specified MGC) during the forward projecting period, the simulations were set up to automatically restart the EM using new (jittered) starting values for mean recruitment and fixed-gear fishing mortality, as this is one action a stock assessment author could take if confronted with this situation. After four attempts, if the EM did not meet the specified convergence criteria in any year of the forward projecting period, the simulation loop continued using the estimated fishing mortality, but the iteration was removed from analyses.

2.3.6 Simulation model iteration in the forward projection period

In the forward projection period (2019–2043), the OM and EM were run as an iterative loop, projecting population dynamics forward in time using the OM, then annually assessing stock status with the EM and specifying harvest levels for the next year using the NPFMC harvest control rules (NPFMC 2018a, NPFMC 2018b). The EM was fit to simulated observations from the population, and the *year+1* projected ABC was extracted and apportioned to the six management areas (Eqn. 17).

2.3.7 Operating model alternative specifications

A central objective of this analysis was to test the sensitivity of EM convergence and the ability to reconstruct demographic trends across a range of potential states of nature with respect to population dynamics described by alternative OM specifications. OM variability was included via the magnitude and distribution of recruitment and via age composition and abundance index ‘sampling’ of the simulated OM population. Many of these attempts resulted in substantial rates of EM non-convergence, where we defined convergence as an EM fit having a maximum gradient threshold of <0.01 during the OM-EM simulation loops in the forward projection period. These model explorations were critical to the final choice of a ‘base’ OM and EM model configuration for later use in exploring alternative apportionment strategies. Alternative OM configurations focused on; 1) OM movement rate assumptions, 2) OM recruitment assumptions, and 3) an OM-EM validation scenario.

We explored different movement rate scenarios to understand the impacts of alternative movement rates on OM and EM population dynamics and on EM performance, given the inherent mismatch between a spatial OM and the panmictic population assumption in the EM. Movement among OM areas was age-specific in the base model, but sensitivity analyses included a no-movement scenario (Tables 2.1, 2.2, and 2.3) where all fish of all ages remain in the area to which they recruited, a hypothetical panmictic or equal movement scenario where fish of every age had an equal probability of moving to any other area as staying in their current area, and a third movement scenario in which the movement of fish of all ages exhibited directional movement towards eastern areas (Table 2.3).

Recruitment was the primary source of process variation in the OM. Therefore, several sensitivity scenarios were conducted regarding the recruitment mean (50% higher or lower than the base specification of 16.5 million recruits) and recruitment variation (σ) 50% higher or lower than the base specification of 0.8 between years (Table 2.1). In addition, a scenario that used the full estimated recruitment for 2018 (163.7 million recruits instead of 16.5 used in the base scenario) was developed to examine the impact of the full value on simulation results and EM convergence rates. This represented an extreme recruitment scenario, reflecting recent observations of several extremely large recruitment events (Goethel et al. 2020).

Validation that the OM could produce population dynamics that were reasonably realistic and that the EM could reproduce the OM population were important steps in the simulation analyses. However, the difference in spatial structure between the OM and EM makes direct validation challenging. There was an inherent mismatch between the six-area OM with movement among areas and the panmictic EM. In model scenario LE_panmictic, we explored a low-error, panmictic OM scenario in which movement rates among areas were equal, recruitment was equally distributed to areas, and with 50% lower observation error for OM age composition and indices (Table 2.2).

2.3.8 Performance metrics

EM convergence rates were calculated for the simulation-estimation analyses because EM convergence (or non-convergence) affects the validity and interpretation of EM output. Convergence was defined as an EM MGC less than 0.01. EM runs in all 25 forward-projecting years of an iteration must have converged for that iteration to be used in any output analyses or calculation of performance metrics. The EM convergence rate was the proportion of fully-converged iterations divided by 500, the number of replication simulations (i.e., iterations) conducted for each scenario. An alternative convergence threshold specified that an EM was ‘converged’ if the MGC was less than 0.1 also was examined.

In the simulation-estimation analyses, comparisons of EM performance measured as agreement with the ‘true’ OM were made for abundance, spawning biomass, survey abundance, fishery CPUE biomass, and recruitment. For each apportionment scenario (*b*),

simulation iteration (i), and year (y), the absolute percentage error (APE) between the converged estimated parameter ($\bar{x}$) and corresponding true value from the OM (x) summed over areas (m) was calculated as,

$$(18) \ APE_{b,y,i} = |(\sum_m x_{b,y,m,i} - \bar{x}_{b,y,i}) / \sum_m x_{b,y,m,i}| * 100.$$

Mean absolute percentage error (MAPE) was then calculated across converged iterations (i) for a given scenario (b),

$$(19) \ MAPE_b = mean \begin{pmatrix} APE_{y=202, i=1,b} & \cdots & APE_{y=202, i=500,b} \\ \vdots & \ddots & \vdots \\ APE_{y=204, i=1,b} & \cdots & APE_{y=2043, i=500,b} \end{pmatrix}.$$

2.4 Results

2.4.1 Operating model and estimation model performance

OM conditioning period

During the conditioning period (1977–2018), the OM produced data and sablefish population dynamics, which, when aggregated across management areas, closely reproduced the Management Model values (Hanselman et al. 2018). The Management Model values were the target for establishing OM sablefish population dynamics and simulating abundance, spawning biomass, catch, and recruitment (Figure 2.2). Base model recruitment for the terminal year of the OM conditioning period (2018) was reduced to the long-term mean (16.5 million recruits) from the extremely high Management Model value [163.7 million recruits based on Goethel et al. (2020)] to improve simulation EM convergence in the transition years between the conditioning period and the forward projection period. These differences were most evident in the abundance and recruitment panels in Figure 2.2. During the conditioning period, abundance was highest in the CG region and generally lowest in the BS (Figure 2.3), which is generally consistent with longline survey observations of relative abundance.

Base scenario EM performance

On average, the Base scenario EM produced low absolute error in predicting the ‘true’ OM population, indicating that the panmictic EM generally performed well in assessing the aggregated OM population dynamics. The mean absolute percentage error (MAPE) across simulation iterations and forward projection years (2019–2043) between the Base scenario EM and ‘true’ OM abundance and spawning biomass was 4.9% and 2.7%, respectively. MAPE for the fishery and survey index were less than 3.1%, and recruitment MAPE was 38.4%. However, the MAPE for each year (across simulation iterations) showed that there was a relatively high error in the abundance and recruitment estimates from the EM for the last few years of simulations (2040–2043) and in the survey index estimates for the first year of the forward projection period (2019; Figure 2.4). Increased uncertainty in estimates of recruitment and abundance at the end of the time series is expected as these year classes are not well informed by data observed through 2043, wherein these year classes have only begun to be observed by the survey and captured in fisheries.

2.4.2 Estimation model convergence

For the Base scenario, 28.4% of the 500 iterations resulted in a converged EM (EM maximum gradient component < 0.01) for every year (2019–2043), resulting in 142 complete iterations retained for analyses (Table 2.4). The proportion of converged Base scenario EMs was 91–99% in any given year across simulation iterations (Figure 2.5). Thus, while some simulation iterations converged every year, there was no year in which every iteration converged. In many iterations, the failed (non-converged) EM was a single year of EM non-convergence. However, some simulation iterations also had multiple years which failed to converge, often consecutively (Figure 2.6).

Convergence rates for model scenarios varied depending on the maximum gradient component value used to determine whether the EM has ‘converged’ in any given year, simulation iteration, and model scenario (Table 2.4, Figure 2.7). Lower values of the convergence criteria (MGC < 0.1 vs. MGC < 0.01) led to fewer simulation iterations converged and fewer iterations retained for analyses. Across all models presented, convergence ranged between 8–209 of 500 possible iterations using the stricter criteria (MGC < 0.01 ; Table 2.4).

We used the Base model scenario to examine whether the selection of $MGC < 0.1$ or $MGC < 0.01$ as the EM convergence criteria affected estimates of SSB, abundance, or ABC. The two alternative criteria used to determine whether an EM converged had little impact on the EM results once non-converged models were discarded (Figure 2.8). The ABC, SSB, and abundance for the Base model had similar median and range regardless of the convergence criteria used to filter simulation iterations for inclusion in analyses. The Base model EM fit survey and fishery indices generated by the OM well in general. Figure 2.9 compares EM and OM fishery and survey indices using either $MGC < 0.01$ as a convergence criterion or showing all simulation iterations (including non-converged). The poorer EM fits to OM data during the conditioning period and the wider range of index values (higher and lower OM index values) during the projection period were frequent in the figure panels, which include non-converged model data and output (Figure 2.9). This highlights that EM non-convergence occurred when the OM simulated indices with extreme observation errors, which should be expected to pose challenges for the EM in reconciling information in the different sources of data to which it is fit.

The convergence rates and percent of retained iterations for each of the sixteen models (Base model and 15 alternative scenarios) varied widely (Table 2.4). In only a small proportion of iterations (0.016–0.418), was EM convergence ($MGC < 0.01$) obtained in all simulated years. However, the total EM convergence rate was relatively high (0.840–0.957) across all year-iteration combinations. The OM scenarios with the highest EM convergence rates and most retained iterations had higher or lower recruitment variability (R_Hvar , R_Lvar) or lower mean recruitment (R_Lmean). This suggests that the EM is more robust to recruitment variability or lower mean recruitment than higher mean recruitment. The lowest EM convergence rates came from the three alternative OM movement rate models, which had no movement between areas (M_none), equal movement (M_eq), or directional movement (M_dir). Only 8.4%, 2.4%, and 16% of models M_none , M_eq , and M_dir converged, respectively, resulting in 42, 12, and 80 retained simulation iterations for these models (Table 2.4). The $LE_panmictic$ scenario, with low observation error and which assumed movement was consistent with the OM, also performed relatively poorly for convergence (13% or 65 iterations retained out of 500). These

results for alternative movement and low error suggest that the panmictic EM may not be a robust alternative in sablefish movement patterns. Alternative or rapidly changing OM population distributions without concurrent consideration of revised EM parameterization, initialization values, or phase of parameter estimation appear to pose problems for the EM, resulting in low convergence rates.

2.4.3 Exploration of alternative scenarios

Movement

Alternative movement rates in the OM were explored to: 1) understand the effect of OM movement when the EM represents a single panmictic stock occupying a single model area that does not explicitly account for fish movement, and 2) understand the effect of movement on the OM population. Movement was the primary driver of area-specific abundance in the OM (Figure 2.10). In the OM population, equal movement rates between all six areas (M_{eq1}) resulted in equal abundance in all areas, and directional movement (M_{dir}) where fish moved from western regions towards eastern regions resulted in greater abundance in the eastern regions of WY and EY (Figure 2.10). Models with no movement between areas (M_{none}) resulted in the proportional abundance of areas nearly equal to the proportional distribution of area recruitment. Recruitment proportions to areas for scenarios M_{none} and Base was [0.14, 0.07, 0.13, 0.43, 0.14, 0.09], and the resulting mean abundance proportions to areas for scenario M_{none} across converged simulation iterations for the forward projection period (2019–2043) was [0.14, 0.06, 0.13, 0.44, 0.14, 0.08].

As stated previously, EM convergence rates were very low for alternative movement scenarios, which affected the interpretation and our ability to draw conclusions. Given this caveat, for converged iterations, there was relatively little difference in the ability of the EM to reproduce the OM abundance, SSB, and fishery and survey indices for the alternative movement scenarios compared to the Base model scenario (Figure 2.11). The EM estimates of population quantities such as abundance were likely more affected by low convergence rates than the movement rates. Using a less stringent criterion for EM convergence ($MGC < 0.1$ instead of our preferred criterion of $MGC < 0.01$) resulted in more retained iterations (100, 98,

or 161 retained iterations for M_none, M_eq, and M_dir, respectively) and greater similarity in abundance estimates between Base model and alternative movement scenarios (Figure 2.12).

Recruitment

Scenarios exploring recruitment mean and variability were important because recruitment can be a key driver of population dynamics. Alternative OM specifications with higher or lower recruitment variance, mean recruitment, and recruitment in 2018 (conditioning year) showed the sensitivity of the EM to this driver of OM population dynamics. When the recruitment variance specified in the OM was 50% higher (R_Hvar) than the value used for the OM Base scenario, the estimated spawning biomass, abundance, and recruitment had slightly lower medians and a larger 95% quantile range across converged simulation iterations (Figure 2.13). Lower recruitment variance (R_Lvar) resulted in spawning biomass and abundance estimates similar to the Base model scenarios but, as expected, with a much narrower range of estimates across simulation iterations. However, the distributions of SSB, abundance, and recruitment estimates from the EM were not symmetrical around the median; the upper bounds representing the 95% confidence intervals were skewed towards outcomes with higher SSB, recruitment, and abundance. This outcome may result from the harvest control rule component of the EM, which is designed to reduce fishing mortality when the estimated population falls below specified biomass thresholds relative to maximum sustainable yield proxies but has no such bound which may constrain the upper range of biomass or abundance.

Model scenarios that increased (R_Hmean) or decreased (R_Lmean) OM mean recruitment by 50% resulted in long-term differences in estimated SSB, recruitment, and abundance (Figure 2.14). After ten years for forward projection, the mean abundance for R_Hmean over 2029–2043 was 48% greater than the Base scenario, and the mean abundance for R_Lmean was 43% lower than the Base scenario. SSB and recruitment had similar results, with 39% and 51% increases, respectively, for R_Hmean, and decreases of 32% and 47% for R_Lmean SSB and recruitment means.

Recruitment for 2018, the terminal year of the conditioning period, was reduced in the Base scenario to improve EM convergence rates, but we explored the original value to illustrate

the effect of that decision. There was a large impact on SSB and abundance when the OM recruitment for 2018 was specified at 163.7 million recruits (R_Alt2018) instead of 16.5 million recruits (Base model). Abundance rose very high in this alternative scenario, but within approximately 13 years, the alternative and the Base scenarios had approached the same median abundance (Figure 2.15). SSB had a similar pattern, but with a delay in the SSB increase for the high 2018 recruitment model while fish grew and reached maturity, and the models took longer (~20 years) to reach a similar median SSB. The EM convergence rate was lower for R_alt2018, with only 18.2% converged EMs compared to 28.4% usable simulation iterations for the Base model (Table 2.4), which underscores the reason for making this change.

Despite the changes to the OM recruitment values for models R_Hvar, R_Lvar, R_Hmean, R_Lmean, and R_Alt2018, there were few differences in the MAPE between the EM and OM for these model scenarios for the Base model (Figure 2.16). This indicated that, for converged models, the EM performed similarly well in estimating OM population parameters and data. However, the difference in convergence rates among the recruitment scenarios (Table 2.4) suggests that the EM configuration was not flexible enough to accommodate the higher mean recruitment assumed in the R_Hmean scenario, as fewer iterations converged compared to the Base scenario. The scenarios with higher mean recruitment variability (R_Hvar) and lower mean recruitment had better convergence rates than the Base scenario.

2.4.4 Simulation validation

Validation that the EM can reproduce the OM population is important in simulation analyses. To examine whether the panmictic EM can reproduce the OM population, we examined a low error, panmictic version of the OM (LE_panmictic) and compared it to the Base model performance. In general, MAPE was slightly lower for the panmictic OM model compared to the Base model (Figure 2.17), indicating small improvements in the ability of the EM to produce estimates closer to the 'true' OM values when the OM values were less variable, recruitment distribution, and movement rates were equal between areas. The SSB, recruitment, and abundance trends were very similar between the Base and LE_panmictic models (Figure 2.18). In combination, these metrics indicate the EM does an adequate job of reproducing the OM population in aggregate. When contrasted against the results from

alternative movement scenarios, these results affirm the idea that altering movement rates without concurrently altering other OM population parameters (selectivity, recruitment distribution to areas, and catchability) creates an OM population that is out of sync with itself and the EM may be too tuned to be flexible in accommodating the OM dynamics.

A recurring result from all simulation scenarios, including the LE_panmictic scenario, was that the EM generally did not perform well at estimating recruitment. MAPE between the OM and EM for the LE_panmictic model was similar to results from the Base scenario (Figure 2.19), with the greatest MAPE occurring in the last three years of the forward projection period. The EM data available to estimate recruitment consists solely of survey age compositions, so this result is not unexpected but may be another contributing factor to EM convergence issues.

2.5 Discussion

When developing paired simulation-estimation analyses, analysts have the flexibility to define the values for demographic parameters in the OM that simulate population dynamics and the complexity of spatial processes such as movement or recruitment. Here we first sought to simulate a 'sablefish-like' population and validate that the simulated population mimicked observed sablefish dynamics. We then explored the sensitivity of estimation model convergence to differences in OM specification and the ability of the EM to estimate the true underlying population dynamics to verify that the model framework would be suitable for future analyses of management procedures.

The sablefish abundance, spawning biomass, catch, and recruitment developed in the simulation conditioning period were close to established stock assessment estimates used for management [i.e., the Management Model described by Hanselman et al. (2019)]. The simulation framework was established based on an understanding of current and historical observed conditions from annual stock assessments. As a result, changing environmental conditions outside of past observed ranges or other dynamics that may affect sablefish distribution, movement, or population dynamics were not explicitly identified or included in establishing the conditioning period population or the forward projection period. Inclusion of potential future conditions is frequently an objective of simulation studies and may help

anticipate management challenges and identify management procedures that are robust to changing climate or environmental conditions (Lehodey et al. 2006, Miller 2007, Palacios-Abrantes et al. 2020) and would be a logical next step for sablefish simulations.

The Base model scenario generally performed well at reproducing the OM population during the forward projection period, as shown by the relatively low error between the OM and EM for abundance, spawning biomass, and abundance indices. However, recruitment estimation in the EM was challenged by a lack of simulated data available to the EM to inform estimation, as only survey age composition data were available to the EM for the terminal year of each simulation-estimation loop during the forward projection period. Simulating length composition data in the OM and providing that as an additional source of information for the EM may have improved recruitment estimation and EM stability (Ono 2014, but also see Lee et al. 2012). However, it would have also increased the complexity and uncertainty of the OM and EM framework.

By developing the OM population in the forward projection period, we identified a few key drivers of the population dynamics for this simulated population and examined the effect of these population characteristics on the EM output and performance. The primary drivers of OM abundance in each spatial area for these analyses were movement rates and patterns among OM areas and the distribution of recruitment to OM areas. The alternative states of nature examined under OM scenarios which increased or decreased recruitment variability (σ_r) and mean recruitment by 50% rescale the OM population and EM population estimates, but the EM was fairly robust to these alternative OM dynamics. However, alternative movement scenarios in the OM resulted in EM convergence failure in many years and simulation iterations, resulting in very low numbers of ‘successful’ simulation iterations available for analyses. This indicates that the EM was sensitive to OM states of nature with alternative movement patterns in its current configuration.

Because a primary intent behind the development of this simulation-estimation framework was to examine a specific management scenario regarding apportionment of catch to management areas (Chapter 3), we maintained strict criteria for simulation inclusion, where

all years of a simulation iteration must converge for the iteration to be retained for analyses. We also wanted to keep the EM as close as possible to the management model used for assessing the Alaska sablefish population and setting harvest regulations. Without these features, it would not have been possible to provide informative management evaluations for the specific question we were addressing. However, comparisons of results stemming from alternative convergence thresholds indicated that 1) the median estimates for abundance and SSB were similar regardless of convergence criteria used to filter EM output, and 2) 'extreme' results were more often removed when using a more conservative threshold for the determination of convergence. The challenge of the OM in fitting to data from these 'extreme' OM simulations is not surprising as these may reflect cases when a very large observation error was encountered which likely set up a conflict with other data informing EM estimates. For example, the fishery and survey indices in Figure 9 show that the individual simulation iterations removed when filtering for EMs that converged with an MGC <0.01 were frequently those with much higher or lower observed and estimated index values. Given that it was often only 1-4 estimation models out of 25 EM runs that failed to converge, but our criteria for iteration selection resulted in the discarding of the entire iteration, the similarity in median EM estimates was not surprising.

To address EM convergence issues while developing and validating the framework, we tried or implemented several procedures to improve convergence and increase the number of retained simulation iterations. We attempted to improve EM convergence by including a 'retry' function to the OM, which restarted a failed EM up to four times by jittering EM initial parameter values. We also implemented, then discarded, an ABC rollover provision, for which the previous year's EM-produced ABC value was 'rolled over' to the consecutive non-converged EM year. These alternative ways to deal with a non-converged EM are options for a stock assessment analyst in a real-world scenario. While this improved convergence, we did not retain this feature for the current simulation-estimation framework because we ultimately intended to use the framework to examine spatial apportionment of ABC and rolling over ABC would have affected the interpretation of apportionment performance metrics. Convergence in an estimation model is not simply defined by a maximum gradient component. A positive

definite Hessian matrix may often signal model convergence but does not guarantee convergence or realism in parameter estimation (see Chapter 1). Inverting the Hessian matrix can be time-consuming and thus prohibitive in simulation-estimation model frameworks when thousands of models may be run. While all of these criteria for model convergence should be evaluated by the stock assessment analyst, the ultimate decision about whether satisfactory convergence has been obtained should be based on a thorough exploration of whether parameters are at or near bounds and if parameter estimates make biological sense, while also considering the maximum gradient component, MCMC simulations, and the Hessian. This level of evaluation for thousands of simulation iterations is not possible and must be simplified. In the real world, rolling over management recommendations such as ABC, perhaps with a higher uncertainty buffer, is a very real outcome in times when the EM fails to converge or is rejected by advisory bodies and in many species that are not assessed annually when management recommendations are set after assessment and held in place until the next assessment cycle. In future simulation analyses, rolling over management decisions such as allowable catch levels may alleviate some of the constraints of EM convergence issues while also providing reasonable population estimates against which to weigh management measures.

Many, if not most, stock assessment models used to manage fisheries are constructed to assess a single stock without spatial areas or spatial parameterization. Moreover, the assessment of a 'stock' or population is sometimes defined by political rather than ecological boundaries. However, recognizing the implications of these critical limitations is growing (Ying et al. 2011, Punt et al. 2015, Berger et al. 2017, Cadrin et al. 2019). Even if the population boundaries match the boundaries of the stock being assessed, population structure may not be uniform across the spatial domain of the population due to fish movement, fishery dynamics, and spawning behavior which lead to spatial recruitment differences, habitat distribution (e.g., thermal condition, benthic habitat structure or distribution; Guan et al. 2013). The end result is a heterogeneous population structure across the modeled stock and a single-area or panmictic EM, which almost always oversimplified the complex spatial dynamics of the true population. Assessments that ignore spatial structure have been shown to perform well in estimating

overall population dynamics, but the risk of localized depletion may remain (Ying et al. 2011, Punt et al. 2015, Bosley et al. 2019, Bosley et al. 2022).

We attempted to model a panmictic OM population with low process and observational error (scenario LE_panmictic) to evaluate the performance of the EM and assumed it would result in high convergence rates and low relative error between the OM and EM. But we found that the relative error between the OM and EM in this scenario was similar to the Base scenario, and EM convergence was low. This also occurred with the alternative movement scenarios, which altered OM movement rates (M_none, M_eq, M_dir). This led us to two conclusions that may have broader implications for others attempting similar analyses: 1) Alaska sablefish are not 'panmictic' in their distribution or population dynamics (Chapter 1), and 2) the EM used for these analyses was tuned to a non-panmictic sablefish population. This occurred because we used a variation of the Management Model as the basis for the simulation-estimation framework EM and conditioned the OM on historical estimates from the model used to manage sablefish. By changing the OM population structure but not concurrently revisiting the EM parameterization and bounds, the EM lacked the flexibility to estimate the OM population dynamics in some cases. Convergence rates in any given year for the EM were quite high, but the ability for the EM to converge every year for the full projection time series was much lower. This represents how an MSE fails to capture the normal management process, in which EMs are adjusted by human analysts as conditions change over time or new data are provided to the model. This is often seen in retrospective analyses that remove sequential years of data from the end of the time-series and re-run the models. Often the EM has been tuned to more recent data, resulting in an identifiable retrospective error pattern.

This also highlights the tradeoffs between OM and EM realism, complexity, and performance, which should be considered when developing and evaluating simulation frameworks intended to test existing EM structures. However, these simulation-estimation analyses can successfully identify core challenges for EM parameterization and structure, including identifying the demographic conditions that make successful EM convergence unlikely.

To avoid this situation when developing simulation-estimation models, analysts could ‘untune’ the EM by widening parameter bounds, considering alternative starting values for model parameters, and estimating as many parameters as possible. If the EM struggles to estimate parameters like selectivity and recruitment when used for management, this may limit the scope of simulated population dynamics that can be reasonably explored through simulations, and it may be reasonable to consider using a simpler EM for simulations, and possibly for management if it is clear that a model may fail when confronted with data outside of the tuned EM. For population estimation models, Punt et al. (2015), with deference to Albert Einstein, said it best: choose an assessment model that is as ‘complicated as needed but not more complicated.’

Finally, the conclusions drawn about management procedures or population from simulation-estimation analyses are only as good as the OM inputs and the range of uncertainties explored; it is possible to gain an understanding of relative risk and tradeoffs but not quantify absolute risk (Butterworth et al. 2010). More realism in the OM is valuable, but if the EM component of the simulation fails as a result, so does the quality of conclusions that can be drawn from alternative configurations. The simulation-estimation framework developed here was suitable for Alaska sablefish research on management procedures conditioned on a historical understanding of population dynamics and movement. We will need to continue the exploration of alternative states of nature (e.g., movement and recruitment) to understand how changing environmental conditions and ecosystem dynamics affect the Alaska sablefish population.

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2.7 Figures

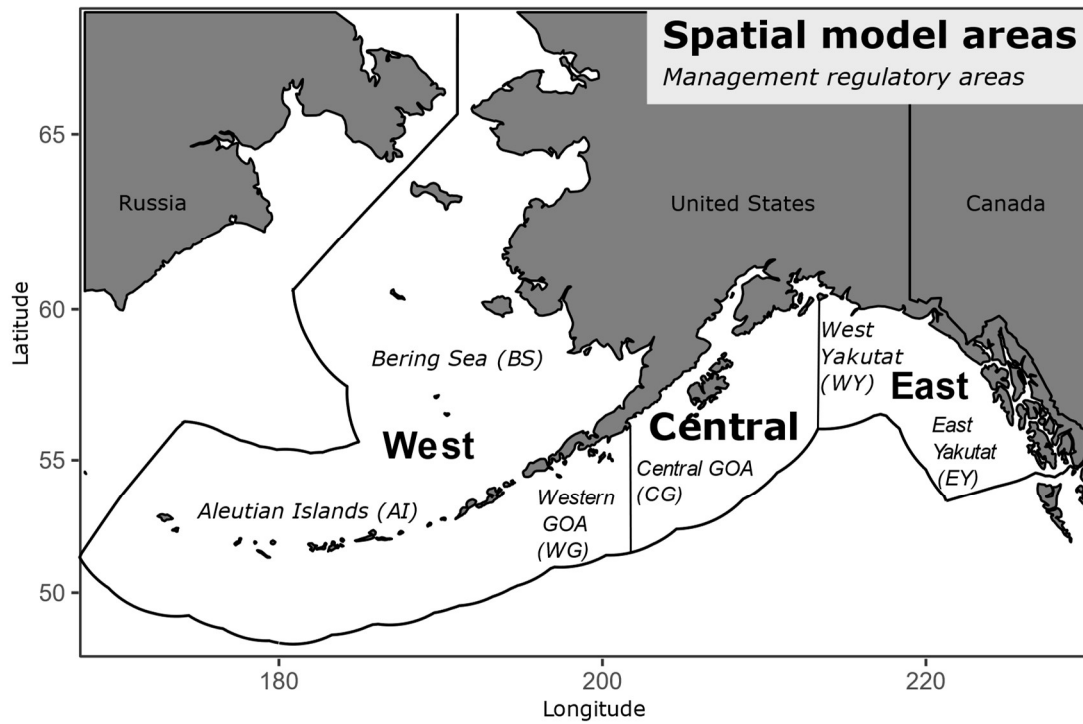


Figure 2.1. Map of Alaska sablefish operating model (OM) and management areas. Six management areas were modeled in the OM; Bering Sea, Aleutian Islands, Western Gulf of Alaska (GOA), Central GOA, West Yakutat, and East Yakutat/Southeast Outside. West, Central, and East designations represent grouping of shared OM parameters for selectivity, and were informed by a spatial sablefish stock assessment model. Management area boundaries are approximate and not intended to be used for management purposes.

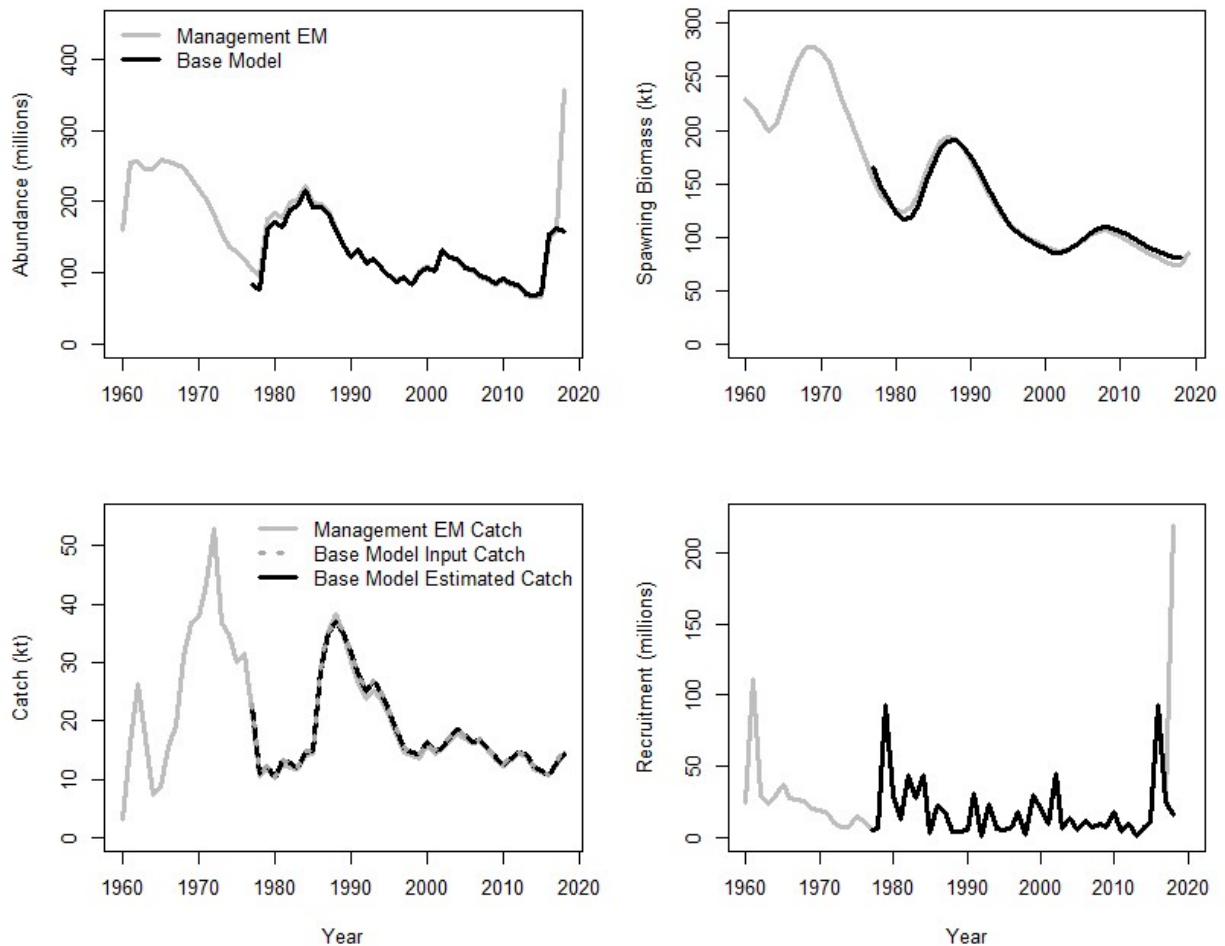


Figure 2.2. Conditioning period estimation model (EM) compared to reference model data. Top left panel: Estimated sablefish abundance in numbers (millions of fish, summed over operating model (OM) areas) generated from the Base Model OM conditioning period (black) for 1977–2018, compared to the Management EM (grey) for 1960–2018. Top right panel: Spawning biomass (kt, summed over OM areas) generated from the Base Model OM conditioning period (black) compared to the Management EM (grey). Bottom left panel: Catch (kt) from the Management EM (grey, 1960–2018) compared to the Base Model conditioning period (1977–2018) input values (black) and estimated values (dashed grey). Bottom right panel: Recruitment inputs to the Base OM (black, 1977–2018) compared to the Management EM estimated values (1960–2018).

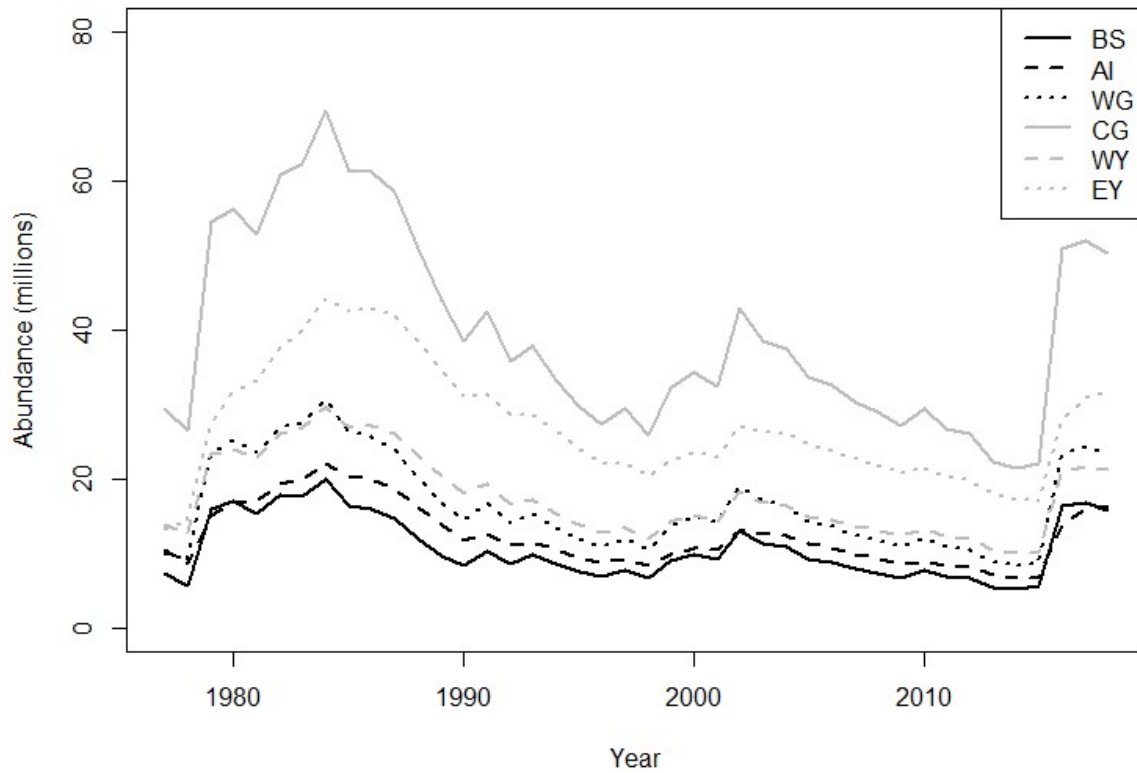


Figure 2.3. Conditioning period abundance by area for Alaska sablefish. Estimated sablefish abundance during the operating model (OM) conditioning period (1977–2018), by management area. Six management areas were modeled in the OM; Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat (WY), and East Yakutat/Southeast Outside (EY).

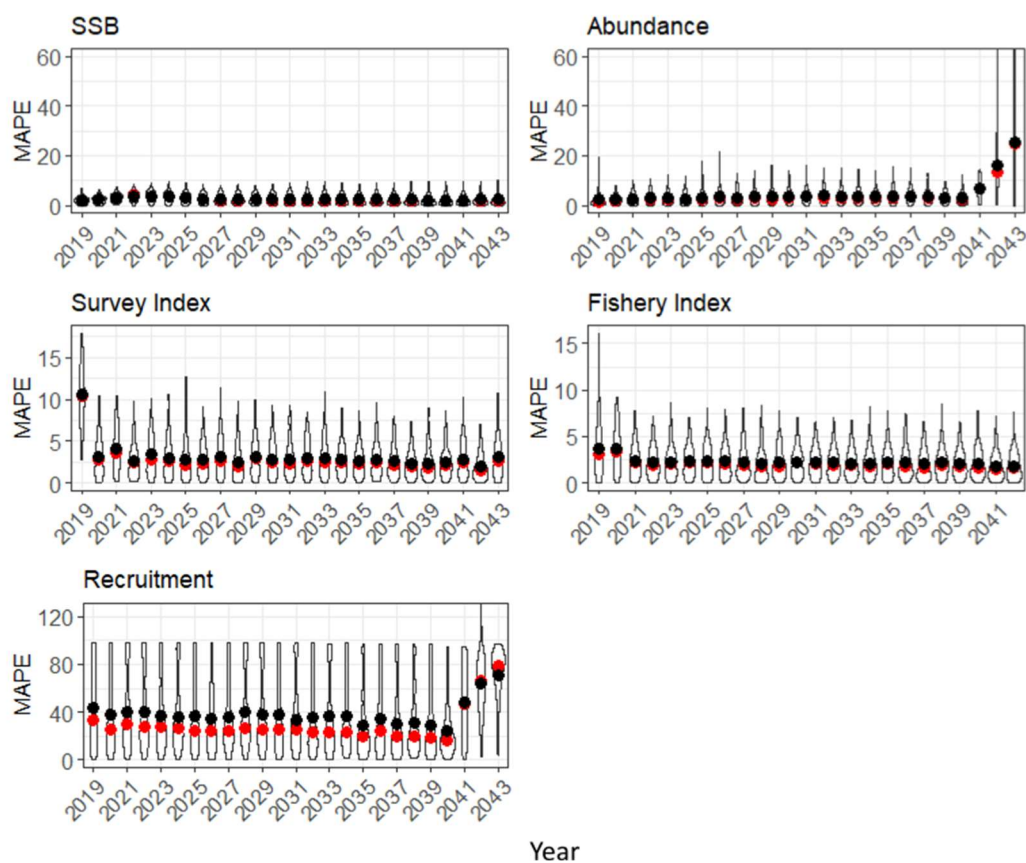


Figure 2.4. Base simulation model performance. Mean absolute percent error (MAPE) between Base model operating model and estimation model for each year 2019–2043, aggregated across modeled areas and for converged simulation iterations (MGC < 0.01 convergence threshold). Black points represent the mean value across iterations, red points represent the median values. Note the different y-axis scales for each row of figure panels.

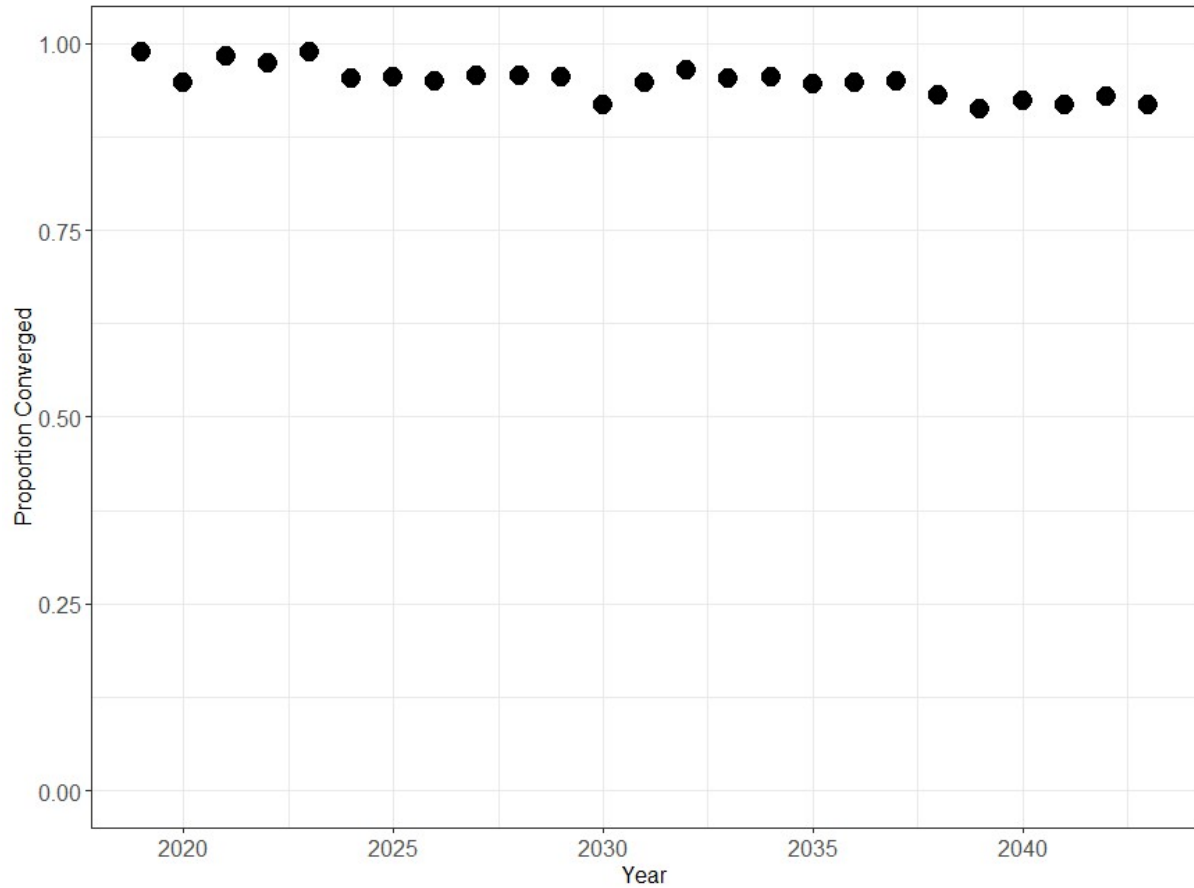


Figure 2.5. Base simulation model convergence rates over time. Base model scenario proportion of estimation models which converged across the 500 simulations iterations for each forward projection year, 2019–2043. EM convergence was defined as a maximum gradient component < 0.01 .

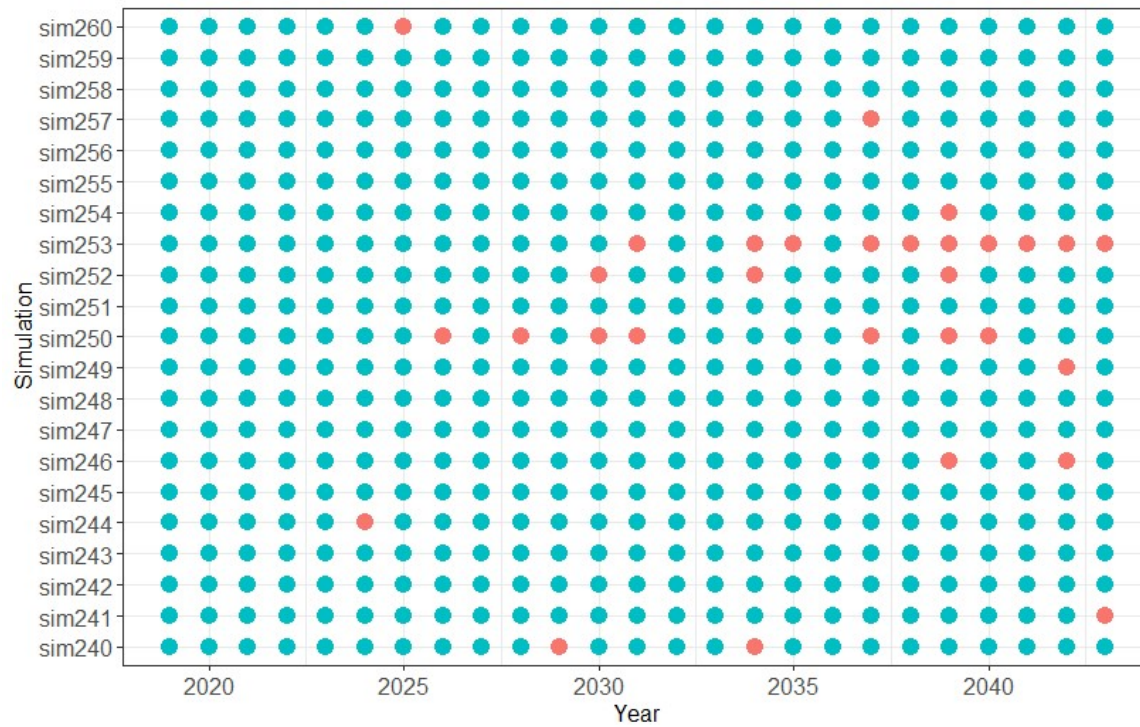


Figure 2.6. Subset of base model convergence rates by year. The estimation model convergence rates for the base simulation model are shown for a subset of simulation iterations (iterations 240–260 of 500). The figure shows converged (blue circle) and non-converged (orange circle) simulation iterations for each year during the forward projection period 2019–2043. A maximum gradient criteria < 0.01 for the EM was used to determine if model convergence had occurred.

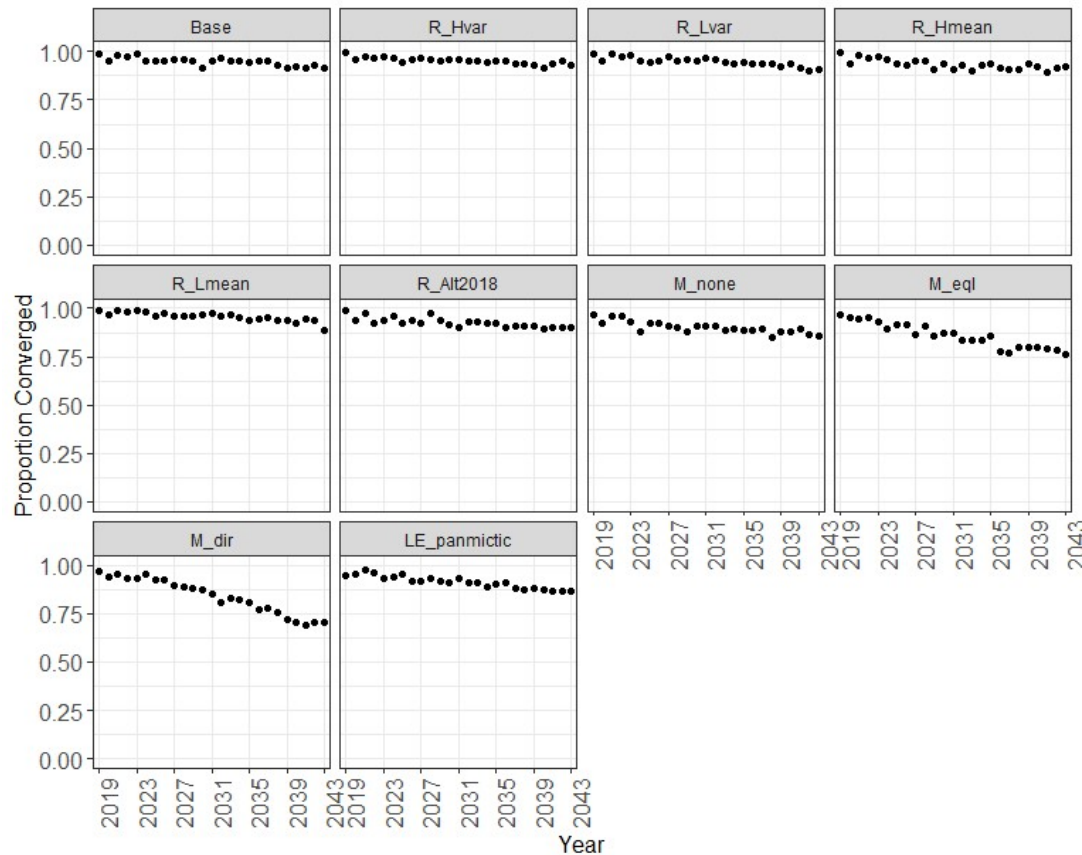


Figure 2.7. Estimation model convergence over time for all simulation scenarios. The figure shows the proportion of estimation model (EM) iterations that converged out of 500 simulation iterations for each forward projection year, 2019–2043, and each scenario examined. EM convergence was defined as having a maximum gradient component < 0.01 . Model scenarios in the figure include the base simulation scenario (Base), model scenarios with 50% higher operating model (OM) recruitment variance (process error; R_Hvar) or 50% lower OM recruitment variance (R_Lvar), increased (R_Hmean) or decreased (R_Lmean) mean recruitment in the operating model by 50% compared to the Base model recruitment mean, and a recruitment value of 163.7 million recruits for the terminal year of the conditioning period (R_alt2018). The M_none scenario assumed no movement among areas, M_eq1 assumed hypothetical movement rates that moved fish in equal proportions to all other areas, and M_dir assumed movement rates were directional with fish predominantly moving from western areas towards eastern areas. The LE_panmictic model indicates a low error, panmictic simulation scenario.

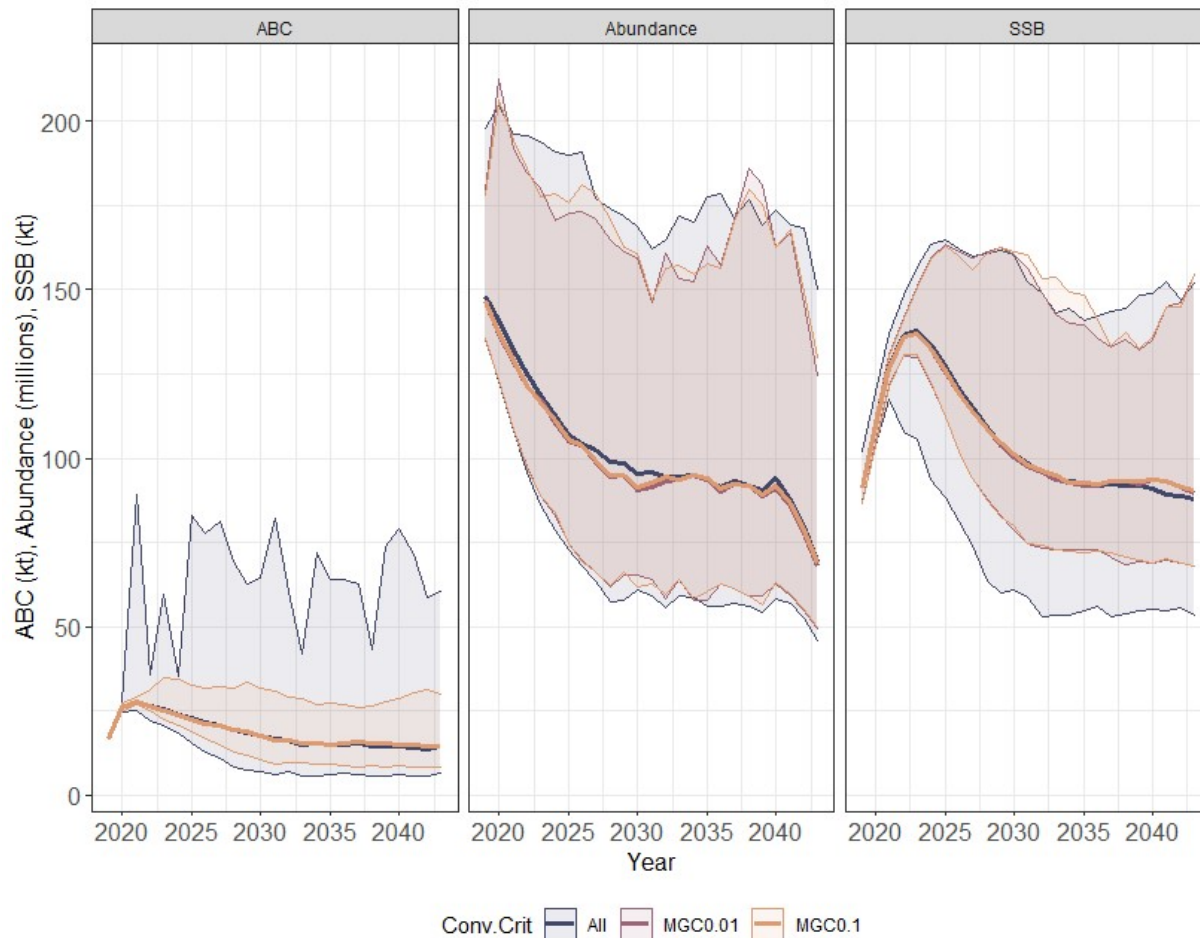


Figure 2.8. Effect of estimation model (EM) convergence criteria on EM population estimates. Base model scenario estimates for allowable biological catch (ABC) (left panel, kt), abundance (center panel, millions), and spawning stock biomass (SSB (kt), right panel) under alternative thresholds for determining EM convergence. The thick colored lines represent the median of ABC, abundance or SSB under the respective convergence criteria, shaded regions indicate the upper and lower 95% quantiles. Median and quantiles for all models (regardless of convergence) are represented by blue lines and shading, red lines and shading represent models that converged based on a maximum gradient component (MGC) < 0.01, and orange lines represent a convergence threshold of MGC > 0.1.

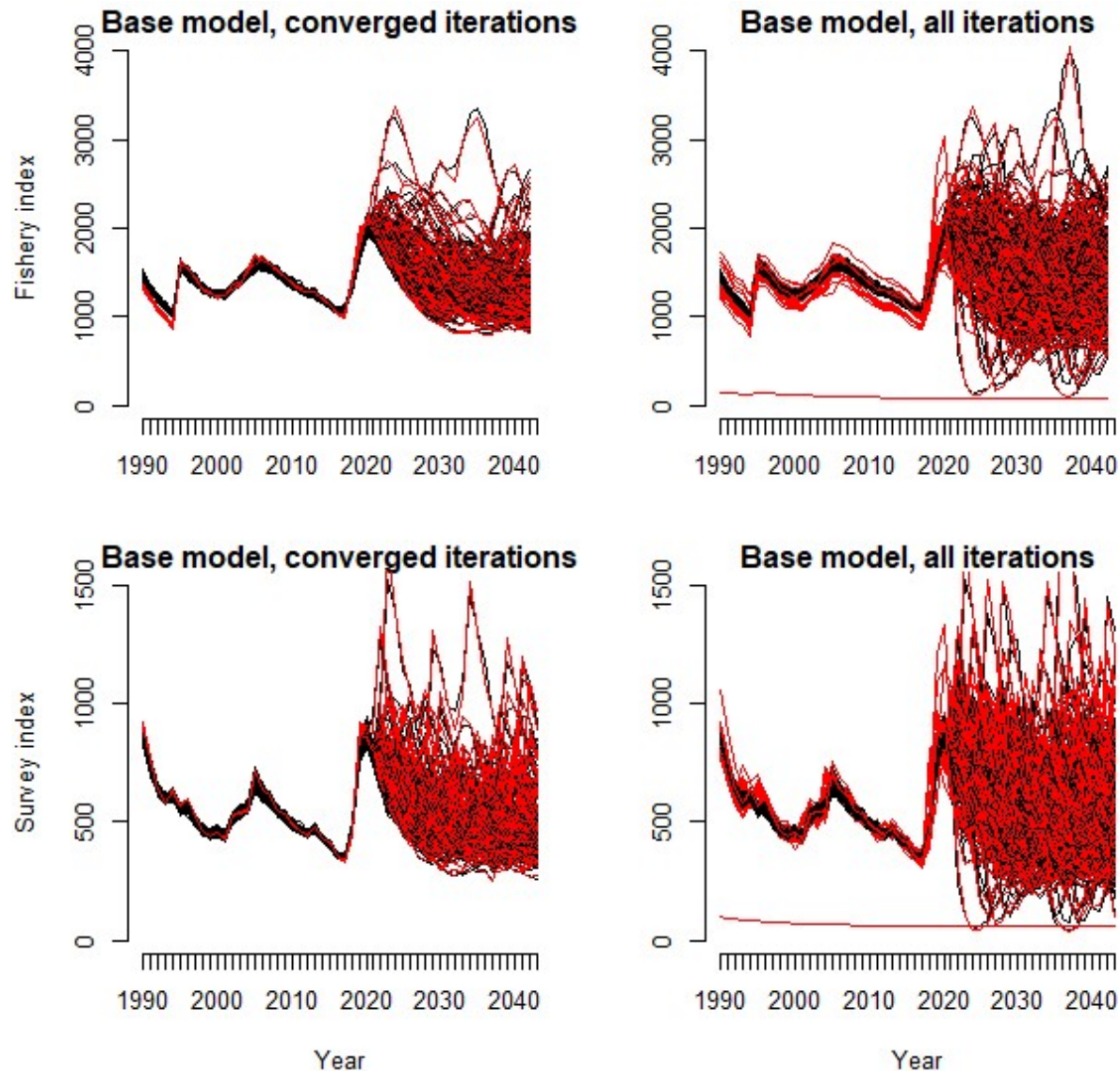


Figure 2.9. Effect of estimation model (EM) convergence criteria on EM fit to operating model (OM) indices. Fishery (top panels) and Survey indices (bottom panels) for Base model scenario. Red lines are EM predicted values and black lines are ‘true’ OM values for 1990–2042 (fishery index) and 1990–2043 (survey index). Left panels are model iterations when the EM converged with a maximum gradient component (MGC) < 0.01. Right panels show all model iterations, including those that failed to meet the MGC < 0.01 convergence criteria. Note that the forward projection period begins in 2019.

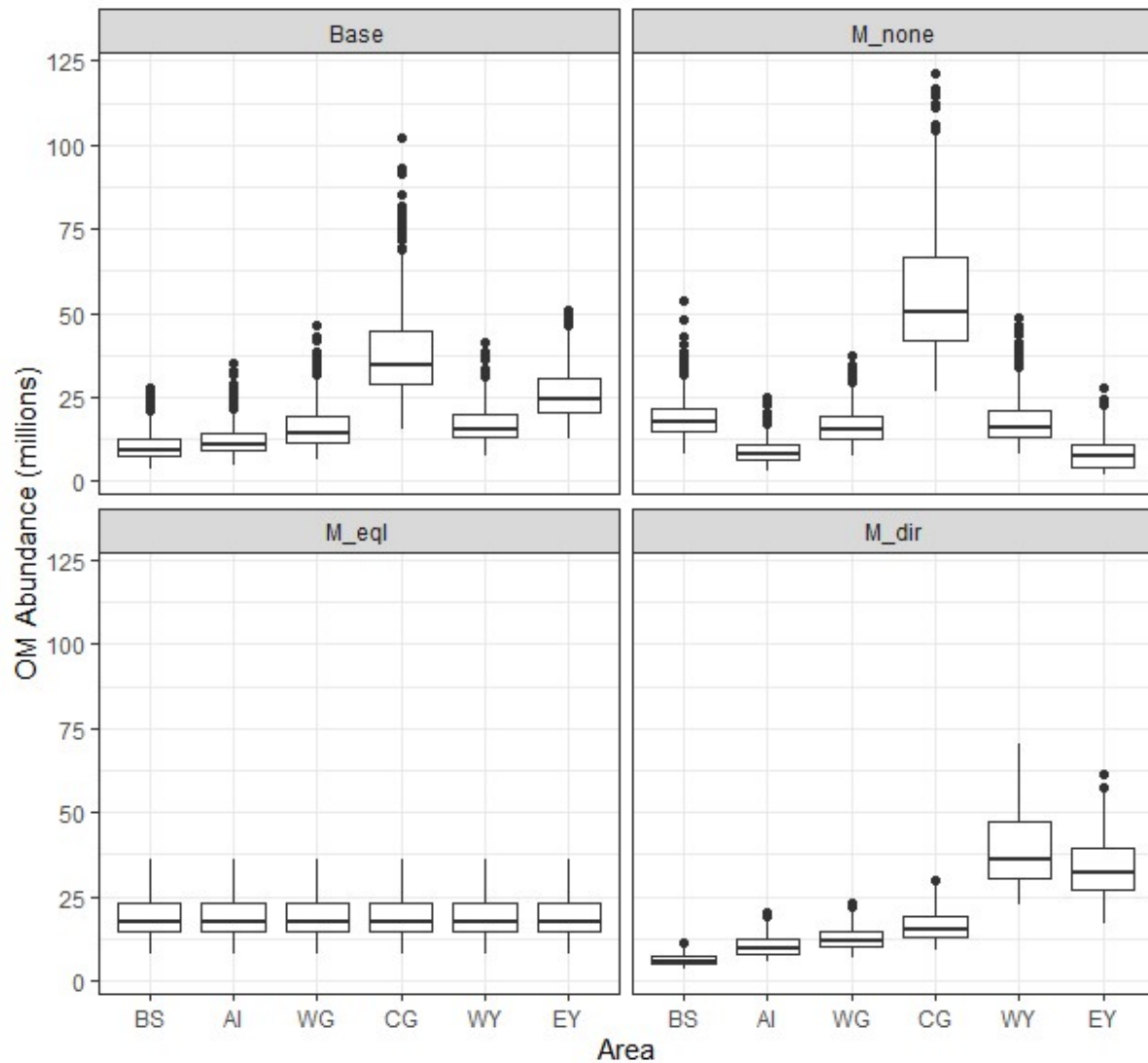


Figure 2.10. Operating model (OM) abundance for alternative movement scenarios. Abundance (millions) simulated in the OM for each spatial area for the Base model scenario and three alternative scenarios with alternative OM movement rates among areas. The M_none scenario assumed no movement occurred between areas, M_eq assumed hypothetical movement rates that moved fish in equal proportions to all other areas, and M_dir assumed movement rates were directional with fish predominantly moving from western areas towards eastern areas. The management areas modeled in the OM were the Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat ((WY), and East Yakutat/Southeast Outside (EY).

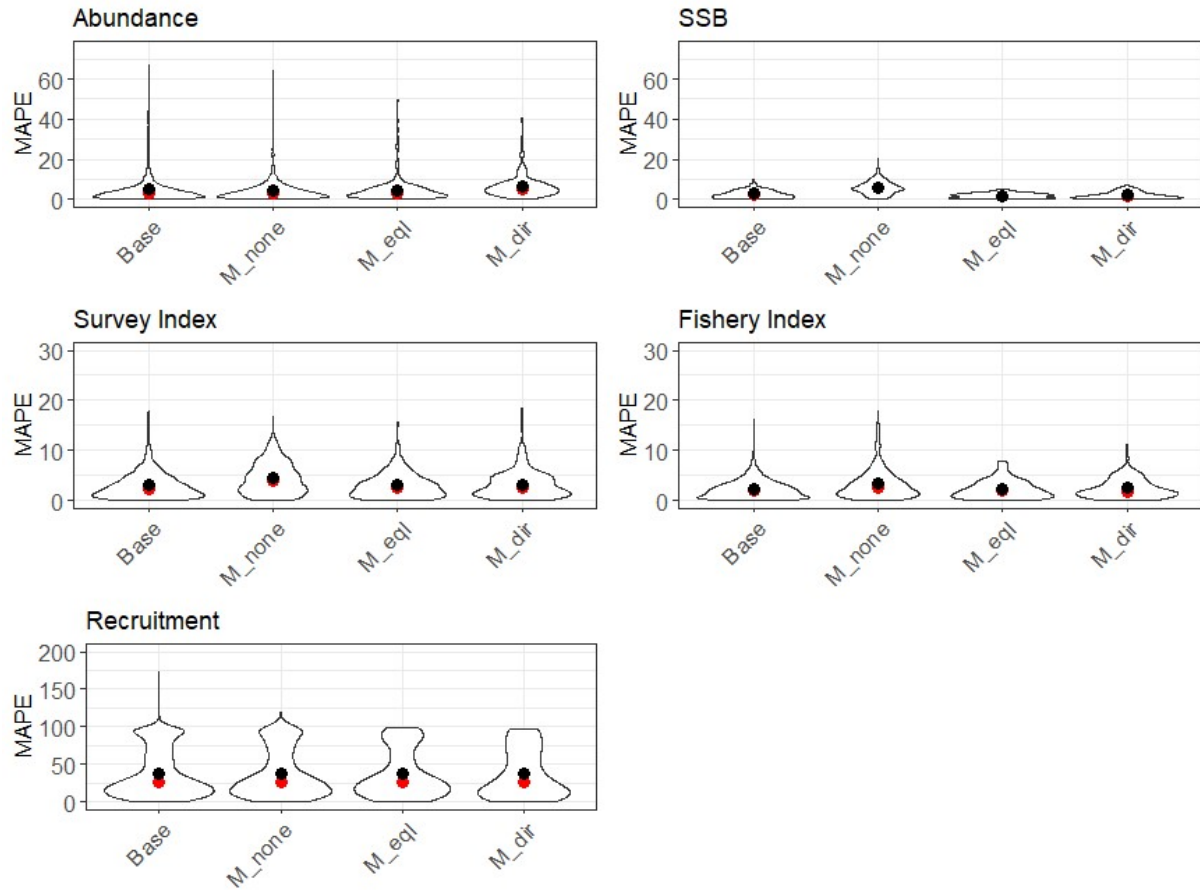


Figure 2.11. Alternative movement model performance. Comparison of mean absolute percent error (MAPE) between the operating model and estimation model abundance, spawning stock biomass (SSB), survey and fishery indices, and recruitment for the Base scenario and three scenarios with alternative movement rates among spatial areas (converged iterations only). M_none assumes no movement occurred between areas, M_eq assumes hypothetical movement rates that moved fish in equal proportions to all other areas, and M_dir assumes movement rates were directional with fish predominantly moving from western areas towards eastern areas. Black points represent the mean value across iterations, red points represent the median values. Note the different y-axis scales for each row of figure panels.

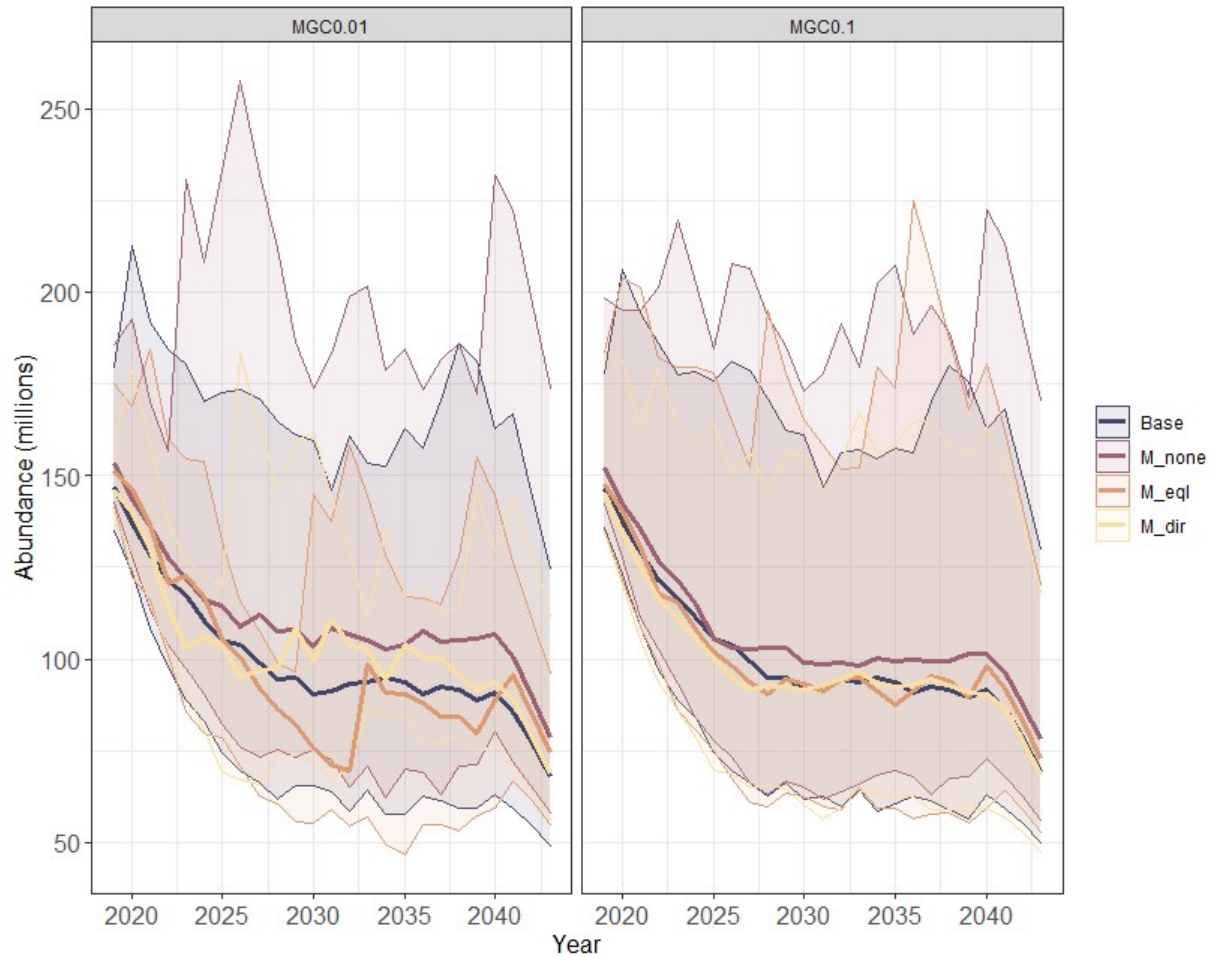


Figure 2.12. Comparison of estimated abundance for alternative movement scenarios alternative convergence criteria. Total abundance (millions) estimated from the estimation model (EM) for the Base scenario and three alternative movement scenarios. Scenario *M_none* assumes no movement occurred between areas, *M_eq* assumes hypothetical movement rates that moved fish in equal proportions to all other areas, and *M_dir* assumes movement rates were directional with fish predominantly moving from western areas towards eastern areas. Solid lines represent the median values across converged simulation iterations, the upper and lower shaded regions represent the 95% quantiles for each scenario. The left panel is model iterations when the EM converged with a maximum gradient component (MGC) < 0.01. The right panel shows model iterations for a MGC < 0.1 convergence criteria.

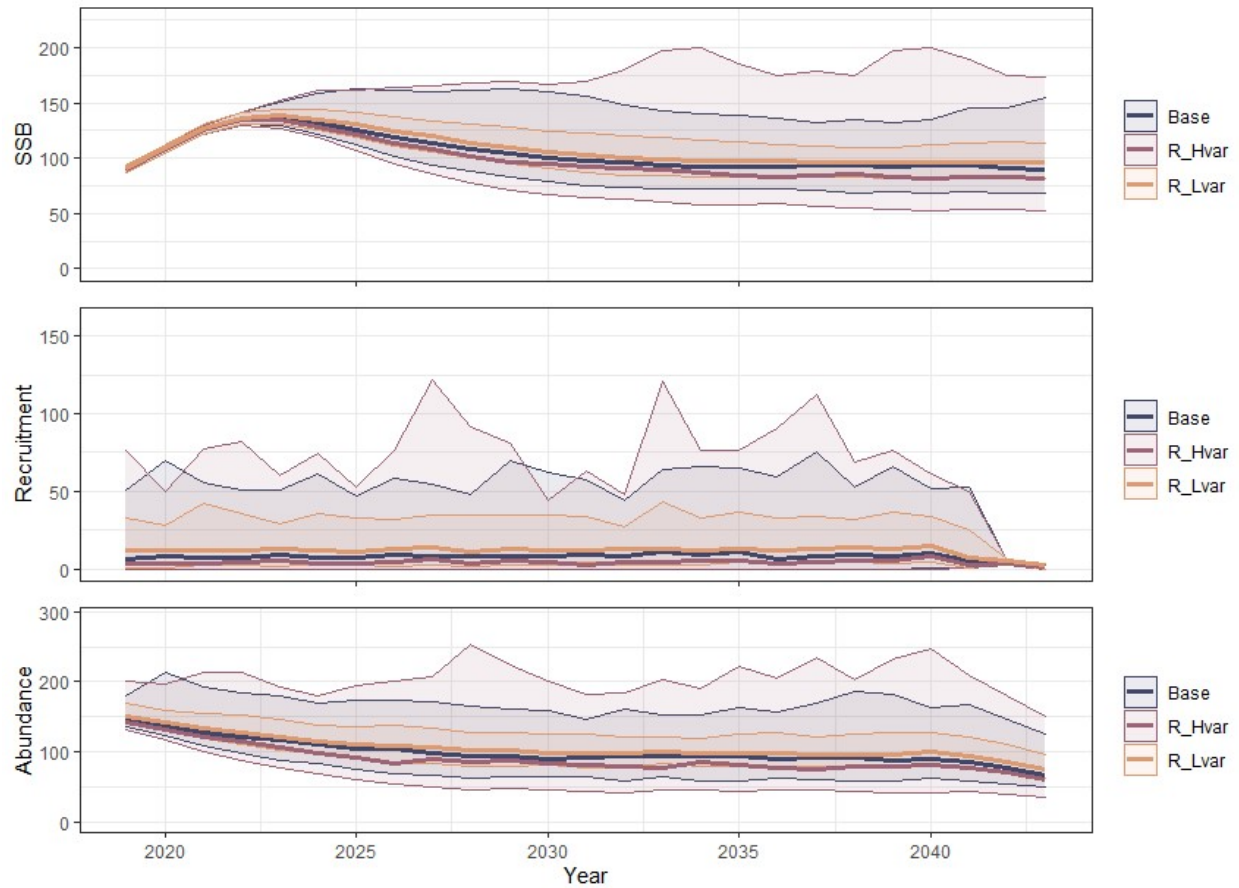


Figure 2.13. Comparison of population estimates for alternative recruitment variability scenarios. Estimation model estimates for spawning stock biomass (SSB, kt), abundance (millions), and recruitment (millions) for Base model scenario and model scenarios with 50% higher operating model (OM) recruitment variance (process error; R_Hvar) or 50% lower OM recruitment variance (R_Lvar). Solid lines represent the median values across converged simulation iterations, the upper and lower shaded regions represent the 95% quantiles for each scenario.

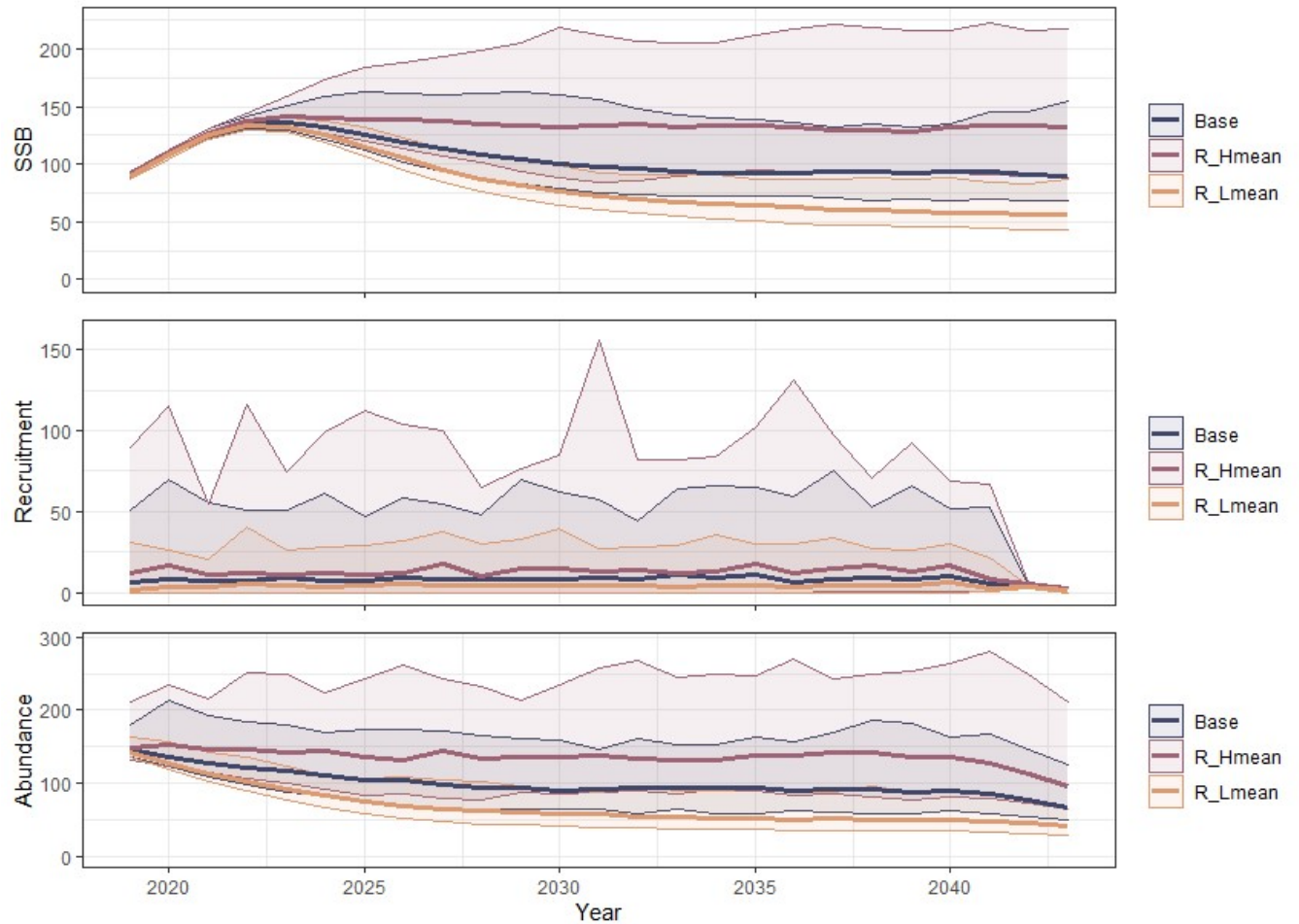


Figure 2.14. Population estimates for alternative mean recruitment scenarios. Estimation model estimates of spawning stock biomass (SSB; kt), abundance (millions), and recruitment (millions) for model scenarios that increased (R_Hmean) or decreased (R_Lmean) mean recruitment in the OM by 50% compared to the Base model recruitment mean. Solid lines represent the median values across converged simulation iterations, the upper and lower shaded regions represent the 95% quantiles for each scenario.

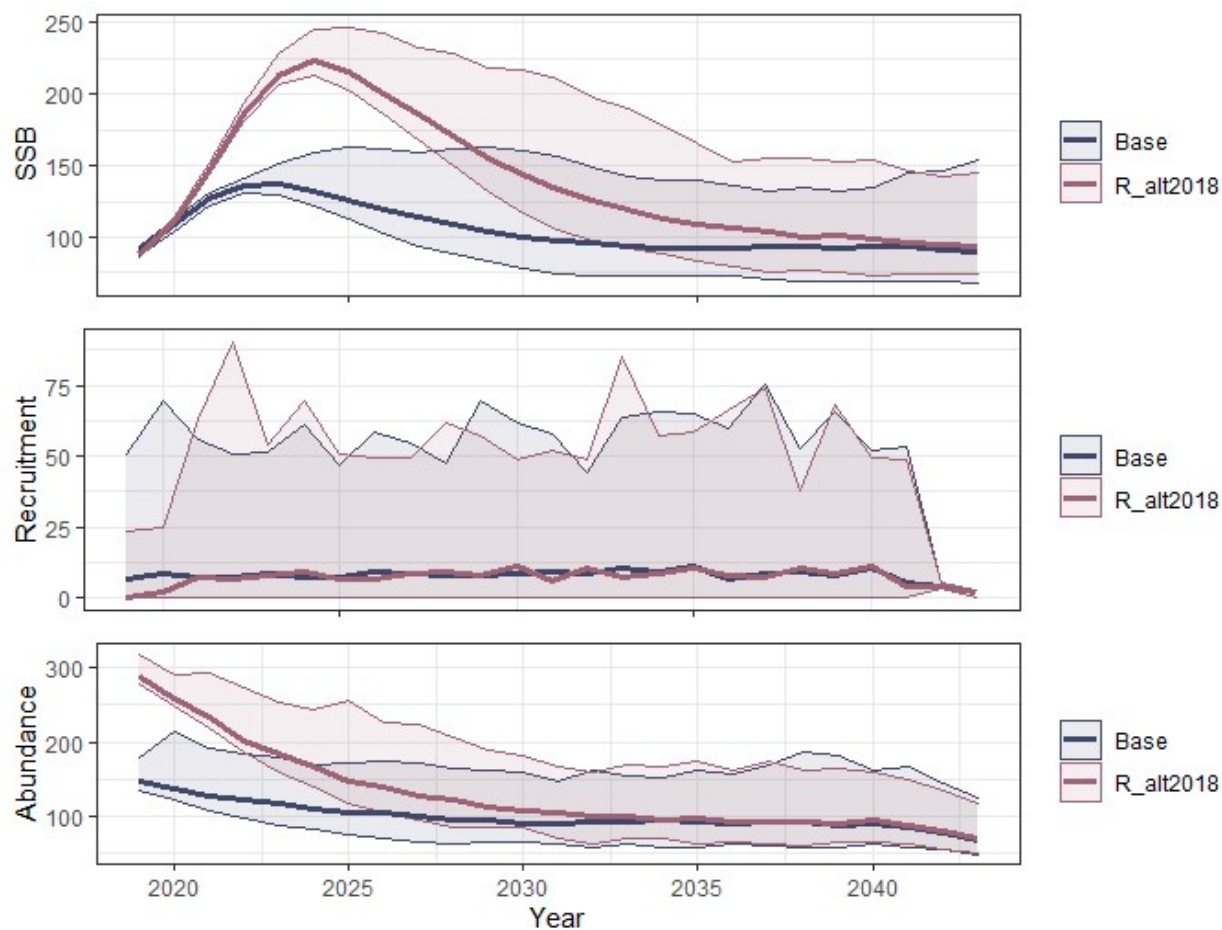


Figure 2.15. Comparison of population estimates for alternative recruitment scenarios. Estimation model estimates of spawning stock biomass (SSB; kt), abundance (millions), and recruitment (millions) for the Base model scenario compared to the R_alt2018 scenario which used a recruitment value of 163.7 million recruits for the terminal year of the conditioning period. Solid lines represent the median values across converged simulation iterations, the upper and lower shaded regions represent the 95% quantiles for each scenario.

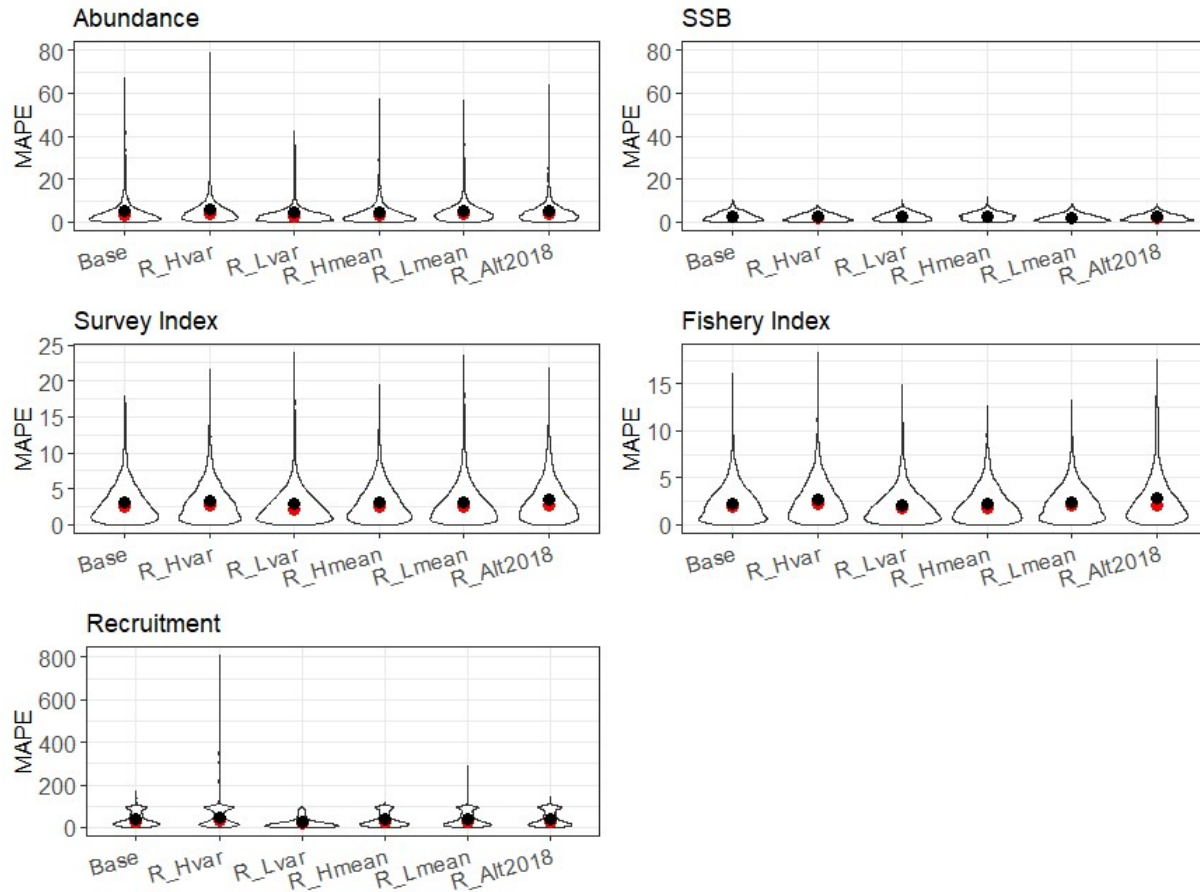


Figure 2.16. Alternative recruitment model performance. Mean absolute percent error (MAPE) between the operating model (OM) and estimation model for model scenarios with alternative OM parameterization of recruitment. Black points represent the mean value across iterations, red points represent the median values. Note the different y-axis scales for each row of figure panels. Model scenarios in the figure include the base simulation scenario (Base), model scenarios with 50% higher OM recruitment variance (process error; R_Hvar) or 50% lower OM recruitment variance (R_Lvar), increased (R_Hmean) or decreased (R_Lmean) mean recruitment in the operating model by 50% compared to the Base model recruitment mean, and a recruitment value of 163.7 million recruits for the terminal year of the conditioning period (R_alt2018).

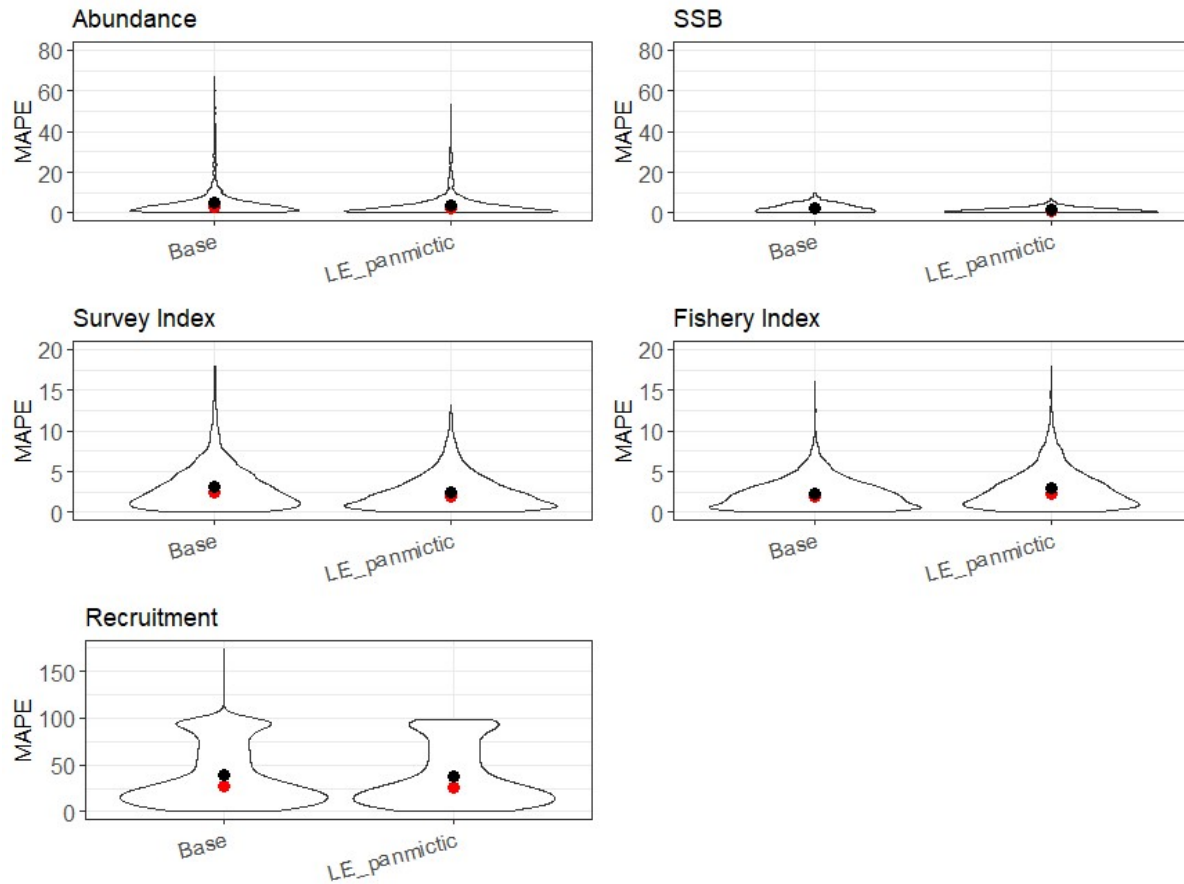


Figure 2.17. Low error, panmictic model performance. Mean absolute percent error (MAPE) between the operating model and estimation model for Base model scenario and the low error, panmictic model (LE_panmictic) scenario. Black points represent the mean value across iterations, red points represent the median values. Note the different y-axis scales for each row of figure panels.

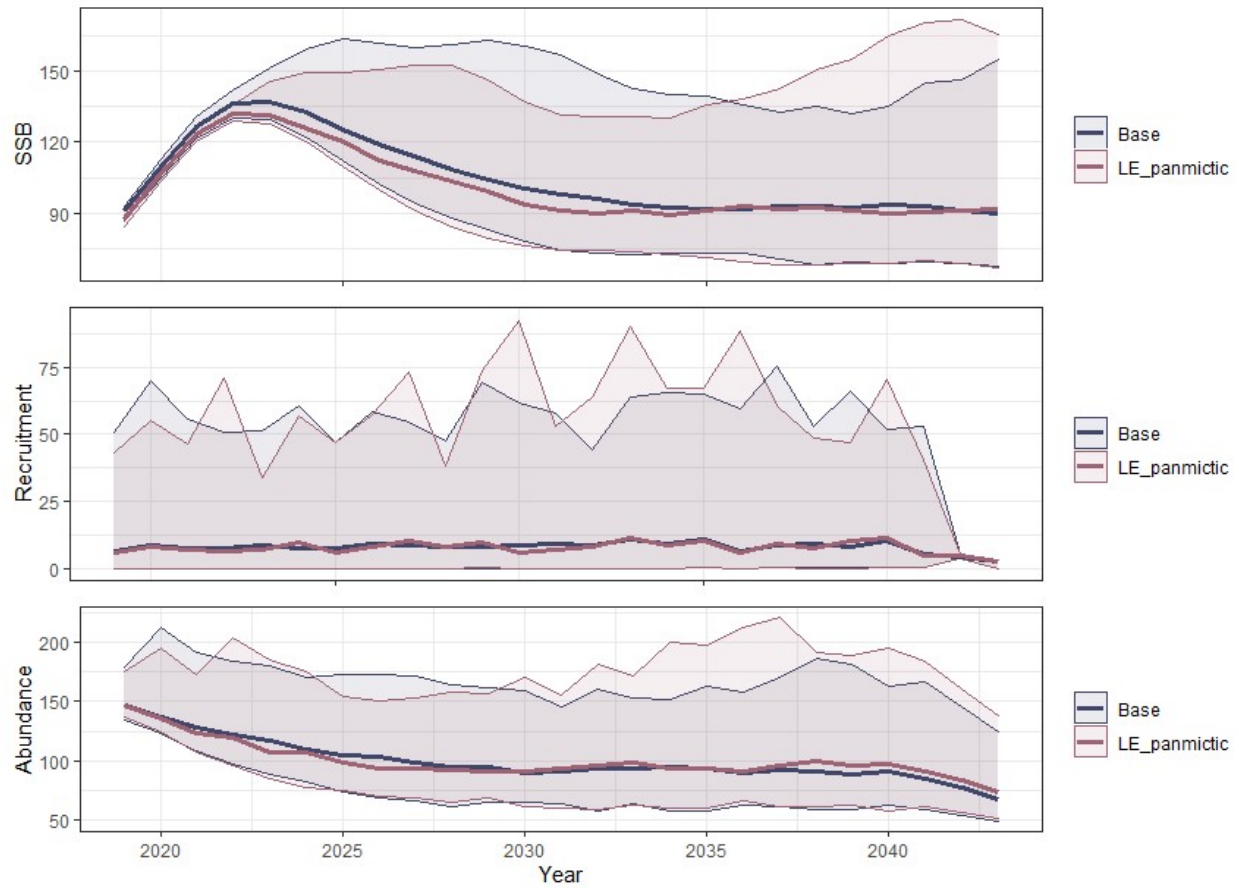


Figure 2.18. Comparison of population estimates for alternative recruitment scenarios. Estimation model estimates of spawning stock biomass (SSB; kt), abundance (millions), and recruitment (millions) for the Base model scenario compared to the low error, panmictic scenario (LE_panmictic). Solid lines represent the median values across converged simulation iterations, the upper and lower shaded regions represent the 95% quantiles for each scenario.

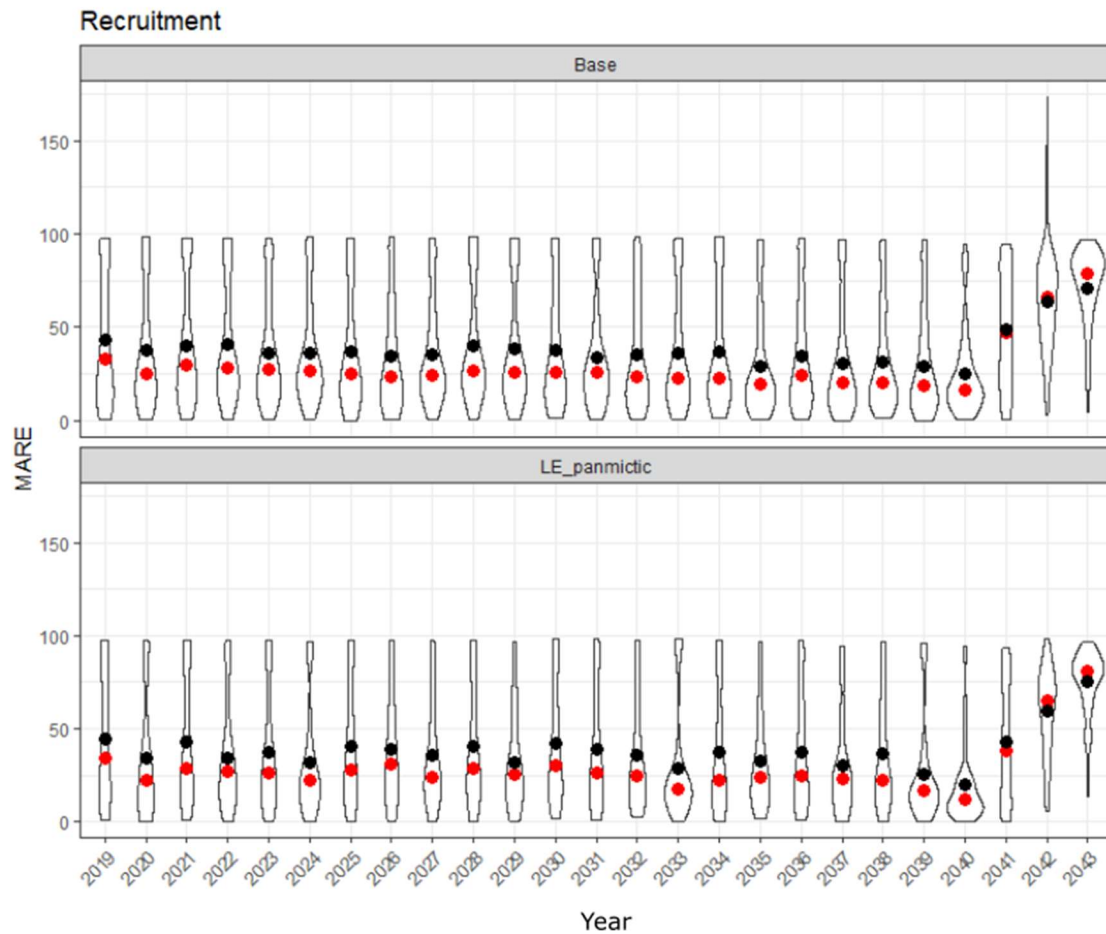


Figure 2.19. Low error, panmictic model performance comparisons over years. Mean absolute percent error (MAPE) between the operating model and estimation model for the Base model scenario and the low error, panmictic model (LE_panmictic) over forward projection years 2019-2043. Black points represent the mean value across iterations, red points represent the median values. Note the different y-axis scales for each row of figure panels.

2.8 Tables

Table 2.1. Operating model (OM) scenarios – movement and recruitment. Alternative OM scenarios deviated from the Base scenario as described below. Blank cells for specific model components indicate they are consistent with the Base OM scenario.

	Recruitment specifications	Recruitment for 2018	Movement
Base model	Mean 16.5, sigma r 0.8; divided into OM regions using mean proportions observed from survey	16.5 million recruits (10 year mean)	Age-specific, base movement rates
<i>Movement Comparisons</i>			
M_none			No movement
M_eq			Hypothetical equal movement
M_dir			Hypothetical directional movement: west to east
<i>Recruitment Comparisons</i>			
R_Hvar	Higher variance: mean 16.5, sigma r=1.2		
R_Lvar	Lower variance: mean 16.5, sigma r=0.4		
R_Hmean	50% Higher mean: mean 16.905, sigma r=0.8		
R_Lmean	50% Lower mean: mean 15.807, sigma r=0.8		
R_alt2018		163.7 million recruits	

Table 2.2. Operating model (OM) and estimation model (EM) validation. The alternative model scenario LE_panmictic deviated from the Base OM scenario for recruitment, observation error, catchability (q), movement, and age at 50% selectivity (a50) as described in the table. The modeled management areas three regional groups; West, Central, and East. The regional groups were comprised of management areas as follows: West region – Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG) management areas, Central region – Central Gulf of Alaska (CG) management area, and East region – West Yakutat (WY), and East Yakutat/Southeast Outside (EY) management areas.

	Recruitment specifications	Observation error; fishery and survey age comps.	Observation error: fishery and survey indices	Post-IFQ fleet catchability	Movement	Post-IFQ fleet selectivity
Base model	Mean 16.5, sigma r 0.8; divided into OM regions using mean observed survey proportions	Multinomial sample size N=200	Sigma=0.3 for BS and AI sigma=0.15 for WG, CG, WY, and EY	Estimated in EM, OM values different for East, West, Central OM area groups	Age-specific, base movement rates	Estimate a50 in EM, OM a50 values different for East, West, Central area groups
LE_panmictic	Same mean and sigma r as Base, but recruitment equally distributed to areas (1/6 to each area)	Lower error, N=500	50% Lower error: sigma=0.15 for BS and AI, sigma=0.075 for WG, CG, WY, EY	Estimated in EM, OM has same q across East, West, Central OM area groups	Age-invariant, hypothetical equal movement between areas	Estimate a50 in EM, OM a50 has same values for East, West, Central area groups

Table 2.3. Operating model (OM) movement rates. Movement rates for the Base OM scenario and alternative scenarios. The specific OM scenarios that use each movement matrix are indicated in parentheses. As subset of age-based movement rates input are provided (see appendix for complete set of matrices for ages 1–16+). Note that rows may not sum to one due to rounding error in the table. Row labels indicate the area from which fish move, column labels indicate the area to which fish move. The management areas modeled in the OM were the Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat ((WY), and East Yakutat/Southeast Outside (EY). Model scenarios in the table include model scenarios with 50% higher operating model recruitment variance (process error; R_Hvar) or 50% lower OM recruitment variance (R_Lvar), increased (R_Hmean) or decreased (R_Lmean) mean recruitment in the operating model by 50% compared to the Base model recruitment mean, and a recruitment value of 163.7 million recruits for the terminal year of the conditioning period (R_alt2018). The M_none scenario assumed no movement among areas, M_eq assumed hypothetical movement rates that moved fish in equal proportions to all other areas, and M_dir assumed movement rates were directional with fish predominantly moving from western areas towards eastern areas. The LE_panmictic model indicates a low error, panmictic simulation scenario.

No movement (M_none)

	BS	AI	WG	CG	WY	EY
BS	1	0	0	0	0	0
AI	0	1	0	0	0	0
WG	0	0	1	0	0	0
CG	0	0	0	1	0	0
WY	0	0	0	0	1	0
EY	0	0	0	0	0	1

Table 2.3, cont.

Equal movement (M_eq, LE_panmictic)

	BS	AI	WG	CG	WY	EY
BS	0.167	0.167	0.167	0.167	0.167	0.167
AI	0.167	0.167	0.167	0.167	0.167	0.167
WG	0.167	0.167	0.167	0.167	0.167	0.167
CG	0.167	0.167	0.167	0.167	0.167	0.167
WY	0.167	0.167	0.167	0.167	0.167	0.167
EY	0.167	0.167	0.167	0.167	0.167	0.167

Hypothetical directional movement towards eastern areas (LE_dir)

	BS	AI	WG	CG	WY	EY
BS	0.150	0.400	0.150	0.100	0.100	0.100
AI	0.100	0.150	0.400	0.200	0.100	0.050
WG	0.050	0.100	0.150	0.400	0.150	0.150
CG	0.050	0.100	0.100	0.150	0.400	0.200
WY	0.025	0.025	0.050	0.100	0.400	0.400
EY	0.050	0.050	0.050	0.050	0.400	0.400

Age-based movement tables (Base, R_Lmean, R_Hmean, R_Hvar, R_Lvar, R_alt2018)

OM Age 1	BS	AI	WG	CG	WY	EY
BS	0.663	0.025	0.244	0.050	0.016	0.002
AI	0.056	0.784	0.140	0.015	0.005	0.000
WG	0.137	0.135	0.307	0.295	0.101	0.026
CG	0.021	0.017	0.155	0.502	0.193	0.111
WY	0.020	0.017	0.147	0.504	0.192	0.120
EY	0.002	0.002	0.024	0.159	0.069	0.744

Table 2.3, cont.

OM Age 5	BS	AI	WG	CG	WY	EY
BS	0.587	0.040	0.203	0.120	0.043	0.007
AI	0.044	0.718	0.129	0.077	0.027	0.005
WG	0.109	0.118	0.247	0.344	0.139	0.043
CG	0.025	0.027	0.144	0.490	0.206	0.108
WY	0.020	0.021	0.134	0.479	0.203	0.142
EY	0.002	0.002	0.022	0.134	0.075	0.765

OM Age 10	BS	AI	WG	CG	WY	EY
BS	0.492	0.058	0.152	0.207	0.076	0.014
AI	0.029	0.636	0.115	0.155	0.055	0.010
WG	0.074	0.098	0.171	0.405	0.187	0.065
CG	0.030	0.039	0.131	0.474	0.223	0.103
WY	0.021	0.027	0.119	0.446	0.217	0.170
EY	0.002	0.002	0.020	0.103	0.083	0.790

OM Age 16+	BS	AI	WG	CG	WY	EY
BS	0.379	0.080	0.092	0.312	0.115	0.022
AI	0.012	0.538	0.098	0.248	0.088	0.016
WG	0.032	0.073	0.081	0.478	0.245	0.091
CG	0.036	0.054	0.114	0.456	0.243	0.098
WY	0.023	0.034	0.100	0.407	0.234	0.203
EY	0.002	0.003	0.017	0.066	0.091	0.821

Table 2.4. Estimation model (EM) convergence rates. Model convergence using two thresholds for determining ‘convergence’; EM maximum gradient component (MGC) < 0.1 (less conservative threshold) and MGC < 0.01 (more conservative threshold). The percent converged columns report the proportion of converged simulations across forward projection years (N=25) and simulation iterations (N=500). The proportion retained iterations is the proportion of simulation iterations for which EM converged (based on the specified threshold) for all of the forward projection years. The number retained iterations is the number of simulation iterations (out of 500) for which the EM converged in all of the forward projection years. Model scenarios in the table include model scenarios with 50% higher operating model recruitment variance (process error; R_Hvar) or 50% lower OM recruitment variance (R_Lvar), increased (R_Hmean) or decreased (R_Lmean) mean recruitment in the operating model by 50% compared to the Base model recruitment mean, and a recruitment value of 163.7 million recruits for the terminal year of the conditioning period (R_alt2018). The M_none scenario assumed no movement among areas, M_eq1 assumed hypothetical movement rates that moved fish in equal proportions to all other areas, and M_dir assumed movement rates were directional with fish predominantly moving from western areas towards eastern areas. The LE_panmictic model indicates a low error, panmictic simulation scenario.

Model	Proportion converged MGC<0.1	Proportion converged MGC<0.01	Prop. retained iterations MGC < 0.01	Num. retained iterations MGC < 0.01
Base model	0.957	0.948	0.284	142
R_Hvar	0.963	0.952	0.340	170
R_Lvar	0.955	0.948	0.298	149
R_Hmean	0.944	0.934	0.184	92
R_Lmean	0.968	0.957	0.418	209
R_Alt2018	0.943	0.926	0.182	91
M_none	0.936	0.901	0.084	42
M_eq1	0.945	0.859	0.024	12
M_dir	0.953	0.840	0.016	8
LE_panmictic	0.948	0.912	0.130	65

2.9 Appendix 2. Additional simulation model details

Table A.1. Movement rates for base operating model (OM). Movement rates for each age (1–16+) used in the Base OM scenario and other scenarios with ‘base’ movement rates among spatial areas. The management areas were as follows: Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG) management area, West Yakutat (WY), and East Yakutat/Southeast Outside (EY).

Age-based movement tables

OM Age 1	BS	AI	WG	CG	WY	EY
BS	0.663	0.025	0.244	0.050	0.016	0.002
AI	0.056	0.784	0.140	0.015	0.005	0.000
WG	0.137	0.135	0.307	0.295	0.101	0.026
CG	0.021	0.017	0.155	0.502	0.193	0.111
WY	0.020	0.017	0.147	0.504	0.192	0.120
EY	0.002	0.002	0.024	0.159	0.069	0.744

OM Age 2	BS	AI	WG	CG	WY	EY
BS	0.644	0.029	0.233	0.068	0.023	0.003
AI	0.053	0.767	0.138	0.031	0.011	0.001
WG	0.130	0.131	0.292	0.307	0.111	0.030
CG	0.022	0.019	0.152	0.499	0.196	0.110
WY	0.020	0.018	0.144	0.498	0.195	0.126
EY	0.002	0.002	0.024	0.153	0.071	0.750

OM Age 3	BS	AI	WG	CG	WY	EY
BS	0.625	0.032	0.223	0.085	0.029	0.005
AI	0.050	0.751	0.135	0.046	0.016	0.002
WG	0.123	0.126	0.277	0.319	0.120	0.034
CG	0.023	0.022	0.150	0.496	0.200	0.109
WY	0.020	0.019	0.140	0.492	0.198	0.131
EY	0.002	0.002	0.023	0.146	0.072	0.755

OM Age 4	BS	AI	WG	CG	WY	EY
BS	0.606	0.036	0.213	0.103	0.036	0.006
AI	0.047	0.734	0.132	0.062	0.022	0.003
WG	0.116	0.122	0.262	0.331	0.130	0.039
CG	0.024	0.024	0.147	0.493	0.203	0.108
WY	0.020	0.020	0.137	0.485	0.201	0.137
EY	0.002	0.002	0.023	0.140	0.074	0.760

App.Table A.1, cont

OM Age 5	BS	AI	WG	CG	WY	EY
BS	0.587	0.040	0.203	0.120	0.043	0.007
AI	0.044	0.718	0.129	0.077	0.027	0.005
WG	0.109	0.118	0.247	0.344	0.139	0.043
CG	0.025	0.027	0.144	0.490	0.206	0.108
WY	0.020	0.021	0.134	0.479	0.203	0.142
EY	0.002	0.002	0.022	0.134	0.075	0.765

OM Age 6	BS	AI	WG	CG	WY	EY
BS	0.568	0.043	0.193	0.138	0.049	0.009
AI	0.041	0.702	0.126	0.093	0.033	0.006
WG	0.102	0.114	0.231	0.356	0.149	0.048
CG	0.026	0.029	0.141	0.487	0.210	0.107
WY	0.021	0.022	0.131	0.472	0.206	0.148
EY	0.002	0.002	0.022	0.128	0.077	0.770

OM Age 7	BS	AI	WG	CG	WY	EY
BS	0.549	0.047	0.183	0.155	0.056	0.010
AI	0.038	0.685	0.123	0.108	0.038	0.007
WG	0.095	0.110	0.216	0.368	0.158	0.052
CG	0.027	0.032	0.139	0.484	0.213	0.106
WY	0.021	0.024	0.128	0.466	0.209	0.153
EY	0.002	0.002	0.021	0.122	0.078	0.775

OM Age 8	BS	AI	WG	CG	WY	EY
BS	0.530	0.051	0.173	0.173	0.062	0.011
AI	0.035	0.669	0.121	0.124	0.044	0.008
WG	0.088	0.106	0.201	0.380	0.168	0.056
CG	0.028	0.034	0.136	0.480	0.216	0.105
WY	0.021	0.025	0.125	0.459	0.212	0.159
EY	0.002	0.002	0.021	0.116	0.080	0.780

OM Age 9	BS	AI	WG	CG	WY	EY
BS	0.511	0.055	0.163	0.190	0.069	0.013
AI	0.032	0.652	0.118	0.139	0.050	0.009
WG	0.081	0.102	0.186	0.392	0.178	0.061
CG	0.029	0.037	0.133	0.477	0.220	0.104
WY	0.021	0.026	0.122	0.453	0.214	0.164
EY	0.002	0.002	0.020	0.109	0.081	0.785

App. Table A.1, cont.

OM Age 10	BS	AI	WG	CG	WY	EY
BS	0.492	0.058	0.152	0.207	0.076	0.014
AI	0.029	0.636	0.115	0.155	0.055	0.010
WG	0.074	0.098	0.171	0.405	0.187	0.065
CG	0.030	0.039	0.131	0.474	0.223	0.103
WY	0.021	0.027	0.119	0.446	0.217	0.170
EY	0.002	0.002	0.020	0.103	0.083	0.790

OM Age 11	BS	AI	WG	CG	WY	EY
BS	0.473	0.062	0.142	0.225	0.082	0.015
AI	0.026	0.620	0.112	0.170	0.061	0.011
WG	0.067	0.094	0.156	0.417	0.197	0.070
CG	0.031	0.041	0.128	0.471	0.226	0.102
WY	0.022	0.028	0.116	0.440	0.220	0.175
EY	0.002	0.002	0.019	0.097	0.084	0.795

OM Age 12	BS	AI	WG	CG	WY	EY
BS	0.454	0.066	0.132	0.242	0.089	0.017
AI	0.024	0.603	0.109	0.186	0.066	0.012
WG	0.060	0.090	0.141	0.429	0.206	0.074
CG	0.032	0.044	0.125	0.468	0.230	0.102
WY	0.022	0.029	0.112	0.433	0.223	0.181
EY	0.002	0.002	0.019	0.091	0.086	0.800

OM Age 13	BS	AI	WG	CG	WY	EY
BS	0.435	0.021	0.106	0.201	0.072	0.013
AI	0.069	0.587	0.122	0.260	0.095	0.018
WG	0.053	0.086	0.126	0.441	0.216	0.078
CG	0.033	0.046	0.122	0.465	0.233	0.101
WY	0.022	0.030	0.109	0.427	0.225	0.186
EY	0.002	0.003	0.018	0.085	0.087	0.805

OM Age 14	BS	AI	WG	CG	WY	EY
BS	0.416	0.073	0.112	0.277	0.102	0.019
AI	0.018	0.571	0.103	0.217	0.077	0.014
WG	0.046	0.082	0.111	0.453	0.226	0.083
CG	0.034	0.049	0.120	0.462	0.236	0.100
WY	0.022	0.031	0.106	0.420	0.228	0.192
EY	0.002	0.003	0.018	0.079	0.088	0.811

App. Table A.1, cont.

OM Age 15	BS	AI	WG	CG	WY	EY
BS	0.397	0.077	0.102	0.295	0.109	0.021
AI	0.015	0.554	0.101	0.232	0.083	0.015
WG	0.039	0.077	0.096	0.466	0.235	0.087
CG	0.035	0.051	0.117	0.459	0.239	0.099
WY	0.022	0.033	0.103	0.414	0.231	0.197
EY	0.002	0.003	0.017	0.072	0.090	0.816

OM Age 16+	BS	AI	WG	CG	WY	EY
BS	0.379	0.080	0.092	0.312	0.115	0.022
AI	0.012	0.538	0.098	0.248	0.088	0.016
WG	0.032	0.073	0.081	0.478	0.245	0.091
CG	0.036	0.054	0.114	0.456	0.243	0.098
WY	0.023	0.034	0.100	0.407	0.234	0.203
EY	0.002	0.003	0.017	0.066	0.091	0.821

Table A.2. Operating model (OM) selectivity values. Selectivity shape parameter values for the logistic selectivity in the OM Base scenario. The estimation model (EM) rows indicated whether selectivity is estimated or fixed in the EM during the projection period of the simulation-estimation analyses. The management areas are: Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat (WY), and East Yakutat/Southeast Outside (EY).

Index years	Fishery, foreign 1977–1981	Fishery, pre-IFQ Fixed 1990–1994	Fishery, post-IFQ 1995–2042	Survey, longline 1990–2043	Survey, U.S.-JP 1979–1994
	<i>*not sex-specific</i>	<i>female value</i>	<i>female value</i>	<i>female value</i>	<i>female value</i>
EM a50 mean	fixed	fixed	estimated	estimated	fixed
OM a50 BS, AI, WG	8.17	2.10	2.35	3.27	2.76
OM a50 CG	8.17	2.10	3.10	3.27	2.76
OM a50 WY, EY	8.17	2.10	4.20	3.27	2.76
EM delta mean	fixed	fixed	estimated	fixed	fixed
OM delta BS, AI, WG	2.20	2.93	2.42	5.75	1.79
OM delta CG	2.20	2.93	2.42	5.75	1.79
OM delta WY, EY	2.20	2.93	2.42	5.75	1.79
		<i>male values</i>	<i>male values</i>	<i>male values</i>	<i>male values</i>
EM a50 mean		fixed	estimated	estimated	fixed
OM a50 BS, AI, WG	-	4.64	2.30	3.19	3.60
OM a50 CG	-	4.64	3.24	3.19	3.60
OM a50 WY, EY	-	4.64	3.66	3.19	3.60
EM delta mean		fixed	estimated	fixed	fixed
OM delta BS, AI, WG	-	2.15	3.15	4.71	1.79
OM delta CG	-	2.15	3.15	4.71	1.79
OM delta WY, EY	-	2.15	3.15	4.71	1.79

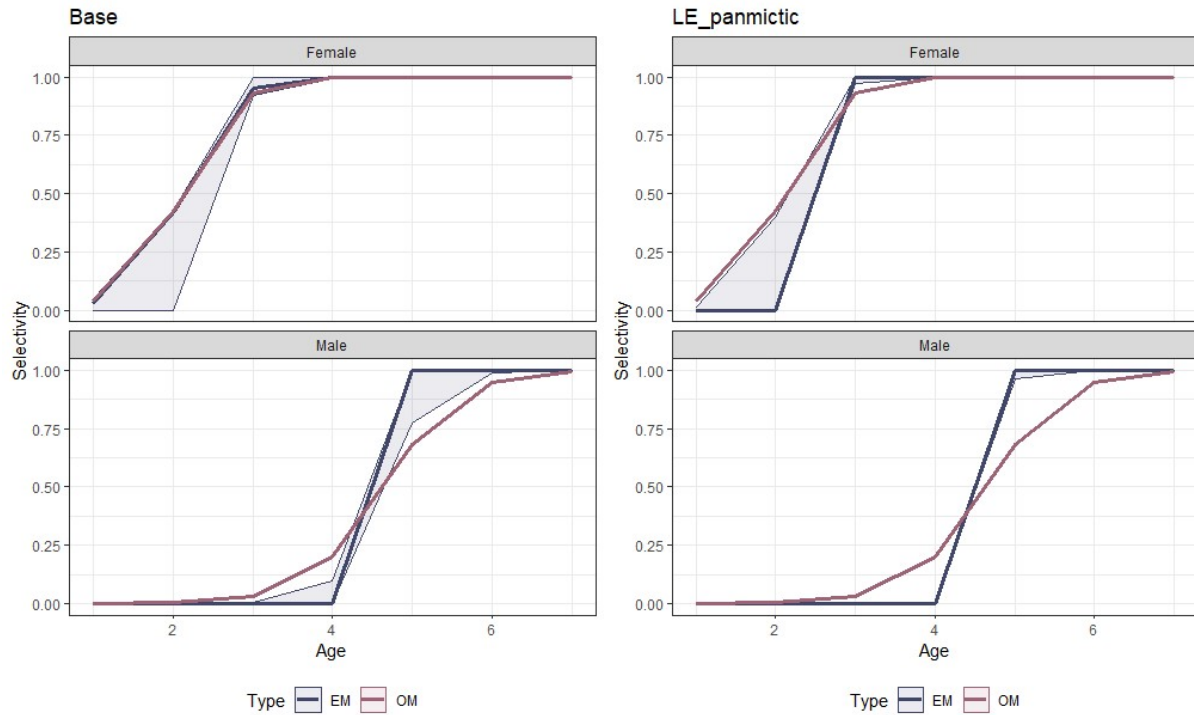


Figure A.1. OM and EM selectivity for Pre-IFQ fishery. EM and OM Pre-IFQ fishery selectivity by sex for the Base model and for the low error 'match' model (LE_panmictic). The OM selectivity value (red) was the same across OM areas. The blue line is the median of EM selectivity estimates for converged models, and the blue shaded region represents the upper and lower 95% quantiles of EM values across converged iterations.

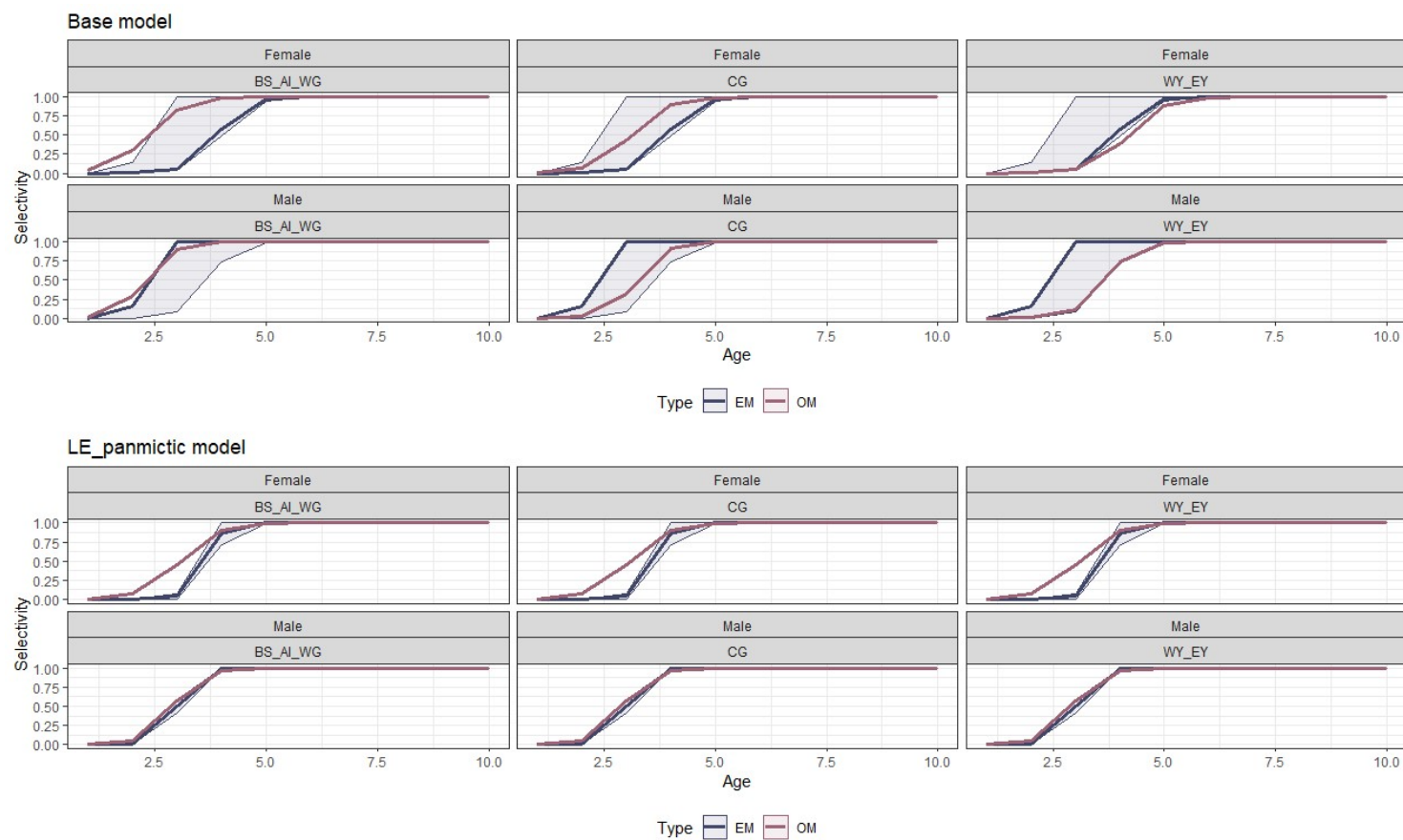


Figure A.2. Estimation model (EM) and operating model (OM) post-IFQ fishery selectivity by area and sex for the Base model and the low error 'match' model (LE_panmictic). The OM selectivity value (red) differs between three area groupings; 1) the Bering Sea (BS), Aleutian Islands (AI), and Western Gulf of Alaska (WG) share OM parameter inputs, 2) the Central Gulf of Alaska (CG), and 3) the West Yakutat (WY) and East Yakutat/Southeast Outside (EY) regions share OM selectivity values. The blue line is the median of EM

selectivity estimates for converged models. The blue shaded region represents the upper and lower 95% of EM values across converged iterations.

Table A.3. Operating model (OM) and estimation model (EM) catchability (q) values. Catchability values assumed for the Base OM scenario and the mean of converged EM iterations for each survey or fishery. EM catchability values were fixed in the model at the value specified or estimated annually. The management areas modeled in the OM were the Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat ((WY), and East Yakutat/Southeast Outside (EY).

Catchability (q)		Fishery, foreign	Fishery, pre-IFQ fixed	Fishery, post-IFQ	Survey,	Survey, U.S.-JP
	Years	1977–1981	1990–1994	1995–2042	1990–2043	1979–1994
	EM mean q	7.02 (fixed)	3.88 (fixed)	6.82 (estimated)	8.74	5.96 (fixed)
	OM areas BS, AI, WG	7.02	5.51	5.49	7.91	5.05
	OM area CG	7.02	3.27	5.06	7.91	5.05
	OM areas WY, EY	7.02	3.64	7.73	7.91	5.05

Chapter 3: Examining tradeoffs to alternative methods of spatial catch apportionment for Alaska sablefish³

3.1 Abstract

The Alaska sablefish stock in the U.S. exclusive economic zone is assessed and managed on an Alaska-wide basis because sablefish movement rates among Alaska management areas are high and exploitation rates are relatively low across the population as a whole, but the annual acceptable biological catch (ABC) limit is allocated between fishing sectors and is apportioned among the six management areas. The apportionment of ABC among areas and the allocation between fleets has potential biological impacts on the sablefish population and social and economic impacts on the fishing fleets and communities in Alaska. We analyzed five sablefish ABC apportionment methods and quantified tradeoffs between alternatives with respect to population sustainability and interannual stability using two techniques: 1) a simulation-based analysis that simulated sablefish population dynamics and used an estimation model similar to the current operational stock assessment model to assess the population annually, and 2) a retrospective approach that apportioned historic total ABCs to illustrate what ABC for each management region would have been had one of the alternative apportionment methods been applied. We found that no single apportionment strategy would address all of the concerns identified by stakeholders, as some desired outcomes conflict with each other. Regardless, there were similar long-term demographic trajectories for the median and range of abundance and spawning biomass estimates across simulation iterations for all five apportionment scenarios. The Terminal Year Survey, North Pacific Fishery Management Council, and Survey Average apportionment alternatives met sustainability objectives more closely by tying ABC to observed biomass among regions. However, this came at the cost of reduced interannual stability in ABC relative to the Fixed apportionment scenario. Age-based ABC apportionment appeared to satisfy neither of the objectives. We found that a significantly larger amount of work was required to develop the simulation analyses than the retrospective analyses, yet the results and conclusions of the two analytical techniques were similar.

³ Fenske, K.H., C. Lunsford, D.H. Hanselman, and C.J. Cunningham.

3.2 Introduction

The division of natural resources among competing entities is fraught with challenges and uncertainties (Charles 1992). Fisheries managers determine the allocation of catch between commercial and recreational fishers and among fishing fleets within the commercial sector. In the U.S., the Magnuson-Stevens Fishery Conservation and Management Act (MSA) details ten National Standards (NS) guiding the management of fisheries in the U.S. exclusive economic zone (EEZ). National Standard 1 specifies the prevention of overfishing while achieving optimal yield. However, fair and equitable allocation of the resource (NS 4) and consideration of fishing communities (NS 8) are also specified in the MSA (NMFS 2006). Tradeoffs are inherent in resource management. For example, the National Standards articulate conflicting objectives regarding conservation, harvest efficiency, bycatch, and socioeconomic outcomes. Variations in resource distribution across space and time and changing environmental conditions that add uncertainty to assessing and managing fish populations make it more important than ever to understand and quantify tradeoffs of alternative management strategies (Kvamsdal et al. 2016), particularly concerning allocative decisions.

In Alaska, sablefish (*Anoplopoma fimbria*) is a valuable commercial species (Fissel et al. 2018) that presents a management challenge due to high movement rates (Heifetz and Fujioka 1991; Hanselman et al. 2015), fluctuations in abundance, and highly variable recruitment. The directed fishery is a fixed gear (pot and longline) fishery managed by the North Pacific Fishery Management Council (hereafter, Council) using catch shares [i.e., Individual Fishing Quotas (IFQs)], with some bycatch allocations to trawl fisheries (NPFMC 2018a, 2018b). Unlike many other broadly distributed species in Alaska (e.g., walleye pollock *Gadus chalcogrammus*, Pacific cod *Gadus macrocephalus*, Pacific ocean perch *Sebastes alutus*), the portion of the Alaska sablefish stock (hereafter ‘Alaska sablefish’) in the U.S. EEZ is managed on an Alaska-wide basis because sablefish movement rates among Alaska management areas are high and exploitation rates are relatively low across the population as a whole (Hanselman et al. 2019). Each year the sablefish stock assessment is updated with new input data. It produces estimates of the acceptable biological catch (ABC, a target reference point for catch limits) and the overfishing limit (OFL, a limit reference point for catch) for the subsequent year. ABC is apportioned among

six management areas; the Aleutian Islands (AI), Bering Sea (BS), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat (WY), and East Yakutat/Southeast Outside (EY). ABC is also allocated between the directed and trawl fleets. The apportionment of ABC among areas and the allocation between fleets has potential biological impacts on the sablefish population and social and economic impacts on the fishing fleets and communities in Alaska.

The Alaska sablefish population has experienced a series of peaks and declines since 1960, when it was first commercially exploited and monitored (Goethel et al. 2020). Infrequent high recruitment events interspersed with periods of low recruitment, coupled with the harvest from fisheries, have led to a pattern of peaks followed by slow declines as the progeny of large recruitment events are progressively harvested or succumb to natural mortality (Goethel et al. 2020). Alaska sablefish biomass declined steadily from the late 1980s until 2016 (Goethel et al. 2020). During 2016–2020, a large influx of small, immature sablefish was observed in annual NOAA Alaska Fisheries Science Center longline surveys (Goethel et al. 2020) and reported by fishers. This influx of fish has created new management challenges.

Resource allocation among stakeholders has become more challenging in the wake of these recent high recruitment events, as the bycatch of small sablefish increased in non-target fisheries in some management regions. In contrast, more valuable large sablefish became scarce. Historically, it is common for above-average recruitment to be first observed in more western regions (BS, AI, WG). These large recruitment events and resulting changes to the abundance and size composition of sablefish across space have led to questions about whether the current methods used for allocating ABC across management areas are adequately balancing the competing ecological and fishery objectives and whether all ABC allocation procedures foster the long-term sustainability of this valuable fishery resource.

The methods for apportioning ABC among management areas have changed over time, in response to biological information and stakeholder concerns. Beginning in December 1999, the Council apportioned the subsequent years' ABC to the six management areas based on the relative abundance among management regions observed in the annual longline survey. A 5-year exponential weighting of the survey index was used to reduce responsiveness to any

specific annual survey data point and provide stability in spatial allocation over time. Later, this ABC apportionment method was updated to use both the survey and IFQ fishery catch per unit (CPUE) abundance indices (hereafter referred to as 'NPFMC apportionment'). This procedure was used from 2000 to 2013. In 2014, spatial ABC apportionment was fixed at the 2013 proportions because of increasing concerns that the interannual variability in apportionment among management areas was too high under the NPFMC apportionment method (Hanselman et al. 2014), highlighting a specific desire by fishery stakeholders to provide greater stability in harvest apportionment. However, it was understood that this stability came at the cost of reduced potential responsiveness to localized depletion or population redistribution. Therefore, analyses were initiated to examine alternative methods of apportioning ABC and understand the tradeoffs associated with each alternative concerning sablefish sustainability, year-to-year stability in area-specific ABC, and other metrics of interest to stakeholders. Analyses took on increased urgency starting in 2016 when the first of several above-average recruitment year classes began showing up in the fishery and survey data, and the spatial distribution of recruits deviated substantially from the spatial distribution of ABC that was based on fixed proportions established in 2014. This resulted in harvest levels that exceeded ABC in some management areas. In spring 2020, a day-long workshop was held involving various sablefish stakeholders. During the workshop, preliminary results from simulation analyses were discussed, ideas and concerns related to the apportionment methods were shared, and suggestions were incorporated into the analyses presented here. Stakeholder concerns were primarily focused on two issues: population sustainability and stability of year-to-year harvest.

The concurrent decline in the availability of fishery data from some management areas needed to calculate the fishery CPUE index has further complicated the application of the NPFMC apportionment method. Fishery-dependent data are available from onboard fishery observers, electronic monitoring, and a voluntary logbook program. Observer coverage has declined in the Bering Sea; the realized observer coverage rate for partial coverage of hook-and-line vessels targeting sablefish in the BSAI in 2019 was negligible and unusable due to data confidentiality restrictions (Ganz et al. 2020). Additionally, voluntary fishery logbook submissions declined in some areas. Further, electronic monitoring has reduced the availability

and representativeness of weight and effort data needed for the fishery index. Low observer or logbook sample sizes have led to increased variability and thus increased instability in apportionment, particularly for the BS and AI, which already has high variability due to biennial survey coverage in these areas. Fishery data can be valuable for tracking spatial trends in abundance-at-age not captured by fishery-independent surveys; however, if current trends continue, there may be insufficient fishery data to use apportionment methods that require fishery data in all areas. In addition, pot gear has rapidly expanded since being legalized in 2017 in the Gulf of Alaska management areas (WG, CG, WY, and EY), and the fishery CPUE index does not yet incorporate catch rate data from the pot fleet. This divergence in gear types further diminishes the quantity of fishery data available until methods are developed to address the shift in gear types.

This study aims to analyze alternative sablefish ABC apportionment methods and quantify tradeoffs between alternatives with respect to population sustainability and interannual stability. We describe two analyses examining ABC apportionment alternatives to management areas. The first was a simulation-based analysis that simulated sablefish population dynamics and used an estimation model similar to the current operational stock assessment model to assess the population annually. This paired simulation-estimation framework simulated population dynamics under alternative apportionment strategies forward through a 25-year trajectory. During each annual assessment, a management procedure consisting of one of five alternative ABC apportionment methods was applied, and variable fishing mortality across space was imposed accordingly. The second analysis adopted a retrospective approach that used the same five methods to apportion historic total ABCs based on observed fishery and survey data to illustrate what ABC for each management region would have been had one of the alternative apportionment methods been applied. These methods are contrasting yet complementary. The simulation analyses allow for a continual feedback loop that incorporates the management procedure's effect (apportionment type) on the population and makes assumptions about future recruitment and fishery utilization of ABC. The retrospective analyses apportion historical ABC values, but each year's ABC available for

apportionment is independent of the particular apportionment type with no demographic feedback.

3.3 Methods

3.3.1 Simulation Operating model and estimation model

The base operating model (OM) developed for the simulation-estimation analyses and the estimation model (EM) used to assess the population are detailed in Chapter 2. To briefly summarize, these apportionment simulation analyses contain two primary components; a six-area OM developed in R (R Core Team 2018) that simulates sex and age-structured spatial population dynamics and a single-area EM developed in AD Model Builder (Fournier et al. 2012) which assumes the population is panmictic over the six management areas. The six OM spatial areas correspond to the six management areas defined by the Council for Alaska sablefish. The EM estimates parameters for Alaska sablefish as a single, panmictic unit using data generated in the OM with observation error and aggregated across management areas. The OM and EM have an annual time step (y), are age-structured, and have two sexes (h). Ages 2-30 are modeled in both the OM and EM, with age 30 serving as a plus group, accumulating fish aged 30 years and older. The simulation EM was based upon the estimation model used to assess the Alaska sablefish population (hereafter referred to as the ‘Management Model’; Hanselman et al. 2018).

An OM conditioning period (1976–2018) established the modeled sablefish population across the six areas. For the years 1977–2017, within the conditioning period, recruitment R_{cond} , or abundance of the first age in the OM (i.e., age-2), was fixed at the values estimated by the Management Model (Hanselman et al. 2019). Recruitment for 2018 was assumed to be approximately equal to the mean recruitment estimated by the Management Model from 1995 to 2017 (16.5 million age-2 recruits).

Annual total recruitment R_{proj} within the OM for the forward projection period was the same for all alternative apportionment scenarios but differed for each year y and simulation replicate in the forward projection period. A vector of annual log-scale recruitment deviations was drawn from a normal distribution (process error) with an arithmetic mean of 16.5 million

recruits ($\overline{Rec}$), standard deviation $\sigma_r = 0.8$ as $dev_y \sim Normal(0, \sigma_r^2)$ and no autocorrelation, including the appropriate log-normal correction,

$$(1) R_{proj,y} = e^{\overline{Rec}} * e^{(dev_y - (\sigma^2/2))}.$$

Total recruitment was then split among the spatial areas in the OM to seed abundance by area for the first age of the OM based on a random draw from a multinomial distribution conditioned on the mean proportions of age-2 fish from the longline survey (1981–2017). The multinomial sample size, controlling the level of variability in recruitment proportions across space, was set at 100, and multinomial draws for recruitment were standardized to sum to one for the sexes combined and then split equally between sexes, assuming a 50:50 sex ratio at recruitment. No stock-recruitment relationship was assumed because some of the largest observed recruitments have come from low spawning biomass (Hanselman et al. 2019). Additionally, little is known regarding spawning behavior, spawning aggregations, or juvenile distributions other than the movement rates of tagged fish among management areas. Thus, all management areas were treated equally regarding spawning importance and juvenile habitat importance.

Age-specific movement between OM areas was based on movement estimates from a long-term (1979–2009) sablefish tagging study (Hanselman et al. 2015, Chapter 2). The tag-recapture program has released more than 360,000 tagged sablefish across Alaska during annual surveys and targeted nearshore tagging events, and has more than 33,000 recaptures from the fishery and survey (Echave et al. 2013). Age-specific movement rates were estimated from the tagging data for fish ages 1–16 (Table 3.1). Fish aged 17 or older were assumed to move at rates equal to those of age 16 fish. In addition, movement rates were assumed to be time-invariant.

Forward simulations were conducted over 2019–2043, with 500 replicate simulation iterations (*i*) of data generated by the spatial OM for each apportionment scenario (*b*). Within the OM for each year of each iteration *i*, the sampling process was simulated with representative levels of observation error added to all data inputs, aggregated across areas, then provided to the EM for stock assessment and harvest specification. For each year 2020–

2043, 1) the EM first produced an estimate of total ABC for the following year, 2) the ABC was then apportioned to management areas using the specified apportionment method, and finally, 3) the OM population (i.e., abundance by sex and age in each area) was updated based on natural and fishing mortality and age-specific movement, sampled, and the simulated values were provided as inputs to the EM in the next year. The EM was similar to the operational stock assessment used for managing Alaska sablefish (Management Model; Hanselman et al. 2018), except that length composition data were not used, and age composition data were available to the EM without the current 1-year lag. This change was made because, without length composition data, there was nothing to inform recruitment estimates in the terminal year. In addition, the assumption of non-lagged age composition data provided a more realistic depiction of the quality of information available for stock assessment in the absence of length compositions.

The Council is responsible for Alaska sablefish management. It has established harvest control rules based on a tier system that specifies a process for determining harvest limits based on stock status, estimation uncertainty, and data availability (NPFMC 2018a). Alaska sablefish are categorized as a tier 3 stock, so ABC is calculated with an F_{ABC} equal to $F_{40\%}$ which is the fishing mortality rate that would lead to B_{40} . B_{40} is the biomass when spawners produced per recruit is at 40% of unfished biomass. If the spawning biomass/ B_{40} ratio falls below 1.0 (Eqn. 2), the harvest control rule reduces fishing mortality F below the projected F_{40} ,

$$(2) F_{ABC,y+1} = F_{40} \left(\frac{SSB_{y+1}}{B_{40}} - 0.05 \right) / (1 - 0.05).$$

ABC for the next year ($y+1$) was projected by the EM as the sum of projected catch in weight (w), by age a and sex s ,

$$(3) ABC_{y+1} = \sum_s ProjCatch_{y+1,s,a} w_{s,a}.$$

Projected catch (*ProjCatch*) for the next year was a function of the EM estimated abundance N , survival, and fishery selectivity Sel ,

$$(4) ProjCatch_{y+1,s,a} = N_{y,s,a} Survival_{y,s,a} (Sel_a F_{ABC,y+1} / (Sel_{s,a} F_{ABC,y+1} + M)) * (1 - \exp(-Sel_{s,a} F_{ABC,y+1} - M)),$$

where M was natural mortality and $Survival_{y,s,a}$ between the current year (y) and projection year ($y+1$) was:

$$(5) \text{ Survival}_{y,s,a} = e^{-(Sel_{s,a} F_{ABC,y} + M)}.$$

3.3.2 Retrospective analyses

To examine retrospective ABC performance under alternative apportionment scenarios, the historic ABC values used for management by the Council for 2000–2020 were compiled and analyzed. The five apportionment scenarios (equations below) were applied to the historic ABC. In the results, the number of years for which ABCs could be apportioned retrospectively differ due to the availability of data needed to inform each apportionment type.

3.3.3 Apportionment data

A longline survey primarily targeting Alaska sablefish is conducted annually by the NOAA Alaska Fisheries Science Center. The relative population weight (survey index, I_{surv}) for each management area observed in the survey feeds into the annual Alaska sablefish Management Model (as an index of abundance in numbers, not weight) and was used for apportioning ABC to management areas under some apportionment scenarios. The survey is annual in the WG, CG, WY, and EY areas, and biennial in the BS and AI areas. For the simulations, the annual survey was generated as in the Chapter 2 base model, using OM abundance at age, time-invariant weight at age, and lognormally-distributed observation error with a CV of 30% for areas BS, AI, and 15% for areas WG, CG, WY, and EY, respectively. The Survey index was available from 2000 to 2020 for retrospective analyses.

A fishery CPUE biomass index (I_{fish}) was calculated annually using logbook and observer data from the Alaska sablefish fishery. The Management Model used the index and was available for apportioning ABC to management areas for 2005-2020 in the retrospective analyses. For the simulations, the annual survey was generated as in the Chapter 2 base model, using OM abundance at age, time-invariant weight at age, and lognormally-distributed observation error with a CV of 30% for areas BS, AI, and 15% for areas WG, CG, WY, and EY, respectively.

3.3.4 Apportionment scenarios

In both the simulation-estimation analyses and the retrospective analyses, a stock assessment was conducted in year y producing an ABC for the next year, $y+1$. Each year, total ABC (ABC_y) was apportioned into ABC ($\psi_{y,m}$) for each management area, m , using one of the five apportionment methods (b) described below and in Table 2, $ABC_y = \sum_m \psi_{y,m}$. For the simulation-estimation models, it was assumed that 100% of the ABC apportioned to each area was caught each year (i.e., full ABC utilization).

Fixed apportionment:

For the Fixed apportionment scenarios, apportionments were determined using $\psi_{b=1,y,m} = ABC_y * \rho_m$, where ρ_m for this apportionment type was a vector equal to [0.10, 0.13, 0.11, 0.34, 0.11, 0.21] for areas (m) BS, AI, WG, CG, WY, and EY. These proportions to areas ρ_m were the proportions of ABC in each management area for 2013 using the NPFMC apportionment method. Fishery managers used the Fixed apportionment method during 2014–2020.

NPFMC apportionment:

In the NPFMC apportionment scenarios, apportionments were determined using a 5-yr exponentially weighted moving average of fishery CPUE and survey biomass indices ($I_{fish,y,m}$ and $I_{surv,y,m}$, respectively) for each year y and area m . In these scenarios, the survey index was given twice the weight ($w_{surv}=2$) of the fishery index ($w_{fish}=1$). This was the method accepted by the Council for apportioning sablefish ABC for 2000–2013.

$$\psi_{b=2,y,m} = ABC_y * 1/(w_{surv} + w_{fish}) * \left[\left(w_{surv} * \sum_{k=1}^5 \vec{\rho}_k * \left(I_{surv,y=y-k+1,m} / \sum_m I_{surv,y=y-k+1,m} \right) \right) + \left(w_{fish} \sum_{k=1}^5 \vec{\rho}_k * \left(I_{fish,y=y-k,m} / \sum_m I_{fish,y=y-k,m} \right) \right) \right],$$

where the exponential weighting factor ($\vec{\rho}_k$) for this apportionment type was a vector equal to [0.5, 0.25, 0.125, 0.0625, 0.0625] for years y , $y-1$, $y-2$, $y-3$, $y-4$ for the survey index and years $y-1$, $y-2$, $y-3$, $y-4$, $y-5$ for the fishery CPUE index.

Survey Average apportionment:

The survey average apportionment scenario was similar to the ‘NPFMC’ apportionment type but only used the survey index and weighted all years equally. $\vec{\rho}_k$ for this apportionment type was a vector equal to [0.2, 0.2, 0.2, 0.2, 0.2] for years y , $y-1$, $y-2$, $y-3$, $y-4$.

$$\psi_{b=3,y,m} = ABC_y * \sum_{k=1}^5 \vec{\rho}_k * \left(I_{surv,y=y-k+1,m} / \sum_m I_{surv,y=y-k+1,m} \right)$$

Terminal Year Survey apportionment:

Under the Terminal Year apportionment scenario, apportionment to area m was based on the proportion of longline survey biomass in area m relative to total survey biomass across all areas.

$$\psi_{b=4,y,m} = ABC_y * I_{surv,y,m} / \sum_m I_{surv,y,m}$$

Age-based apportionment:

For the Age-based apportionment scenario, ABC is apportioned to area m based on the biomass of fish greater than the ~50% age at maturity (age 6 for these analyses; Sasaki 1985) as estimated by the longline survey data for area m relative to the total biomass across all areas and ages. Areas with a greater proportion of fish age-6 or older received more ABC.

$$\psi_{b=5,y,m} = ABC_y * I_{surv,y,m,a=6+} / \sum_{m,a} I_{surv,y,m,a}$$

3.3.5 Performance metrics

In the simulation-estimation analyses, EM performance was compared to the ‘true’ OM values for abundance, spawning biomass, survey abundance, fishery CPUE biomass, and recruitment. For each apportionment scenario (b), simulation iteration (i), and year (y), the absolute percentage error (APE) between the converged EM estimated parameter ($\bar{x}$) and corresponding ‘true’ value from the OM (x) summed over areas (m) was calculated as,

$$(6) APE_{b,y,i} = |(\sum_m x_{b,y,m,i} - \bar{x}_{b,y,i}) / \sum_m x_{b,y,m,i}| * 100.$$

Mean absolute percentage error (MAPE) was calculated across iterations (i) and years (y) for a given apportionment scenario (b),

$$(7) \text{MAPE}_b = \text{mean} \begin{pmatrix} APE_{y=202, i=1, b} & \cdots & APE_{y=202, i=500, b} \\ \vdots & \ddots & \vdots \\ APE_{y=204, i=1, b} & \cdots & APE_{y=204, i=500, b} \end{pmatrix}.$$

We also examined the magnitude of difference in proportions (Δ) of apportioned ABC ($\psi_{y,m}$) by area and the proportions of x , when x was either observed survey biomass (for retrospective analyses) or the ‘true’ OM spawning biomass (simulation analyses) by area, for each apportionment scenario (b),

$$(8) \Delta_{b,y,m,i} = (\psi_{b,y,m,i} / \sum_m \psi_{b,y,m,i}) - (x_{b,y,m,i} / \sum_m x_{b,y,m,i}).$$

In these analyses, stability referred to year-to-year changes in total ABC and in the apportioned ABC to management areas. To examine interannual change in total ABC ($\delta Total_ABC_{b,y,i}$) for each apportionment scenario (b) and year (y), and iteration (i), we calculated the difference in ABC for year $y+1$ from the previous year y ,

$$(9) \delta Total_ABC_{b,y,i} = ABC_{b,y,i} - ABC_{b,y+1,i}.$$

Similarly, for interannual change in area-specific ABC ($\psi_{b,y,m,i}$) for each apportionment scenario, year, and area, the difference in area-specific proportion of ABC for year $y+1$ from the previous year y was calculated as,

$$(10) \delta Area_ABC_{b,y+1,m,i} = \psi_{b,y,m,i} - \psi_{b,y+1,m,i}.$$

Stability in ABC also was considered in the context of being above or below specific ABC thresholds (in tons of apportioned catch) in each management area. For each apportionment method, we calculated the proportion of converged simulation iteration-year combinations in which the simulated ABC was lower (less than the 25% quartile) or higher (greater than the 75% quartile) than realized area-specific ABC in past years. These upper and lower thresholds were based on the ABCs used for sablefish management between 2000 and 2020.

Because the sustainability of the sablefish population was a primary concern, we evaluated how different apportionment scenarios performed with respect to a B_{40} biological reference point for the simulation analysis. Because the EM was a single-area assessment model, it produced estimates of the biological reference point B_{40} , the biomass at 40% of unfished biomass, for the whole modeled population for the simulation analyses. We calculated the ratio of terminal year SSB/ B_{40} from the EM estimates and compared results among apportionment scenarios for each year and simulation iteration.

3.4 Results

3.4.1 Simulation-estimation analyses

There were similar long-term demographic trajectories for the median and range of abundance and spawning biomass estimates across simulation iterations for all five apportionment scenarios (Figures 3.1, 3.2). The mean absolute relative deviations (MAPE) between the EM and ‘true’ OM population were generally low across years and converged iterations (Figure 3.3). Across years and simulation iterations, MAPE was less than 0.05 (<5%) for all apportionment methods for abundance, spawning biomass, survey abundance index, and fishery CPUE index. Recruitment estimates from the EM deviated from the OM values to a greater extent than other values, with mean MAPE ranging from 0.37 to 0.39 for apportionment scenarios (Figure 3.3). This indicates that the EM generally does an adequate job of reconstructing the ‘true’ biomass and abundance trends simulated by the OM. The median EM estimates for abundance are lower than the OM population median in the last two years of the forward projection period (Figure 3.1). There was relatively little data available in the terminal model year to inform the EM recruitment estimates, only survey age compositions and the longline survey abundance index. Spawning biomass (Figure 3.2) does not show the same pattern, suggesting that recruitment, or abundance of immature fish, is driving the deviations between the EM and OM.

3.4.2 Performance metric outcomes for apportionment from simulation-estimation and retrospective analyses

ABC to management areas and total ABCs

As total ABC changed from year to year, the amount apportioned to each management area also fluctuated depending upon the apportionment scenario and the observed abundance of sablefish among areas. For each retrospective apportionment scenario, the proportions of ABC apportioned to each management area are shown in Figure 3.4a. The actual ABC apportioned to each area under the five retrospective apportionment scenarios are shown in the supplemental appendix. During the retrospective period, the CG received the largest proportion of ABC in a majority of years for all five apportionment methods (mean: 33.5%). The EY region generally was apportioned the second-highest proportion of total ABC (mean: 20.2%, Figure 3.4) across most years and apportionment methods. However, after 2015 under the NPFMC, Survey Average, and Terminal Year apportionment alternatives, the AI area received a larger fraction of the total ABC than the EY area. After 2015, reduced apportionment to the EY and CG areas and increased apportionment to the AI were observed, and extreme recruitment events were observed by the longline survey, which resulted in a shift in harvest opportunity in response to observed changes in relative abundance across regions.

The same pattern of CG and EY receiving the largest proportion of ABC was also observed across all apportionment scenarios during forward simulation (Figure 3.5). For all but the Fixed apportionment scenario, the rank order of ABC proportions to areas for the simulation was, from highest to lowest proportions, generally CG, EY, WY, and WG, with AI and BS showing considerable overlap. However, the Fixed apportionment scenario was frozen during a period when the AI proportion of ABC was high, so for this method, AI received the third highest proportion of ABC instead of the 5th highest (Figure 3.6). The Age-based apportionment scenario resulted in ABC being nearly equal among management areas during the forward projecting period (Figure 3.6). This result was likely driven by movement rates in the OM, the fixed selectivity values in the OM and estimated in the EM, the areal distribution of recruitment in the OM, and the fact that apportionment was age-based instead of size/length-based. Due to dimorphic growth, the age-based apportionment would be less effective at the objective of apportioning more catch to where the largest fish are.

Total ABC from the retrospective analyses ranged between 11,795 t and 23,000 t for 2000–2020 (Table 3.3, see Appendix 3 for area-specific ABC, $\delta Area_ABC_{b,y+1,m,i}$). Mean ABC, summed over management areas, during 2000–2020 was 17,122 t.

Despite the range of area-specific ABC values, all five apportionment scenarios had similar means for total ABC across years and simulation iterations for converged simulations. The mean ABC (2020–2043) across converged simulation iterations and years was 19.2, 19.3, 19.5, 19.4, and 19.7 thousand t for Fixed, NPFMC, Age-based, Terminal year survey, and Survey average apportionment scenarios. These values result from a combination of the unique set of iterations that converged for each apportionment scenario, the feedback loop inherent in the simulation, and the OM parameterization of recruitment mean and variability.

Biological sustainability

To maximize potential catches while maintaining biological sustainability and minimize the risk of localized depletion for a stock, harvest pressure was optimally allocated in proportion to realized population abundance across space. Figure 3.7 shows the distribution among years of the difference between ABC proportions by area and the relative biomass indexed by the longline survey across regions from the retrospective analysis. The Terminal year survey apportionment scenario was, by definition, apportioning ABC based on the most recent estimates of survey biomass distribution and thus had zero deviations. The Survey average and NPFMC apportionment scenarios also performed well, with low median deviations across time between area-apportioned ABC and estimated regional biomass, indicating a consistent match between allowable harvest and observed biomass among areas. In contrast, the Age-based apportionment method showed the greatest median difference between apportioned ABC and relative survey biomass estimates across areas. Under this apportionment scenario, the CG and WY regions were routinely allocated a higher proportion of ABC compared with the distribution of biomass across space observed by the longline survey, and the AI, BS, and WG areas received a lower proportion of ABC relative to the observed biomass in each region. The Fixed apportionment scenario showed a similar pattern but was less pronounced with ABC apportioned closer to relative biomass.

Recall that since the EM is a single-area assessment model it only produced biological reference points such as B_{40} , the biomass at 40% of unfished biomass, estimates for the whole modeled population for the simulation analyses. Therefore, the biological sustainability of apportionment alternatives was examined in terms of the EM reference point B_{40} and compared the B_{40} estimated in each year of converged iterations to the EM spawning biomass (Figure 3.8). SSB/B_{40} increased steadily for the first five years of simulations as the high recruitment in 2016 of the OM conditioning period entered the population. Then SSB/B_{40} decreased to an equilibrium level for all apportionment types, with a mean range of 0.90-0.93 (across converged simulation iterations) and a median range of 0.88-0.90 among apportionment alternatives for the terminal year of simulations (the year 2043; Figure 3.8). The five apportionment types performed similarly for SSB/B_{40} , with no apportionment type resulting in a mean or median SSB/B_{40} value of 1.

Similar to the analyses conducted on the retrospective ABC apportionment, the difference in proportions of true OM SSB in management areas was compared to the proportions of ABC apportioned for each area for the five scenarios examined (Figure 3.9). For apportionment types NPFMC, Terminal Year Survey, and Survey average, the median difference between OM SSB and apportioned ABC was less than 5% for all six areas. The Age-based apportionment type had the greatest deviations between OM SSB and ABC proportions, indicating that this apportionment type did not match 'true' OM SSB as closely as the other apportionment types. However, despite variation in the extent of mismatch between ABC apportionment and OM SSB, none of the apportionment alternatives led to localized population crashes (Figure 3.10) in the 'true' OM population for converged models.

Interannual ABC 'Stability' performance metric

Stability in interannual ABC may be important for IFQ shareholders' business planning and ensuring sufficient processing capacity at shore-based processors. Large interannual changes in ABC were the impetus for freezing ABC apportionment proportions in 2014. In the retrospective analysis, the year-to-year changes in total ABC ranged between 0 and 38% from 2000 through 2020 (Table 3.3). Area-specific ABC changes varied depending on the

apportionment scenario applied. However, there were often more extreme changes for a given area than the total interannual change in ABC for a given year.

Accounting for the annual change in total ABC, the retrospective apportionment analyses identified the stability tradeoffs among the five apportionment types. Figure 3.11 illustrates how the area-specific change in ABC compares to the total change in ABC between years. The Fixed apportionment method used time-invariant proportions to apportion ABC. The Age-based and Terminal year survey apportionment methods had the greatest interannual change for any given management area (Figure 3.11). In some years, the area ABC changed as much as ten times more than the total ABC under these scenarios.

The same analyses of interannual stability of apportionment scenarios were conducted for the simulation analyses (Figure 3.12) with similar outcomes as observed in the retrospective analysis (Figure 3.11). The fixed apportionment of ABC to management areas was the most stable (least variable). In contrast, Age-based and Terminal year survey apportionment were the least stable (most variable) for the BS and AI regions, which frequently showed a much larger percentage change in area-ABC than the total change in ABC. The unusual pattern of high variability in the BS for Age-based apportionment is due to consistent (across simulation iterations) high variability between 2020 and 2021 in the forward projections. The OM is configured to first 'sample' BS with the longline survey in 2021 and use that data to inform Age-based apportionment in the alternating year setup of the BS and AI surveys, leading to a large difference in ABC compared to the ABC used in the conditioning period. Table 4 details the mean absolute percent difference in ABC for each management area and total ABC. Apportioned ABC was less stable (more variable) on average for the BS and AI regions for all apportionment scenarios and was particularly variable for the Age-based and Terminal Year Survey apportionment scenarios. Fixed apportionment was the most stable scenario but also had the greatest deviations in proportional ABC to areas compared to estimated (EM) and 'true' OM spawning biomass.

The simulation analyses were used to determine how frequently ABC in a given area was below or above the 25% and 75% quartile thresholds of historic (years 2000-2020) ABC used for

Alaska sablefish management. Table 3.5 shows the ‘winners’ and ‘losers’ under each apportionment method and area, in which a lower proportion of simulations falling below the 25% quartile threshold or a higher proportion above the 75% quartile may be seen as a favorable situation for stakeholders in that area. Conversely, a higher proportion of simulation iterations below the 25% or a lower proportion above the 75% quartile may make that apportionment method less desirable. The Fixed apportionment method was the most similar among management areas for the lower threshold; simulated ABC fell below the 25% quartile 23-34% of the simulated years and iterations (Table 3.5). The Age-based apportionment was the most heterogeneous among areas for the proportion of ABC below the 25% quartile threshold; the CG region simulated ABC was below the lower threshold in 86% of years and iterations, whereas the WY region only fell below that in 3% of years and iterations. If a higher probability of high ABC is important, the apportionment method favored would depend on the management area in which a fisher held quota shares or fished most frequently, i.e., fishers in the BS and AI would likely favor Age-based apportionment, while fishers in CG might favor anything but Age-based, as the other alternatives were similar and more favorable for the simulation-based analyses.

3.5 Discussion

Stakeholder objectives and model outcomes

Discussions with stakeholders illuminated four primary and potentially conflicting stakeholder objectives for apportionment of ABC for Alaska sablefish management: 1) Achieve inter-annual stability in area-specific ABC; 2) Identify apportionment methods that do not deviate too far from observed sablefish abundance by area over time. This was seen as a proxy for sustainability and as a mechanism to avoid ABC overages in areas with high abundance but low proportion of ABC under Fixed apportionment; 3) Avoid the potential to harvest too many immature sablefish; and 4) Address concerns about the relative scarcity of large fish in catches in recent years. These four concerns can be summarized in terms of two primary issues: population sustainability and catch stability.

Results from these retrospective and simulation analyses suggest that no single apportionment strategy will address all of the concerns identified by stakeholders, as some desired outcomes conflict with each other, and concerns and objectives varied across stakeholder groups. Regardless, in the absence of a six-area spatial stock assessment model, which would remove the need for post hoc apportionment of ABC to management areas and estimate area-specific reference points, current management requires apportioning of ABC to management areas. The Terminal Year Survey, NPFMC, and Survey Average apportionment alternatives met sustainability objectives more closely by tying ABC to observed biomass among regions. In addition, the Terminal Year Survey method ignores measurement error and in practice may not follow actual changes in abundance. However, this came at a cost in terms of interannual stability in ABC relative to the Fixed apportionment scenario. Age-based ABC apportionment appears to satisfy neither of these objectives, but did have a higher proportion of simulation iterations and years when the simulated ABC was greater than the 75th quartile of historical ABCs in all regions but the CG.

The analysis of simulated ABC values with respect to thresholds relative to historic ABC is a simple, intuitive metric that allows direct comparison to past ABC values. Stakeholders can see how each apportionment type performs in simulations relative to their on-the-ground experience with harvest limits over the past 20 years. Much like the other performance metrics, the perception of which apportionment type is ‘best’ depends on which area or areas one fishes. For example, simulated ABC under the Age-based apportionment was very similar in every management area (Figure 3.6). Thus, for areas with historically lower ABC, the simulated Age-based ABC was much more likely to be higher than the 75th quartile of historic values because it acted as an equalizer among areas. Age-based apportionment was the most different of the apportionment types, as it was not based on the historical fishery or survey abundance indices. This may be why it performed so differently from the others, which share similarities in the data underlying apportionment.

All five apportionment scenarios performed similarly for the long-term median trajectory of SSB/B_{40} across spatial areas in the simulations. While there was high variability in realizations from the individual simulation iterations within each apportionment alternative,

the central tendency (mean and median across simulation iterations) was similar across apportionment strategies. This suggests that the Council's harvest control rule, which adjusts fishing mortality rates when the total spawning biomass on aggregate across all areas falls below threshold values (NPFMC 2018a, 2018b), may be more influential on the population dynamics than any of the apportionment alternatives explored in these simulation analyses. In addition, sablefish are assessed annually, and this, coupled with the harvest control rule used in the tier system, has been shown through simulations to be effective in minimizing the risk of overfishing (Wiedenmann et al. 2016). High movement rates of sablefish among management areas (Heifetz and Fujioka 1991; Hanselman et al. 2015) may also have contributed to this outcome, as high movement rates among management areas have been shown to result in a wider range of quota apportionment scenarios that still resulted in maximized yield (Bosley et al. 2019). Finally, the OM distribution of recruitment was based on the observed proportional abundance of age-2 fish in the longline survey, an explicit assumption with no coupled feedback loop between recruitment and spawner biomass. If one management area is more critical for spawners, or spawners in some areas contribute more to recruitment, apportionment scenarios may perform less similar to each other for SSB/B_{40} as some may be better suited to protect important spawners.

Spatial recruitment uncertainty

Failure to recognize and account for spatial population structure and biological processes can negatively impact the effectiveness of fishery management actions or lead to stock decline and reduced yields (Ricker 1958; Smedbol and Stephenson 2001; Kerr et al. 2014; Takashima and Baskett 2016). Mature sablefish are observed in all management areas (Goethel et al. 2020). However, we do not sufficiently understand whether the Alaska sablefish population depends on one or more productive spawning locations to sustain the population. Given the lack of understanding about the drivers of Alaska sablefish recruitment success and spawning habits, apportionment scenarios that distribute fishing mortality across spatial areas may reduce the potential to overharvest a critical portion of the population. This approach of conserving a diversity of fish biological traits, migratory behaviors, or habitats has been shown through simulations to contribute to population stability, resilience and sustainability (Hilborn

et al. 2003; Kerr et al. 2010; Schindler et al. 2010). Without better information on recruitment, we chose to distribute OM recruitment in proportion to the average observed abundance of age-2 sablefish in the longline survey. However, further research focused on sablefish spawning locations, drivers of recruitment success, the impact of harvesting large year classes from a single management area, and understanding whether spawning aggregations or natal homing exists would be beneficial. Such research could contribute to a greater understanding of potential conservation concerns associated with alternative harvest apportionment or management procedures.

Small, newly-recruited sablefish are frequently observed in some management areas before others, typically showing up in the BS and WG regions, before appearing in other regions. However, if recruitment is sufficiently high, age-1 and age-2 sablefish have been observed simultaneously in most management areas (Echave et al. 2013; Goethel et al. 2020). Because of the observed distribution of younger fish appearing first in western areas (BS, AI, WG) and older mature fish being more prevalent in eastern areas (CG, WY, EY), the distribution of apportioned ABC in periods of high or low recruitment can have an impact on different portions of the sablefish population-at-age. High catches in western areas (BS, AI, WG) may lead to higher mortality on younger fish when year-classes are above average. However, given the extreme magnitude of recent large year classes, it is unlikely that moderate increases in the catch of young fish will jeopardize fishery sustainability or sablefish stock resilience. Similarly, it is likely that the higher natural mortality associated with younger fish (based on basic life-history strategies and potential density-dependent mortality) is being shifted to fishing mortality. Conversely, purposely avoiding mortality on areas with younger fish may inadvertently lead to an increased mortality on larger, mature fish in other areas, where small fish may be less prevalent. Given the shift in the age structure of the current population towards younger ages and the reliance on a few older age-classes, the result could be a further reduction of the spawning stock, particularly if recent year classes do not materialize at the strength estimated by the assessment (e.g., due to retrospective bias or increased natural mortality).

Tradeoffs in model complexity and stakeholder involvement

Models can be useful tools to make assumptions transparent and to explore the sensitivity of outcomes to those assumptions. However, models can create barriers to effective communication between stakeholders and analysts because of the lack of a shared vocabulary, the technical nature of analyses, and other issues (Goethel et al. 2018, Runnebaum et al. 2018). A significantly larger amount of work was required to develop the simulation analyses than the retrospective analyses, yet the results and conclusions were similar. Communicating to stakeholders the simple methods and caveats to the retrospective analyses was much easier, despite many participating stakeholders being highly experienced with stock assessment and management strategy evaluations. Clear communication of technical information can be a hurdle to transparent fisheries management, so the development of more complicated models that are more challenging to convey to stakeholders should not be undertaken without full consideration of the potential drawbacks and rewards to stakeholder buy-in and involvement (Walters 1986, Peterman 2004). However, there were limits to the simpler retrospective modeling approach, which lacked a feedback loop in which the alternative management action could impact the population. Without a simulated population, we do not know the unobserved population status, so there is no ‘true’ population for comparison.

For these sablefish analyses, there is also the possibility that the two methods for examining apportionment resulted in similar conclusions because the simulation analyses were limited by the characterization of process and observation error and hampered by a relatively complex EM chosen to replicate the current assessment and management procedures. Further development of the simulation-estimation framework should focus on expanding utility to address environment changes and the effect of spawner-recruitment relationships or spawning migrations on population dynamics and be developed outside the constraint of an applied management question and a narrowly defined estimation model. In particular, understanding potential linkages between SSB and recruitment in specific management areas may allow for a more representative distribution of recruitment in the model and a better understanding of the impact of ABC apportionment on biological processes.

Recommendations for management

After examining the performance of the alternative apportionment scenarios, the Survey Average apportionment was chosen for Alaska sablefish management beginning in 2021 (Goethel et al. 2020). This scenario was chosen because of its relative stability and efficacy at tracking biomass across management areas, and because it does not rely on increasingly sparse fishery-dependent data like the previously used NPFMC apportionment method. There are limitless options and possibilities for apportionment strategies, but our results show that overall biological outcomes are similar under the fishing mortality recommended for the stock. What is not addressed in this paper are the possible social and economic advantages of possible apportionments. For example, the age-based apportionment could be tuned for improved stability by using multi-year age compositions or a different set of age-classes. This could result in harvesting more valuable large fish and avoiding low-value small, young fish. However, such a strategy would likely need to be coupled with changes to the ABC calculation, which could result in total ABCs that may be lower overall, but more valuable. Continued monitoring of the stability of Alaska sablefish ABC, model-based assessment of the population abundance, and communication with stakeholders could be developed into an intentional adaptive management process that benefits fishery managers and stakeholders.

3.6 References

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3.7 Figures

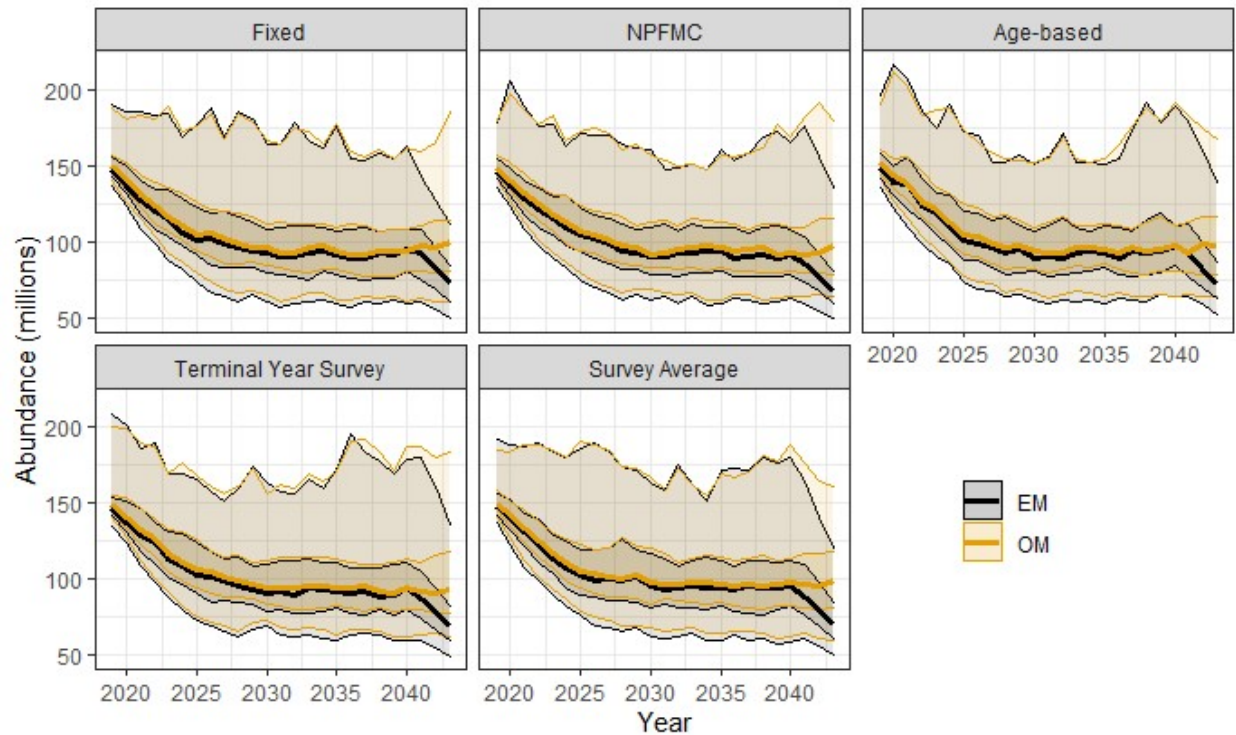


Figure 3.1. Simulation analyses abundance. Total abundance (in millions of age 2+ fish) of Alaska sablefish from the operating model (OM, black line and shading) aggregated over the six modeled areas and estimated in the estimation model (EM, orange line and shading) for each catch apportionment method and year of the simulation forward projecting period, 2019–2043. The thickest center line center represents the median abundance across simulation iterations, the inner shaded regions represent the 50th percentile intervals, and outer shaded regions represent the 95th percentile intervals for all converged results.

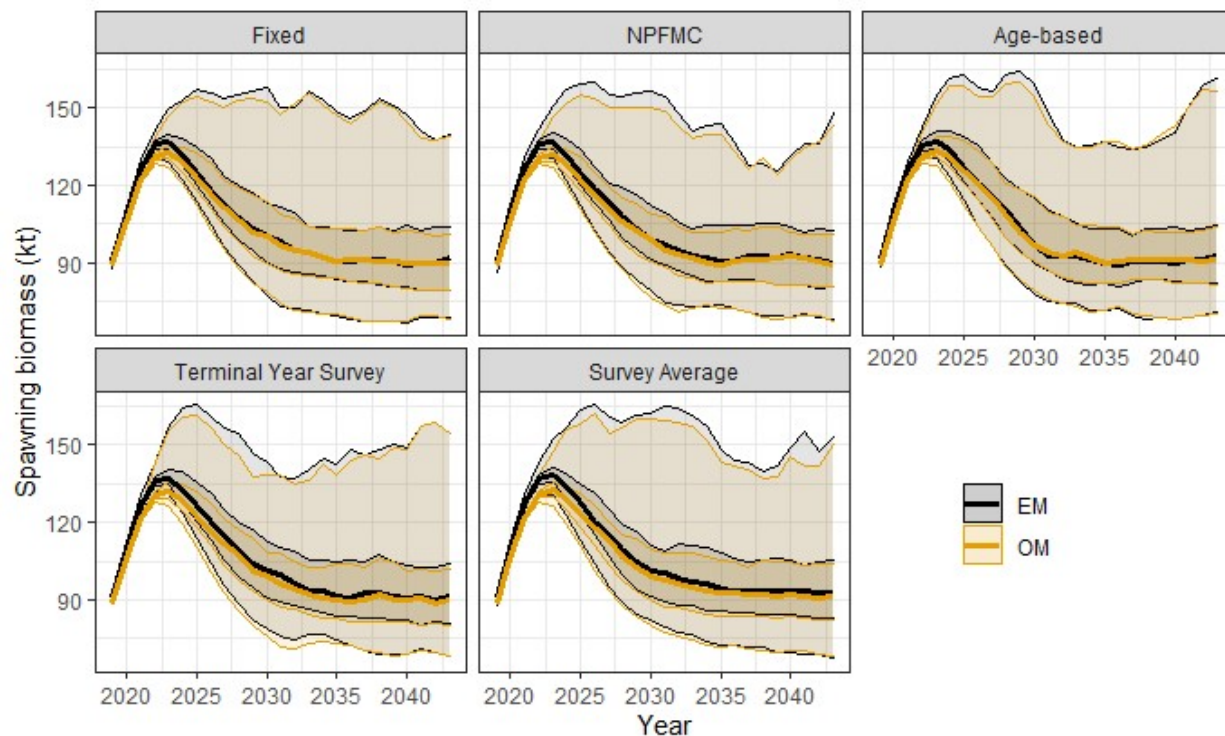


Figure 3.2. Simulation analyses spawning biomass. Spawning biomass (kt) of Alaska sablefish from the operating model (OM, black line and shading) aggregated over the six modeled areas and estimated in the estimation model (EM, orange line and shading) for each catch apportionment method and year of the simulation forward projecting period, 2019– 2043. The thickest center line represents the median spawning biomass across simulation iterations, the inner shaded regions represent the 50th percentile intervals, and outer shaded regions represent the 95th percentile intervals for all converged results.

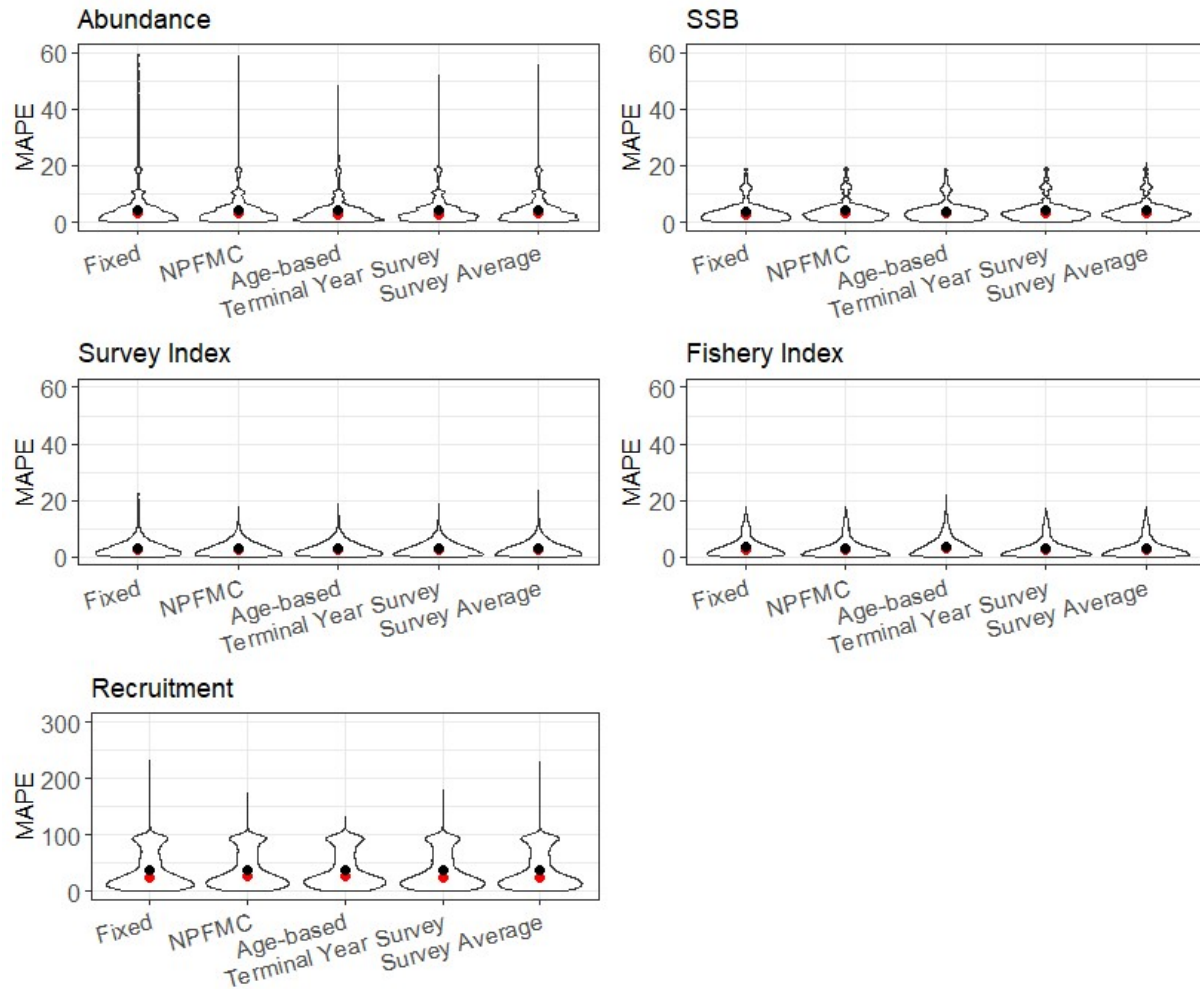


Figure 3.3. Simulation mean absolute percentage error (MAPE) for Alaska sablefish abundance, spawning biomass (SSB), survey and fishery indices of abundance, and recruitment. Violin plots show the MAPE between operating model (OM) and estimation model across forward projecting simulation years (2019–2043) and simulation iterations for each catch apportionment method. Black points represent the mean, red points the median. OM values were summed across the six modeled spatial areas. Note all y-axis are the same except for recruitment.

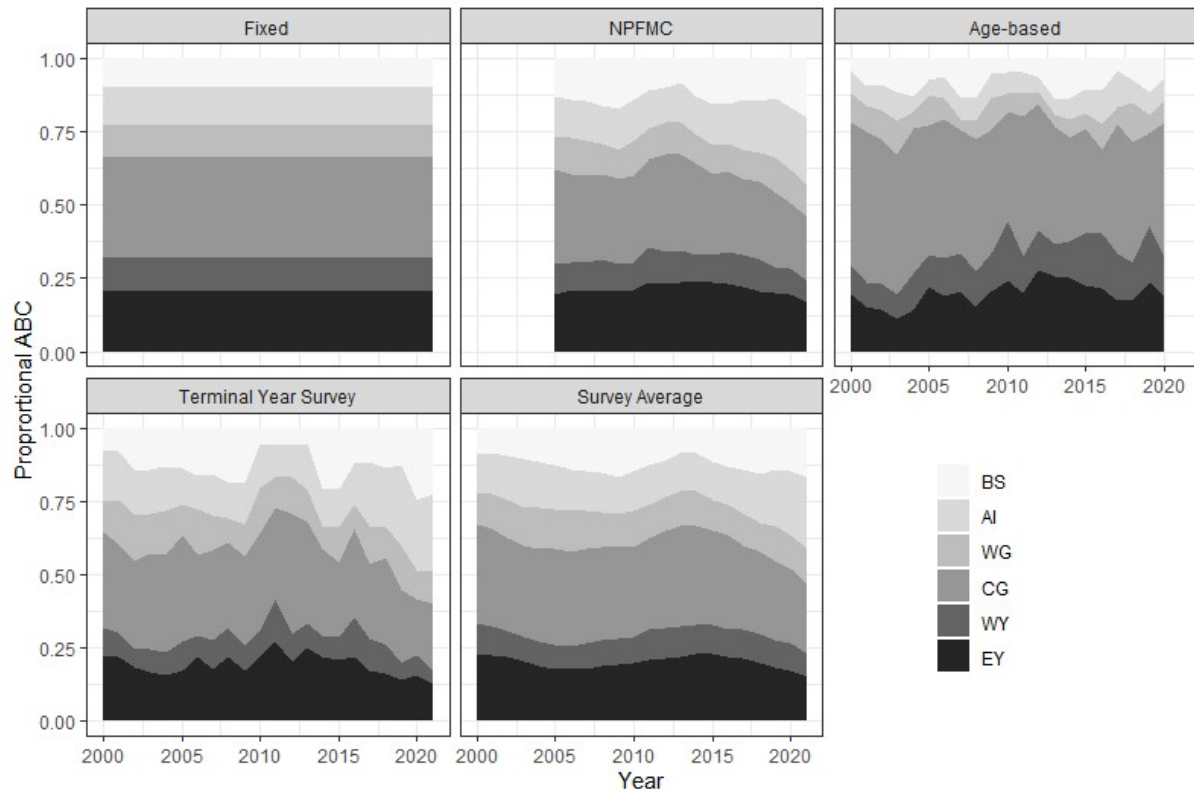


Figure 3.4. Retrospective allowable biological catch (ABC) apportioned to management areas. Each panel shows the proportions of apportioned ABC to each of the six management areas during 2000– 2021 for each retrospective apportionment method. Catch was apportioned among six management areas; Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat (WY), and East Yakutat/Southeast Outside (EY). Note that the time series of data available differ depending on the apportionment type and the data needed for that method. The NPFMC apportionment method uses fishery catch per unit effort data that have a shorter time series, so the time series begins in 2005.

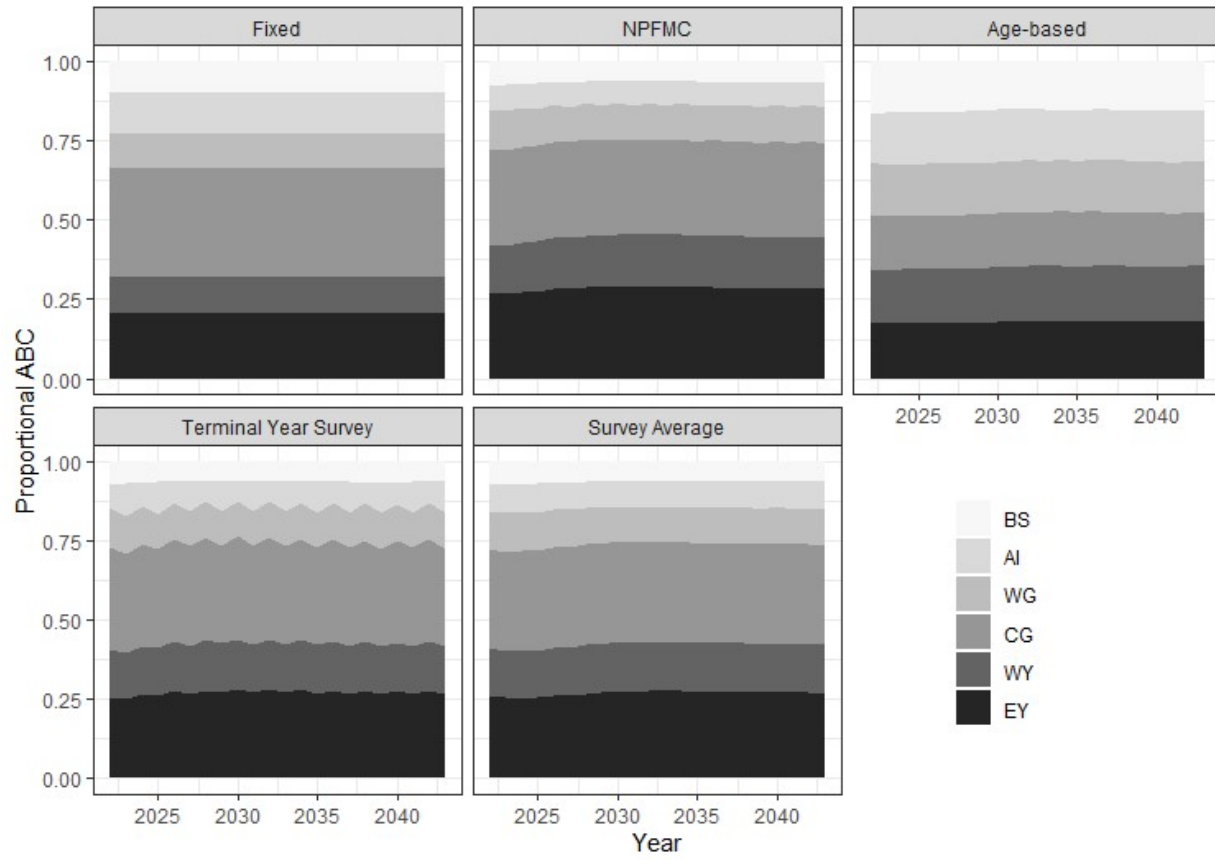


Figure 3.5. Simulation proportions of allowable biological catch (ABC) to management areas. Mean annual apportionment proportions to each management area across converged simulation iterations, for each apportionment scenario for forward projection period, 2019–2043. Six management areas were modeled in the OM; Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat (WY), and East Yakutat/Southeast Outside (EY).

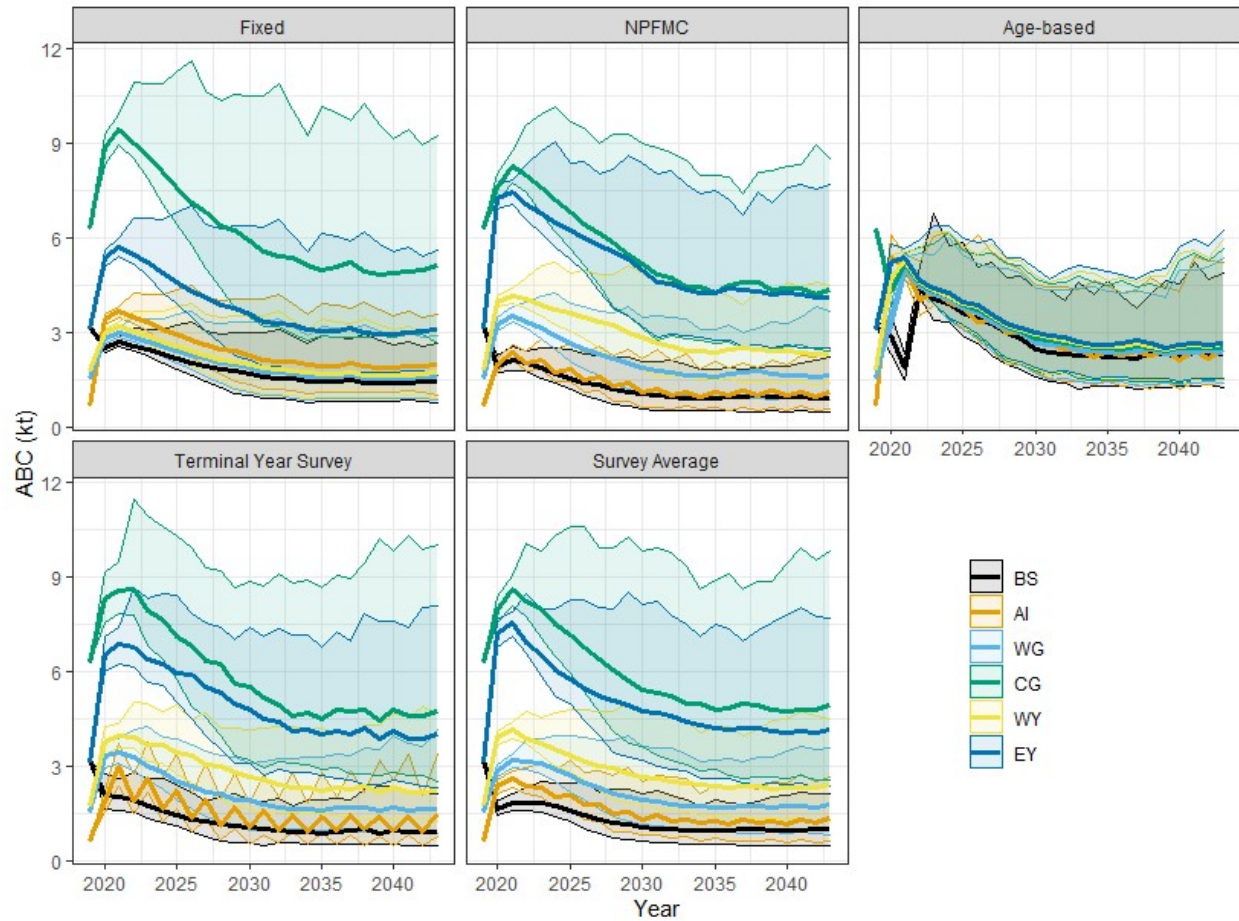


Figure 3.6. Simulation proportions allowable biological catch (ABC) to management areas. ABC apportioned to six management areas in the simulation analyses: Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat (WY), and East Yakutat/Southeast Outside (EY). The thickest line represents the median of ABC apportioned to management areas across converged simulation iterations for each year (2019–2043) for each apportionment method (panels). The outer shaded regions represent the 95th percentile intervals for all converged results.

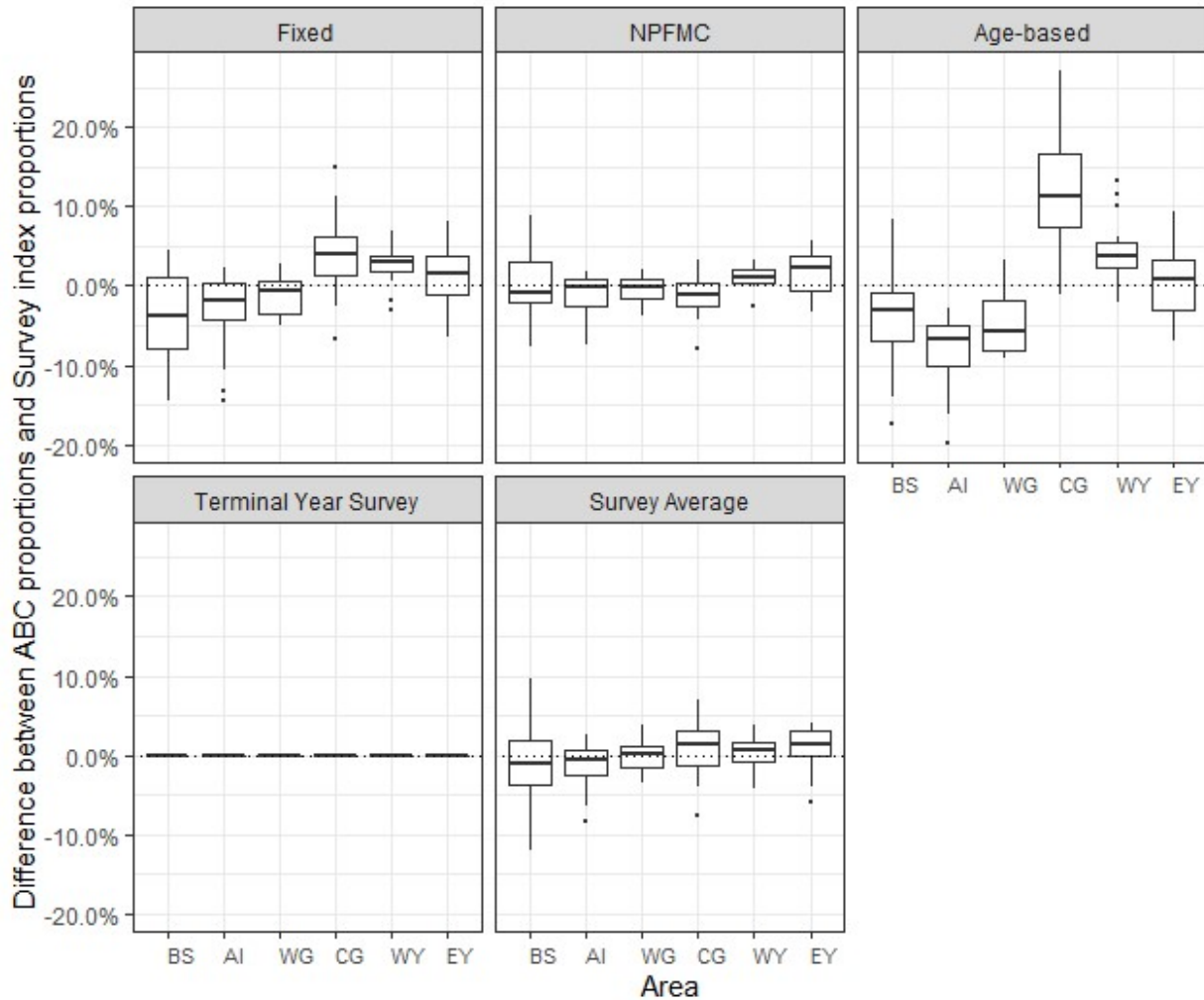


Figure 3.7. Retrospective biological sustainability. Difference between allowable biological catch (ABC) proportions to area and proportions of biomass estimated by the longline survey for each area and apportionment scenario for the retrospective analyses over years 2000–2019 for all apportionment types (panels) except NPFMC, which contains years 2005–2019. Six areas management areas receive apportioned catch: Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat (WY), and East Yakutat/Southeast Outside (EY). Boxplot center line represents the median percent difference between each management area proportion of ABC in year y and area-specific biomass proportions as estimated from the longline survey in year y . The upper and lower hinges of the boxes represent the 75th and 25th percentiles of the difference, and the upper and lower whiskers represent the values at 150% of the interquartile range of the upper and lower hinges,

respectively. Values close to zero indicate that the apportionment method more closely matches the survey index biomass in an area.

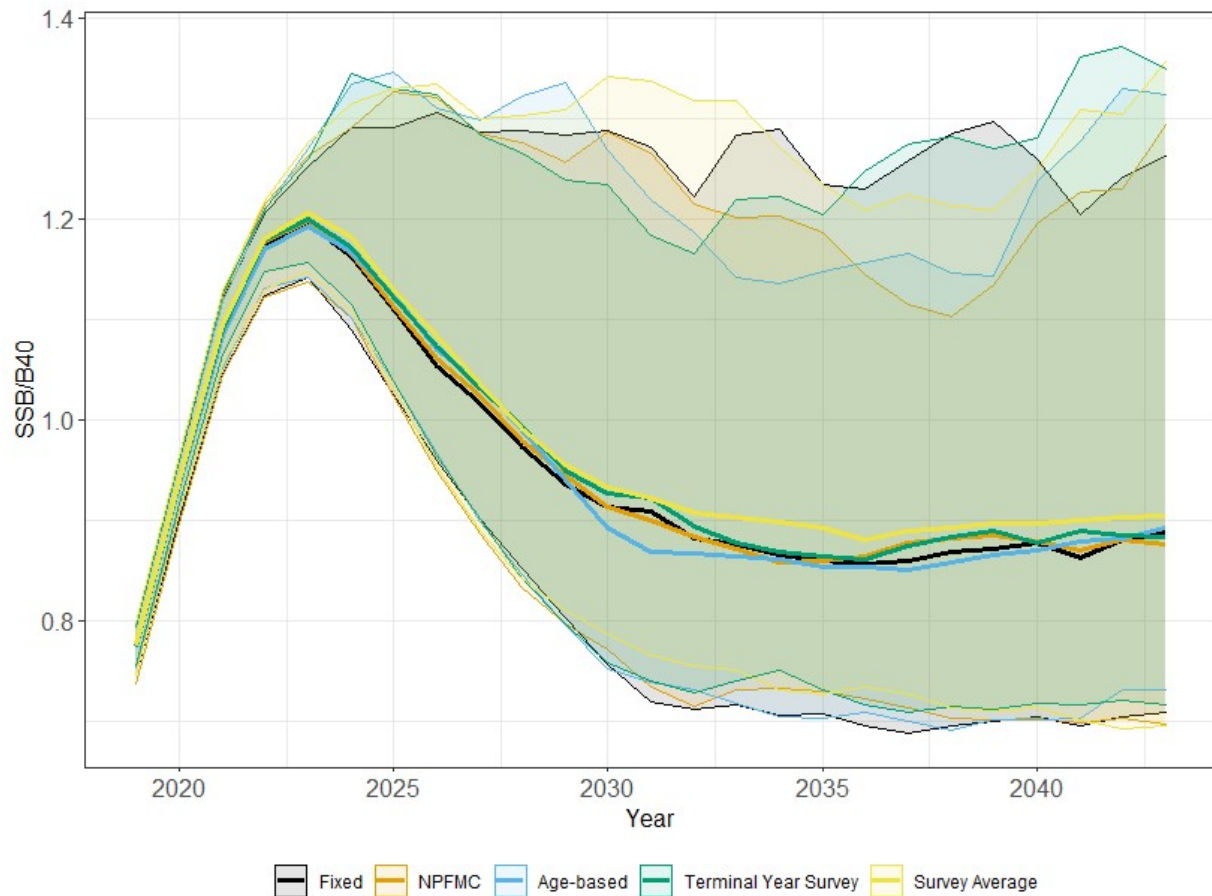


Figure 3.8. Simulation spawning stock biomass (SSB)/ B_{40} biological sustainability performance metric. Ratio of EM spawning biomass to the B_{40} reference point estimated in the EM for each apportionment method for the converged forward projection simulations, 2019–2043. The thickest line represents the median of SSB/ B_{40} across simulation iterations for each model across years. The outer shaded regions represent the 95th percentile intervals for all converged results.

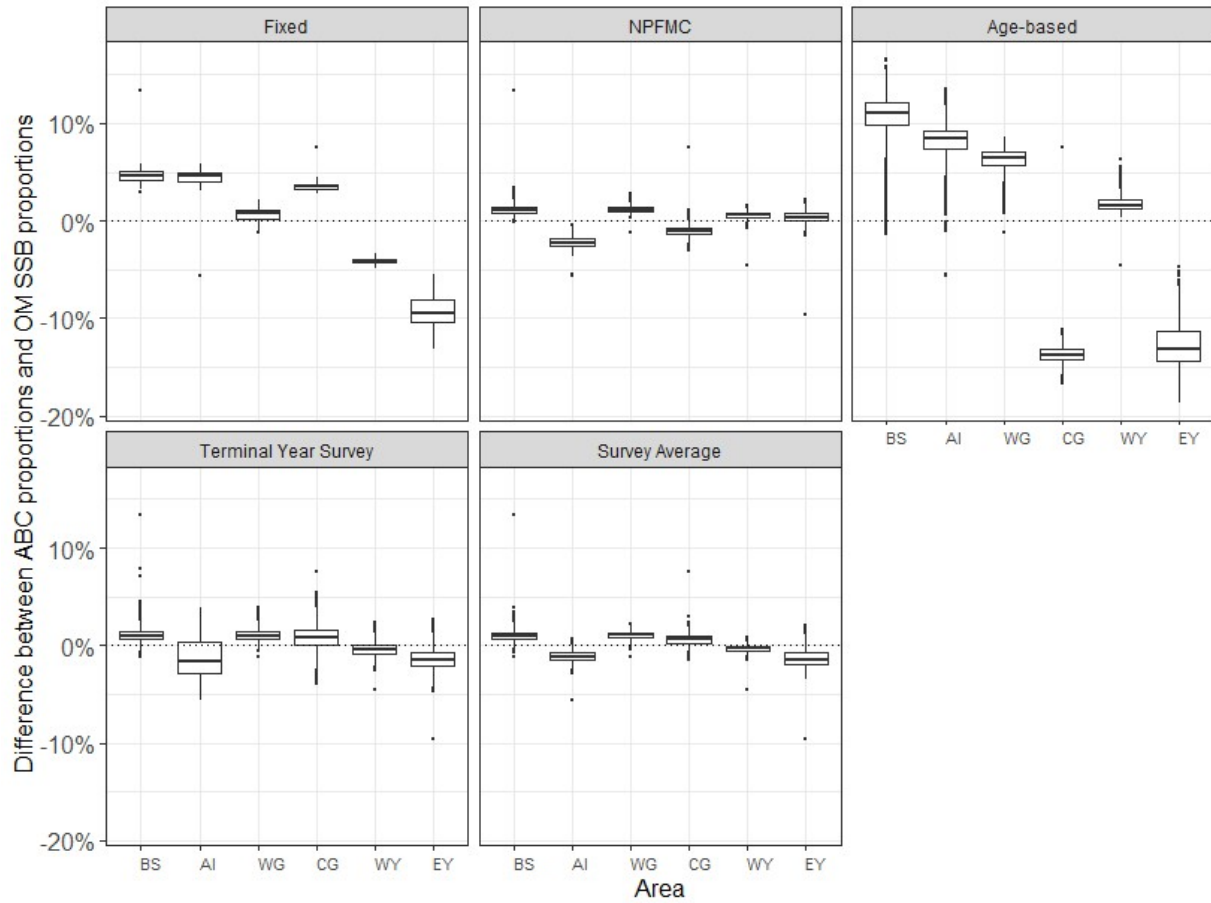


Figure 3.9. Simulation sustainability by area. Difference between simulated allowable biological catch (ABC) proportions to area and ‘true’ OM spawning biomass proportions for each area and apportionment scenario across years 2020–2043. Six areas management areas receive apportioned catch: Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat (WY), and East Yakutat/Southeast Outside (EY). Boxplot center line represents the median percent difference. The upper and lower hinges of the box represent the 75th and 25th percentiles of the difference, and the upper and lower whiskers represent the values at 150% of the interquartile range of the upper and lower hinges, respectively. Values close to zero indicate that the apportionment method more closely matches the underlying population spawning biomass by area.

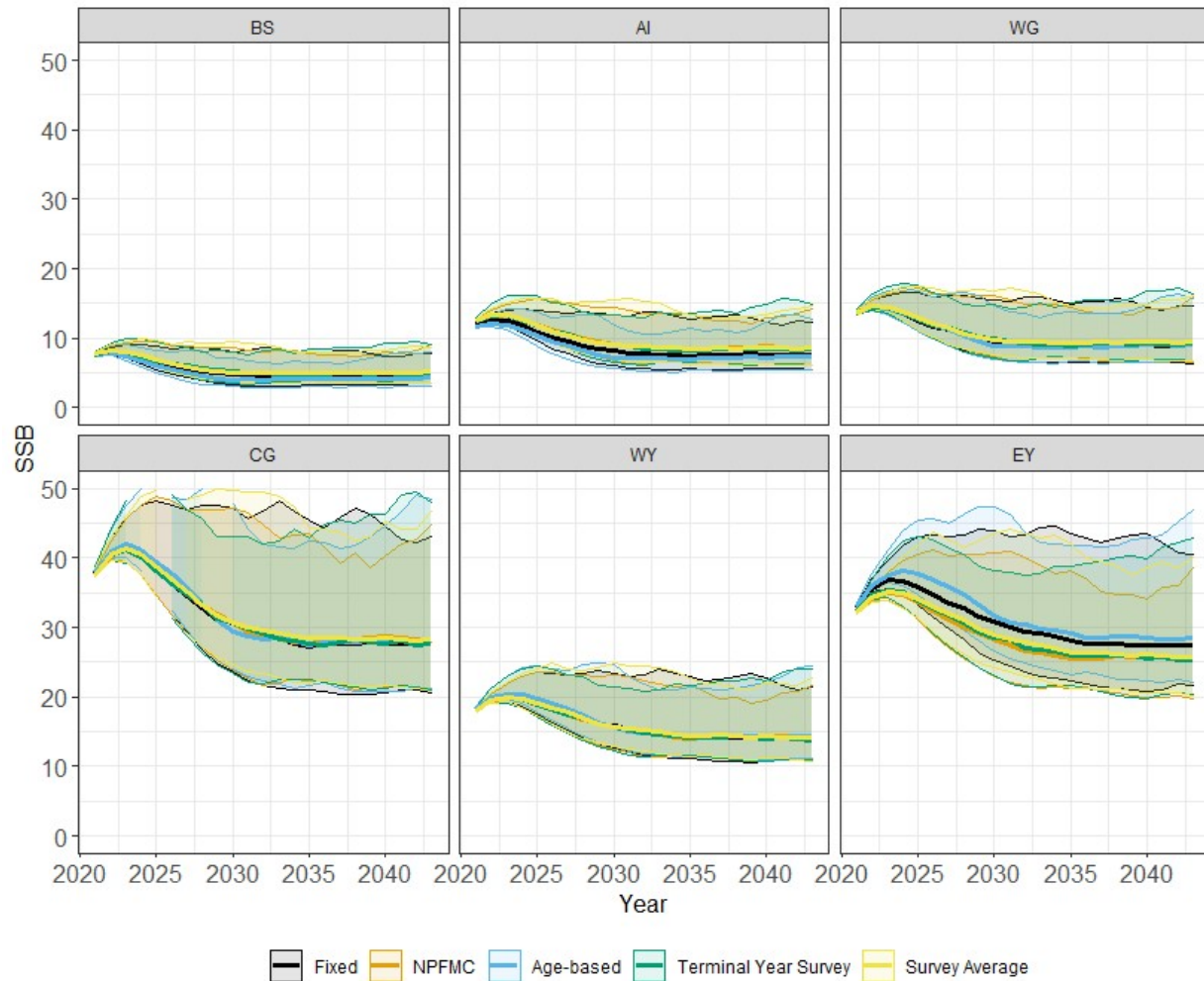


Figure 3.10. Simulated operating model spawning biomass. The thickest line represents the median spawning biomass (SSB) across simulations for each year (2020–2043). Panels represent each of the six areas management areas modeled: Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat (WY), and East Yakutat/Southeast Outside (EY). The outer shaded regions represent the 95th percentile intervals for all converged results.

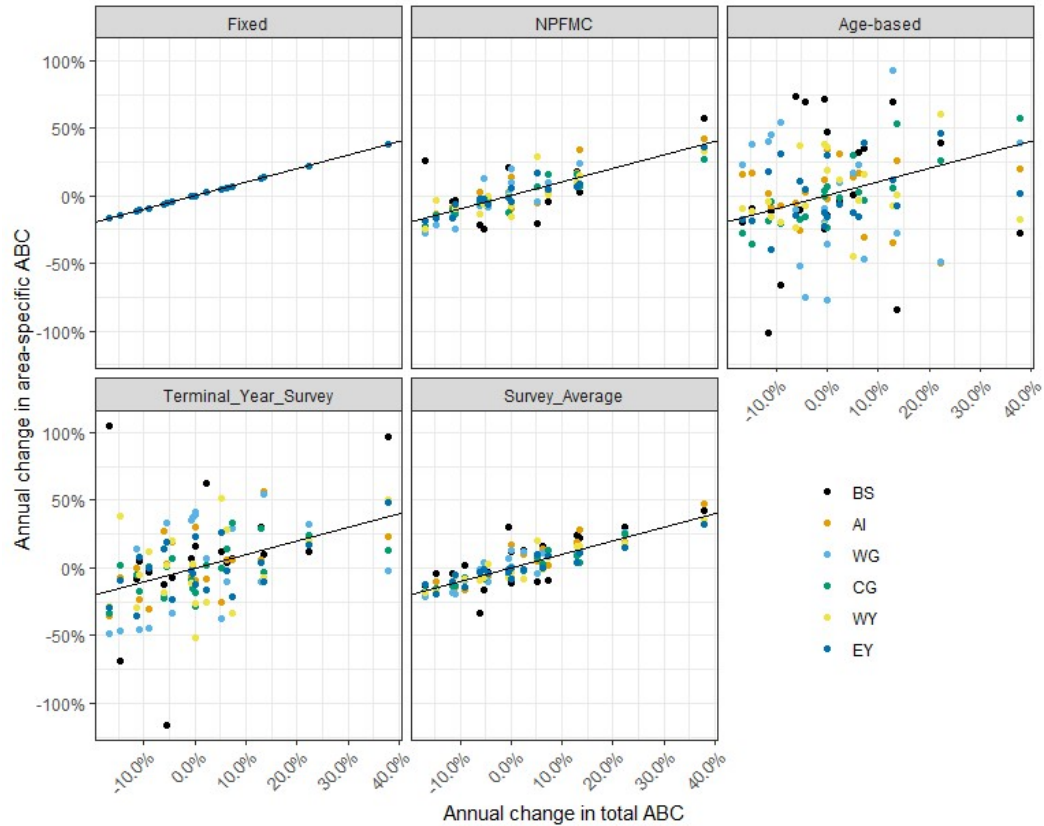


Figure 3.11. Retrospective stability by area, accounting for total allowable biological catch (ABC) stability. Retrospective analyses of the year-to-year area-specific ABC, $\delta Area_ABC_{b,y+1,m,i}$, compared to year-to-year change in total ABC, $\delta Total_ABC_{b,y,i}$. The x-axis is the percent difference in year $y+1$ to year y total ABC and the y-axis is the corresponding percent difference in year $y+1$ to year y ABC apportioned to an area. Each panel represents an apportionment scenario, and colored circles represent the six different management areas; Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat (WY), and East Yakutat/Southeast Outside (EY). The black line is the 1:1 line; colored circles falling above the line indicate that the area's percent change in apportioned ABC was greater than the overall percent change in ABC, circles below the line indicate that the area ABC change less than the overall ABC.

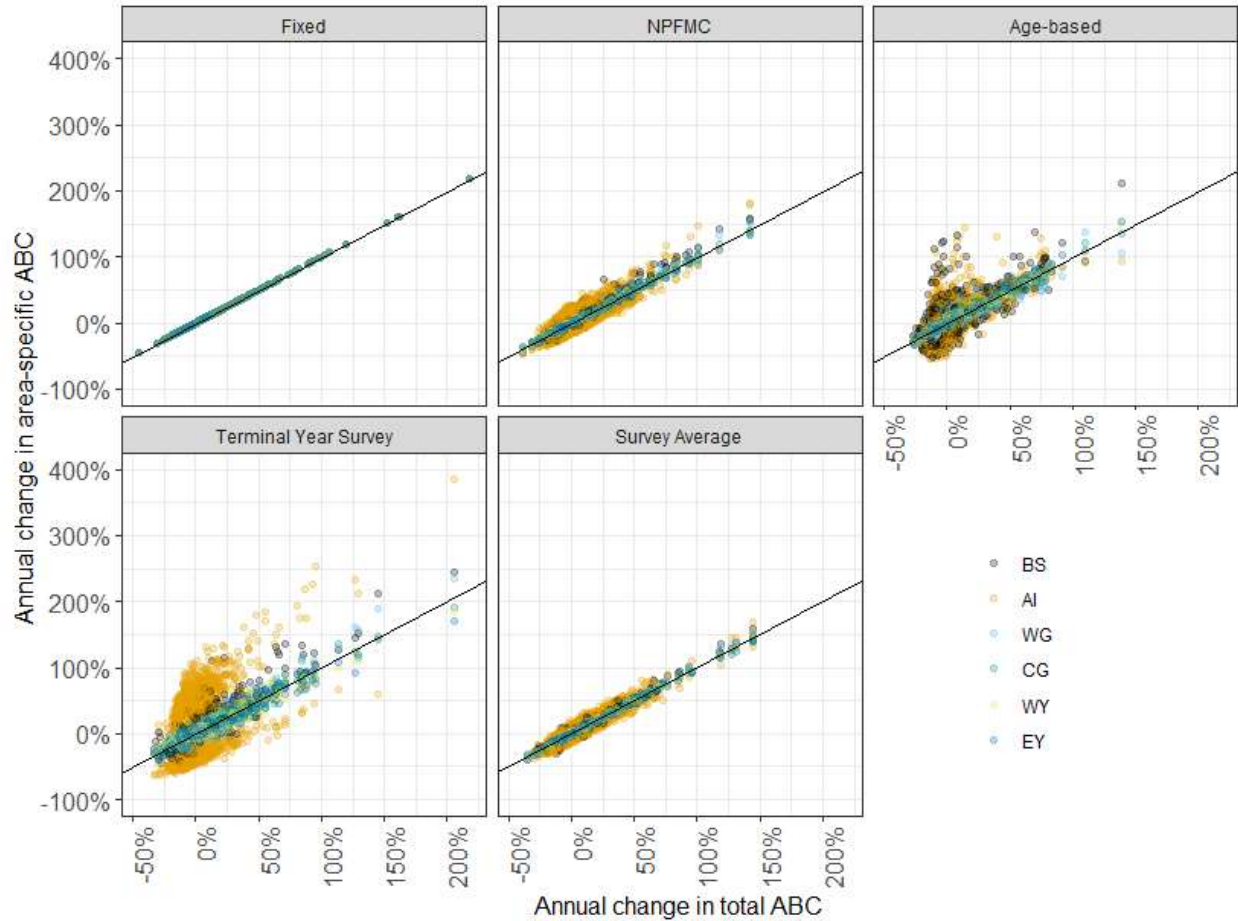


Figure 3.12. Simulation stability accounting for total allowable biological catch (ABC) stability. Simulation analyses of the year-to-year area-specific ABC, $\delta Area_ABC_{b,y+1,m,i}$, compared to year-to-year change in total ABC, $\delta Total_ABC_{b,y,i}$. The x-axis is the percent difference in year $y+1$ to year y total ABC and the y-axis is the corresponding percent difference in year $y+1$ to year y ABC apportioned to an area. Each panel represents an apportionment scenario, and colored circles represent the six different management areas; Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat (WY), and East Yakutat/Southeast Outside (EY). The black line is the 1:1 line; colored circles falling above the line indicate that the area's percent change in apportioned ABC was greater than the overall percent change in ABC, circles below the line indicate that the area ABC change less than the overall ABC.

3.8 Tables

Table 3.1. Sablefish movement rates for simulation analyses operating model (OM). Movement rates among each of the six modeled regions in the OM were age-specific for operating model ages 1–15, and then the same for ages 16 and greater. The six areas were the Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat (WY), and East Yakutat/Southeast Outside (EY). A subset of movement matrices is provided below and a full set is available in the appendix. Row labels indicate region from which fish moved and columns indicate region to which fish moved in each annual time step. Note that values in the table may not sum to 1 across rows due to rounding error.

OM Age 1	BS	AI	WG	CG	WY	EY
BS	0.663	0.025	0.244	0.050	0.016	0.002
AI	0.056	0.784	0.140	0.015	0.005	0.000
WG	0.137	0.135	0.307	0.295	0.101	0.026
CG	0.021	0.017	0.155	0.502	0.193	0.111
WY	0.020	0.017	0.147	0.504	0.192	0.120
EY	0.002	0.002	0.024	0.159	0.069	0.744
OM Age 2	BS	AI	WG	CG	WY	EY
BS	0.644	0.029	0.233	0.068	0.023	0.003
AI	0.053	0.767	0.138	0.031	0.011	0.001
WG	0.130	0.131	0.292	0.307	0.111	0.030
CG	0.022	0.019	0.152	0.499	0.196	0.110
WY	0.020	0.018	0.144	0.498	0.195	0.126
EY	0.002	0.002	0.024	0.153	0.071	0.750
OM Age 5	BS	AI	WG	CG	WY	EY
BS	0.587	0.040	0.203	0.120	0.043	0.007
AI	0.044	0.718	0.129	0.077	0.027	0.005
WG	0.109	0.118	0.247	0.344	0.139	0.043
CG	0.025	0.027	0.144	0.490	0.206	0.108
WY	0.020	0.021	0.134	0.479	0.203	0.142
EY	0.002	0.002	0.022	0.134	0.075	0.765

Table 3.1, cont.

OM Age

10	BS	AI	WG	CG	WY	EY
BS	0.492	0.058	0.152	0.207	0.076	0.014
AI	0.029	0.636	0.115	0.155	0.055	0.010
WG	0.074	0.098	0.171	0.405	0.187	0.065
CG	0.030	0.039	0.131	0.474	0.223	0.103
WY	0.021	0.027	0.119	0.446	0.217	0.170
EY	0.002	0.002	0.020	0.103	0.083	0.790

OM Age

15	BS	AI	WG	CG	WY	EY
BS	0.397	0.077	0.102	0.295	0.109	0.021
AI	0.015	0.554	0.101	0.232	0.083	0.015
WG	0.039	0.077	0.096	0.466	0.235	0.087
CG	0.035	0.051	0.117	0.459	0.239	0.099
WY	0.022	0.033	0.103	0.414	0.231	0.197
EY	0.002	0.003	0.017	0.072	0.090	0.816

Table 3.2. Summary of catch apportionment types. List of each apportionment type, the plain-language description of the method or data for calculation, and reason for inclusion in the analyses.

Apportionment type	Characteristics
Fixed	Proportions of allowable biological catch (ABC) to each area were fixed at the 2013 proportions used in management of Alaska sablefish. Inherently stable, but doesn't change if sablefish biomass shifts spatially.
NPFMC	Combined fishery and survey index data, with survey index given twice the weight of the fishery index. The most recent five years of data were used for ABC calculations, with values exponentially weighted to recent years have more weight. Exponential weighting was less stabilizing than just average across years, but tracked spatial changes in index values relatively more closely.
Survey average	Used survey index data only; the most recent five years of data are used for ABC calculations, with all years weighted equally. Fishery index data in some years was sparse or had high variability, so this method only used survey data.
Terminal year survey	Used survey index data only, and only the most recent year of data. Tracked spatial changes in biomass closely, but high variability from year to year for any given area.
Age-based	Used survey biomass of age-6 and older fish in each region as a proportion of the total (area-aggregated) mature biomass. Age six is the approximate age of 50% maturity. Intended to spread fishing mortality across regions based on maturity and avoid immature fish.

Table 3.3. Retrospective total allowable biological catch (ABC) and interannual stability. Total acceptable biological catch (t, summed across management areas) and percent difference in year $y+1$ to year y total ABC, $\delta Total_ABC_{b,y,i}$, for the retrospective analyses.

Year	Total ABC	Difference
2000	17,000	-
2001	16,900	-0.6%
2002	17,300	2.3%
2003	18,400	6.2%
2004	23,000	22.2%
2005	21,000	-9.1%
2006	21,000	0.0%
2007	20,100	-4.4%
2008	18,030	-10.9%
2009	16,080	-11.4%
2010	15,230	-5.4%
2011	16,040	5.2%
2012	17,250	7.3%
2013	16,230	-6.1%
2014	13,722	-16.7%
2015	13,657	-0.5%
2016	11,795	-14.6%
2017	13,509	13.5%
2018	15,380	13.0%
2019	15,380	0.0%
2020	22,551	37.8%

Table 3.4. Simulation mean absolute stability for apportionment scenarios. Mean absolute percent difference in allowable biological catch (ABC), between year $y+1$ and year y , by area and for total ABC (summed over areas), across years 2021–2043 and converged simulation iterations. The six areas were the Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat (WY), and East Yakutat/Southeast Outside (EY).

	BS	AI	WG	CG	WY	EY
Fixed	9.8%	9.8%	9.8%	9.8%	9.8%	9.8%
NPFMC	11.2%	14.6%	10.3%	9.4%	9.0%	9.0%
Age-based	20.8%	16.0%	11.8%	11.0%	10.9%	11.1%
Terminal Year Survey	13.8%	42.2%	11.5%	10.1%	10.2%	9.5%
Survey Average	10.4%	12.1%	9.9%	9.5%	9.4%	9.4%

Table 3.5. Proportion of simulated years and iterations above and below allowable biological catch (ABC) thresholds. The proportion of converged simulation years and iterations which were above the 75% quartile and below the 25% quartile for each apportionment method and management area. Threshold quartiles were based on the area-specific ABC values used for Alaska sablefish management during years 2000–2020. The six management areas were the Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat (WY), and East Yakutat/Southeast Outside (EY).

	BS	AI	WG	CG	WY	EY
Threshold (t)	2,790	2,620	2,260	5,921	2,080	3,540
Proportion above 75% ABC quartile						
Fixed	0.09	0.44	0.38	0.68	0.52	0.56
NPFMC	0.04	0.01	0.45	0.59	0.81	0.84
Age-based	0.51	0.60	0.72	0.07	0.84	0.44
Term Year Surv	0.05	0.09	0.43	0.63	0.75	0.79
Surv Avg	0.04	0.06	0.47	0.65	0.79	0.82

	BS	AI	WG	CG	WY	EY
Threshold (t)	1,464	1,988	1,581	4,740	1,640	2,890
Proportion below 25% ABC quartile						
Fixed	0.29	0.34	0.32	0.23	0.25	0.24
NPFMC	0.61	0.82	0.29	0.33	0.04	0.03
Age-based	0.04	0.19	0.07	0.86	0.03	0.35
Term Year Surv	0.63	0.75	0.29	0.29	0.06	0.05
Surv Avg	0.60	0.70	0.26	0.26	0.05	0.04

3.9 Appendix 3. Additional apportionment tables

Table A.1. The total allowable biological catch (ABC, t) values used in the management of Alaska sablefish for 2000–2020, apportioned using the five alternative apportionment scenarios. The six management areas were the Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat (WY), and East Yakutat/Southeast Outside (EY).

Age-based

Year	BS	AI	WG	CG	WY	EY	Total ABC
2000	730	1251	1695	8322	1715	3287	17000
2001	1542	1137	1543	8680	1381	2617	16900
2002	1482	1557	1698	8563	1549	2453	17300
2003	2046	1850	2129	8765	1505	2105	18400
2004	3031	1107	1294	11420	2797	3352	23000
2005	1523	1033	2262	9286	2295	4601	21000
2006	1316	1456	1573	9925	2786	3945	21000
2007	2710	1493	714	8471	2588	4124	20100
2008	2423	1368	1135	8137	2221	2746	18030
2009	793	1387	1713	6782	2136	3270	16080
2010	715	1067	1008	5673	3115	3652	15230
2011	720	1227	1196	7702	1983	3212	16040
2012	1028	902	739	7475	2321	4785	17250
2013	2229	854	664	6498	1828	4157	16230
2014	1825	1001	838	4914	1657	3489	13722
2015	1433	1126	690	4902	2435	3071	13657
2016	1299	1337	1010	3412	2172	2564	11795
2017	529	1741	766	5909	2185	2379	13509
2018	1088	1230	2083	6267	2036	2676	15380
2019	1759	1207	921	4932	2945	3617	15380
2020	NA	NA	NA	NA	NA	NA	

Fixed

Year	BS	AI	WG	CG	WY	EY	Total ABC
2000	1659	2243	1834	5799	1951	3514	17000
2001	1649	2230	1823	5765	1939	3493	16900
2002	1688	2283	1866	5901	1985	3576	17300
2003	1795	2428	1985	6277	2111	3803	18400
2004	2244	3035	2481	7846	2639	4754	23000
2005	2049	2771	2265	7163	2410	4341	21000
2006	2049	2771	2265	7163	2410	4341	21000
2007	1961	2652	2168	6856	2306	4155	20100

App. Table A.1, cont.

2008	1759	2379	1945	6150	2069	3727	18030
2009	1569	2122	1735	5485	1845	3324	16080
2010	1486	2010	1643	5195	1748	3148	15230
2011	1565	2117	1730	5472	1841	3316	16040
2012	1683	2276	1861	5884	1979	3566	17250
2013	1584	2142	1751	5536	1862	3355	16230
2014	1339	1811	1480	4681	1575	2836	13722
2015	1333	1802	1473	4659	1567	2823	13657
2016	1151	1557	1272	4023	1353	2438	11795
2017	1318	1783	1457	4608	1550	2792	13509
2018	1501	2030	1659	5246	1765	3179	15380
2019	1501	2030	1659	5246	1765	3179	15380
2020	2201	2976	2433	7693	2588	4662	22551

NPFMC

Year	BS	AI	WG	CG	WY	EY	Total ABC
2000	NA	NA	NA	NA	NA	NA	0
2001	NA	NA	NA	NA	NA	NA	0
2002	NA	NA	NA	NA	NA	NA	0
2003	NA	NA	NA	NA	NA	NA	0
2004	NA	NA	NA	NA	NA	NA	0
2005	2678	2958	2324	6797	2128	4115	21000
2006	2959	2705	2573	6396	1986	4381	21000
2007	2954	2673	2360	6044	1976	4094	20100
2008	2866	2399	1850	5279	1799	3839	18030
2009	2750	2216	1587	4734	1524	3269	16080
2010	2144	2156	1802	4494	1435	3198	15230
2011	1747	2050	1720	4810	1917	3797	16040
2012	1672	2116	1890	5664	1939	3968	17250
2013	1346	2172	1761	5363	1702	3886	16230
2014	1751	1809	1330	4263	1327	3242	13722
2015	2152	1858	1358	3765	1321	3203	13657
2016	1823	1626	1090	3252	1281	2723	11795
2017	1873	2302	1388	3484	1493	2969	13509
2018	2230	2714	1499	4102	1648	3187	15380
2019	2086	3125	1833	3862	1410	3064	15380
2020	3745	4787	2555	5068	1965	4431	22551

App. Table A.1, cont.

Survey Average

Year	BS	AI	WG	CG	WY	EY	Total ABC
2000	1503	2233	1809	5730	1808	3915	17000
2001	1433	2390	1945	5649	1679	3804	16900
2002	1626	2641	2184	5530	1554	3766	17300
2003	1914	3043	2440	5704	1542	3758	18400
2004	2607	3681	3185	7322	1843	4360	23000
2005	2651	3136	2900	6843	1694	3774	21000
2006	3001	2908	2930	6732	1651	3779	21000
2007	2917	2748	2640	6480	1703	3611	20100
2008	2765	2371	2185	5686	1601	3422	18030
2009	2662	2067	1818	4997	1445	3092	16080
2010	2265	2044	1885	4621	1348	3066	15230
2011	2051	2149	1808	4989	1649	3394	16040
2012	1862	2201	1977	5703	1780	3728	17250
2013	1329	2166	1953	5550	1625	3607	16230
2014	1162	1799	1578	4649	1352	3182	13722
2015	1583	1731	1462	4422	1312	3147	13657
2016	1513	1563	1222	3788	1106	2603	11795
2017	1888	2087	1406	3922	1298	2908	13509
2018	2404	2529	1574	4312	1521	3040	15380
2019	2157	2989	1795	4169	1469	2801	15380
2020	3318	4874	2561	5829	2102	3867	22551

Terminal Year Survey

Year	BS	AI	WG	CG	WY	EY	Total ABC
2000	1236	3014	1780	5593	1601	3777	17000
2001	1320	2821	2531	5120	1426	3682	16900
2002	2508	2612	2727	5203	1109	3141	17300
2003	2617	2775	2452	6000	1472	3085	18400
2004	2964	3547	3383	7659	1797	3649	23000
2005	2879	2599	2160	7698	2033	3631	21000
2006	3387	2366	3292	5804	1554	4596	21000
2007	3139	2855	2349	6231	1904	3622	20100
2008	3308	2251	1482	5245	1808	3936	18030
2009	3051	2240	1712	4982	1342	2753	16080
2010	808	2318	2384	5025	1374	3322	15230
2011	912	1791	1629	5042	2331	4335	16040
2012	975	1896	2177	7037	1665	3499	17250

App. Table A.1, cont.

2013	865	2504	1801	5643	1380	4037	16230
2014	2793	1750	1095	4038	1036	3010	13722
2015	2858	1781	1593	3478	1066	2881	13657
2016	1399	1657	996	3552	1576	2615	11795
2017	1541	2970	1735	3427	1463	2373	13509
2018	2090	3138	1574	4583	1525	2470	15380
2019	1898	4262	2334	3812	897	2176	15380
2020	5492	5393	2278	4325	1505	3559	22551

Table A.2. The proportion of allowable biological catch (ABC) to each management area and apportionment scenario for the retrospective analyses. The tables below show the proportion of total ABC apportioned to each management area for each scenario in the retrospective analyses. Summed row values may not sum to one due to rounding error. Data availability for calculating apportionment differs among apportionment scenarios, 'NA' denotes apportioned ABC not available due to data limitations. The six management areas were the Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat (WY), and East Yakutat/Southeast Outside (EY).

Age-based

Year	BS	AI	WG	CG	WY	EY
2000	0.043	0.074	0.100	0.490	0.101	0.193
2001	0.091	0.067	0.091	0.514	0.082	0.155
2002	0.086	0.090	0.098	0.495	0.090	0.142
2003	0.111	0.101	0.116	0.476	0.082	0.114
2004	0.132	0.048	0.056	0.497	0.122	0.146
2005	0.073	0.049	0.108	0.442	0.109	0.219
2006	0.063	0.069	0.075	0.473	0.133	0.188
2007	0.135	0.074	0.036	0.421	0.129	0.205
2008	0.134	0.076	0.063	0.451	0.123	0.152
2009	0.049	0.086	0.107	0.422	0.133	0.203
2010	0.047	0.070	0.066	0.372	0.205	0.240
2011	0.045	0.077	0.075	0.480	0.124	0.200
2012	0.060	0.052	0.043	0.433	0.135	0.277
2013	0.137	0.053	0.041	0.400	0.113	0.256
2014	0.133	0.073	0.061	0.358	0.121	0.254
2015	0.105	0.082	0.051	0.359	0.178	0.225
2016	0.110	0.113	0.086	0.289	0.184	0.217
2017	0.039	0.129	0.057	0.437	0.162	0.176
2018	0.071	0.080	0.135	0.407	0.132	0.174
2019	0.114	0.078	0.060	0.321	0.191	0.235
2020	NA	NA	NA	NA	NA	NA

Fixed

Year	BS	AI	WG	CG	WY	EY
2000	0.098	0.132	0.108	0.341	0.115	0.207
2001	0.098	0.132	0.108	0.341	0.115	0.207
2002	0.098	0.132	0.108	0.341	0.115	0.207
2003	0.098	0.132	0.108	0.341	0.115	0.207

App. Table A.2, cont.

2004	0.098	0.132	0.108	0.341	0.115	0.207
2005	0.098	0.132	0.108	0.341	0.115	0.207
2006	0.098	0.132	0.108	0.341	0.115	0.207
2007	0.098	0.132	0.108	0.341	0.115	0.207
2008	0.098	0.132	0.108	0.341	0.115	0.207
2009	0.098	0.132	0.108	0.341	0.115	0.207
2010	0.098	0.132	0.108	0.341	0.115	0.207
2011	0.098	0.132	0.108	0.341	0.115	0.207
2012	0.098	0.132	0.108	0.341	0.115	0.207
2013	0.098	0.132	0.108	0.341	0.115	0.207
2014	0.098	0.132	0.108	0.341	0.115	0.207
2015	0.098	0.132	0.108	0.341	0.115	0.207
2016	0.098	0.132	0.108	0.341	0.115	0.207
2017	0.098	0.132	0.108	0.341	0.115	0.207
2018	0.098	0.132	0.108	0.341	0.115	0.207
2019	0.098	0.132	0.108	0.341	0.115	0.207
2020	0.098	0.132	0.108	0.341	0.115	0.207

NPFMC

Year	BS	AI	WG	CG	WY	EY
2000	NA	NA	NA	NA	NA	NA
2001	NA	NA	NA	NA	NA	NA
2002	NA	NA	NA	NA	NA	NA
2003	NA	NA	NA	NA	NA	NA
2004	NA	NA	NA	NA	NA	NA
2005	0.128	0.141	0.111	0.324	0.101	0.196
2006	0.141	0.129	0.123	0.305	0.095	0.209
2007	0.147	0.133	0.117	0.301	0.098	0.204
2008	0.159	0.133	0.103	0.293	0.100	0.213
2009	0.171	0.138	0.099	0.294	0.095	0.203
2010	0.141	0.142	0.118	0.295	0.094	0.210
2011	0.109	0.128	0.107	0.300	0.119	0.237
2012	0.097	0.123	0.110	0.328	0.112	0.230
2013	0.083	0.134	0.109	0.330	0.105	0.239
2014	0.128	0.132	0.097	0.311	0.097	0.236
2015	0.158	0.136	0.099	0.276	0.097	0.235
2016	0.155	0.138	0.092	0.276	0.109	0.231
2017	0.139	0.170	0.103	0.258	0.111	0.220
2018	0.145	0.176	0.097	0.267	0.107	0.207
2019	0.136	0.203	0.119	0.251	0.092	0.199
2020	0.166	0.212	0.113	0.225	0.087	0.196

App. Table A.2, cont.

Survey Average

Year	BS	AI	WG	CG	WY	EY
2000	0.088	0.131	0.106	0.337	0.106	0.230
2001	0.085	0.141	0.115	0.334	0.099	0.225
2002	0.094	0.153	0.126	0.320	0.090	0.218
2003	0.104	0.165	0.133	0.310	0.084	0.204
2004	0.113	0.160	0.138	0.318	0.080	0.190
2005	0.126	0.149	0.138	0.326	0.081	0.180
2006	0.143	0.138	0.140	0.321	0.079	0.180
2007	0.145	0.137	0.131	0.322	0.085	0.180
2008	0.153	0.132	0.121	0.315	0.089	0.190
2009	0.166	0.129	0.113	0.311	0.090	0.192
2010	0.149	0.134	0.124	0.303	0.089	0.201
2011	0.128	0.134	0.113	0.311	0.103	0.212
2012	0.108	0.128	0.115	0.331	0.103	0.216
2013	0.082	0.133	0.120	0.342	0.100	0.222
2014	0.085	0.131	0.115	0.339	0.099	0.232
2015	0.116	0.127	0.107	0.324	0.096	0.230
2016	0.128	0.133	0.104	0.321	0.094	0.221
2017	0.140	0.155	0.104	0.290	0.096	0.215
2018	0.156	0.164	0.102	0.280	0.099	0.198
2019	0.140	0.194	0.117	0.271	0.095	0.182
2020	0.147	0.216	0.114	0.259	0.093	0.171

Terminal Year Survey

Year	BS	AI	WG	CG	WY	EY
2000	0.073	0.177	0.105	0.329	0.094	0.222
2001	0.078	0.167	0.150	0.303	0.084	0.218
2002	0.145	0.151	0.158	0.301	0.064	0.182
2003	0.142	0.151	0.133	0.326	0.080	0.168
2004	0.129	0.154	0.147	0.333	0.078	0.159
2005	0.137	0.124	0.103	0.367	0.097	0.173
2006	0.161	0.113	0.157	0.276	0.074	0.219
2007	0.156	0.142	0.117	0.310	0.095	0.180
2008	0.183	0.125	0.082	0.291	0.100	0.218
2009	0.190	0.139	0.106	0.310	0.083	0.171
2010	0.053	0.152	0.157	0.330	0.090	0.218
2011	0.057	0.112	0.102	0.314	0.145	0.270
2012	0.057	0.110	0.126	0.408	0.097	0.203
2013	0.053	0.154	0.111	0.348	0.085	0.249
2014	0.204	0.128	0.080	0.294	0.076	0.219
2015	0.209	0.130	0.117	0.255	0.078	0.211

App. Table A.2, cont.

2016	0.119	0.140	0.084	0.301	0.134	0.222
2017	0.114	0.220	0.128	0.254	0.108	0.176
2018	0.136	0.204	0.102	0.298	0.099	0.161
2019	0.123	0.277	0.152	0.248	0.058	0.141
2020	0.244	0.239	0.101	0.192	0.067	0.158

Table A.3. Mean proportion of allowable biological catch apportioned by area and apportionment scenario for the simulation forward projection period (2019–2043) for converged simulations. Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WG), Central Gulf of Alaska (CG), West Yakutat (WY), and East Yakutat/Southeast Outside (EY).

Age-based

Year	BS	AI	WG	CG	WY	EY
2020	0.112	0.206	0.133	0.165	0.183	0.201
2021	0.067	0.184	0.181	0.186	0.189	0.193
2022	0.163	0.152	0.166	0.171	0.173	0.175
2023	0.161	0.166	0.163	0.168	0.170	0.172
2024	0.159	0.161	0.164	0.170	0.172	0.174
2025	0.160	0.165	0.162	0.168	0.171	0.174
2026	0.159	0.157	0.165	0.171	0.173	0.176
2027	0.159	0.165	0.162	0.169	0.171	0.174
2028	0.156	0.159	0.164	0.170	0.174	0.177
2029	0.156	0.165	0.162	0.169	0.173	0.176
2030	0.152	0.162	0.161	0.171	0.175	0.179
2031	0.151	0.164	0.162	0.170	0.175	0.178
2032	0.151	0.157	0.162	0.172	0.177	0.181
2033	0.153	0.163	0.160	0.170	0.175	0.180
2034	0.153	0.152	0.164	0.172	0.177	0.182
2035	0.155	0.163	0.159	0.169	0.174	0.179
2036	0.152	0.154	0.162	0.172	0.177	0.182
2037	0.148	0.165	0.162	0.171	0.175	0.179
2038	0.154	0.157	0.162	0.171	0.176	0.180
2039	0.154	0.164	0.160	0.169	0.174	0.179
2040	0.156	0.153	0.164	0.172	0.176	0.180
2041	0.156	0.163	0.160	0.169	0.173	0.178
2042	0.151	0.159	0.161	0.171	0.176	0.182
2043	0.150	0.164	0.161	0.170	0.175	0.180

Fixed

Year	BS	AI	WG	CG	WY	EY
2020	0.098	0.132	0.108	0.341	0.115	0.207
2021	0.098	0.132	0.108	0.341	0.115	0.207
2022	0.098	0.132	0.108	0.341	0.115	0.207
2023	0.098	0.132	0.108	0.341	0.115	0.207
2024	0.098	0.132	0.108	0.341	0.115	0.207
2025	0.098	0.132	0.108	0.341	0.115	0.207
2026	0.098	0.132	0.108	0.341	0.115	0.207
2027	0.098	0.132	0.108	0.341	0.115	0.207

App. Table A.3, cont.

2028	0.098	0.132	0.108	0.341	0.115	0.207
2029	0.098	0.132	0.108	0.341	0.115	0.207
2030	0.098	0.132	0.108	0.341	0.115	0.207
2031	0.098	0.132	0.108	0.341	0.115	0.207
2032	0.098	0.132	0.108	0.341	0.115	0.207
2033	0.098	0.132	0.108	0.341	0.115	0.207
2034	0.098	0.132	0.108	0.341	0.115	0.207
2035	0.098	0.132	0.108	0.341	0.115	0.207
2036	0.098	0.132	0.108	0.341	0.115	0.207
2037	0.098	0.132	0.108	0.341	0.115	0.207
2038	0.098	0.132	0.108	0.341	0.115	0.207
2039	0.098	0.132	0.108	0.341	0.115	0.207
2040	0.098	0.132	0.108	0.341	0.115	0.207
2041	0.098	0.132	0.108	0.341	0.115	0.207
2042	0.098	0.132	0.108	0.341	0.115	0.207
2043	0.098	0.132	0.108	0.341	0.115	0.207

NPFMC

Year	BS	AI	WG	CG	WY	EY
2020	0.075	0.076	0.125	0.292	0.153	0.279
2021	0.077	0.085	0.127	0.296	0.149	0.266
2022	0.076	0.076	0.127	0.301	0.152	0.267
2023	0.073	0.084	0.124	0.299	0.154	0.267
2024	0.072	0.074	0.122	0.302	0.158	0.272
2025	0.068	0.081	0.117	0.300	0.159	0.274
2026	0.067	0.070	0.116	0.303	0.163	0.282
2027	0.065	0.078	0.113	0.300	0.163	0.282
2028	0.064	0.068	0.113	0.303	0.165	0.287
2029	0.063	0.076	0.111	0.300	0.164	0.287
2030	0.063	0.067	0.112	0.302	0.166	0.291
2031	0.063	0.075	0.111	0.298	0.164	0.289
2032	0.063	0.067	0.112	0.300	0.166	0.292
2033	0.063	0.075	0.111	0.297	0.164	0.290
2034	0.064	0.067	0.112	0.300	0.165	0.292
2035	0.064	0.076	0.112	0.297	0.163	0.289
2036	0.065	0.067	0.114	0.301	0.164	0.290
2037	0.065	0.076	0.113	0.298	0.162	0.286
2038	0.066	0.068	0.115	0.301	0.163	0.287
2039	0.066	0.077	0.114	0.298	0.161	0.284
2040	0.066	0.069	0.115	0.301	0.163	0.286
2041	0.066	0.077	0.114	0.297	0.162	0.284
2042	0.066	0.069	0.115	0.301	0.163	0.286
2043	0.066	0.077	0.114	0.298	0.162	0.283

App. Table A.3, cont.

Survey Average

Year	BS	AI	WG	CG	WY	EY
2020	0.064	0.091	0.110	0.307	0.151	0.278
2021	0.067	0.094	0.114	0.307	0.149	0.270
2022	0.070	0.090	0.119	0.311	0.148	0.263
2023	0.073	0.092	0.122	0.313	0.146	0.254
2024	0.073	0.086	0.123	0.317	0.147	0.253
2025	0.070	0.091	0.120	0.316	0.148	0.255
2026	0.068	0.083	0.118	0.319	0.151	0.260
2027	0.066	0.088	0.115	0.317	0.151	0.263
2028	0.064	0.080	0.114	0.319	0.154	0.269
2029	0.063	0.085	0.112	0.317	0.154	0.270
2030	0.062	0.077	0.112	0.319	0.155	0.275
2031	0.062	0.083	0.110	0.316	0.155	0.274
2032	0.062	0.076	0.111	0.317	0.156	0.277
2033	0.062	0.083	0.111	0.315	0.154	0.275
2034	0.063	0.077	0.112	0.317	0.155	0.277
2035	0.063	0.084	0.111	0.314	0.154	0.274
2036	0.064	0.077	0.112	0.316	0.154	0.275
2037	0.064	0.084	0.112	0.315	0.153	0.273
2038	0.064	0.078	0.113	0.317	0.154	0.274
2039	0.064	0.085	0.112	0.315	0.153	0.271
2040	0.065	0.079	0.114	0.317	0.154	0.272
2041	0.065	0.086	0.113	0.315	0.153	0.270
2042	0.065	0.079	0.114	0.317	0.154	0.271
2043	0.065	0.086	0.113	0.315	0.152	0.269

Terminal Year Survey

Year	BS	AI	WG	CG	WY	EY
2020	0.081	0.073	0.129	0.320	0.146	0.251
2021	0.074	0.108	0.124	0.308	0.142	0.245
2022	0.074	0.072	0.125	0.325	0.149	0.256
2023	0.069	0.104	0.118	0.313	0.145	0.252
2024	0.069	0.071	0.119	0.323	0.153	0.265
2025	0.066	0.100	0.112	0.312	0.149	0.261
2026	0.064	0.066	0.114	0.324	0.156	0.275
2027	0.062	0.096	0.110	0.313	0.152	0.268
2028	0.063	0.062	0.113	0.326	0.158	0.278
2029	0.060	0.095	0.108	0.311	0.153	0.272
2030	0.063	0.063	0.112	0.323	0.158	0.282
2031	0.061	0.095	0.109	0.312	0.153	0.271
2032	0.063	0.064	0.113	0.321	0.158	0.282
2033	0.061	0.097	0.109	0.310	0.152	0.271

App. Table A.3, cont.

2034	0.065	0.065	0.113	0.320	0.157	0.281
2035	0.062	0.097	0.110	0.310	0.151	0.269
2036	0.065	0.065	0.115	0.322	0.156	0.277
2037	0.063	0.097	0.111	0.311	0.150	0.267
2038	0.067	0.066	0.116	0.321	0.155	0.276
2039	0.064	0.098	0.111	0.310	0.150	0.266
2040	0.067	0.066	0.115	0.321	0.155	0.275
2041	0.065	0.097	0.111	0.311	0.150	0.266
2042	0.065	0.066	0.116	0.321	0.156	0.276
2043	0.063	0.099	0.112	0.311	0.150	0.265

General conclusions

After hundreds of spatial model configurations have been explored, and hundreds of thousands of simulation loops run and analyzed, what have I learned about Alaska sablefish population dynamics and management? Over the next few pages, I repeatedly return to these overarching questions in the context of each chapter: What did I learn about sablefish population dynamics and uncertainties, and how do these affect our management of sablefish? How can we improve management and better ensure the sustainability of Alaska sablefish? What still needs to be answered, and where do we go from here for the management of Alaska sablefish?

The sablefish spatial stock assessment models explored in Chapter 1 illustrate how consideration of spatial structure within estimation models can reveal potential risks of overexploitation on smaller regional scales, and can be useful in quantifying source-sink dynamics within a population. The myriad spatial model variants developed indicated that the East region, comprised of the West Yakutat and East Yakutat/Southeast Outside management regions, were below target reference points. In contrast, the population as a whole was estimated to be within or just below targets in the panmictic stock assessment model. While the overall population estimates from panmictic models may be unbiased (Li et al. 2015, Li et al. 2018, Bosley et al. 2021), these spatial dynamics are critical to convey to sablefish fishery managers and stakeholders for two reasons. First, if the East region is a critical component of future recruitment success (i.e., spawning occurs predominantly here, or spawners here contribute more to recruitment), over-exploitation and commensurate localized depletion of these critical spawning areas may negatively affect future recruitment success and ultimately, the long-term sustainability of the population. Much is unknown about sablefish spawner-recruit relationships, and their high movement rates (if not tied solely to spawning) might be a buffer against these negative effects of spatial depletion. However, as a precaution, management measures should attempt to reduce fishing mortality in areas that are below target reference points. Second, Alaska sablefish are commercially targeted in longline and pot fisheries, and are incidentally taken in trawl fisheries. Because the longline and pot fisheries are

managed through a catch shares program, and quota shares are tied to specific management areas, spatial depletion or overexploitation can have real economic consequences for shareholders and other stakeholders. My second and third chapters addressed this, where apportionment of allowable catch limits to management areas was explored and weighed against sustainability performance metrics. Apportionment of sablefish allowable harvest is the primary tool available to address the issue of localized depletion.

The development of a spatial stock assessment model for Alaska sablefish started with explorations of the available fishery dependent and fishery independent data. In aggregate, across Alaska, sablefish are relatively data rich, with a time series of fishery dependent data beginning in the 1960s, and fishery independent data starting in the late 1970s. However, parsing historical data into their spatial components was challenging for two reasons: 1) there was not always sufficient information retained in the historical datasets to split data into management areas, particularly prior to 1977, and 2) even when there was sufficient spatial information available to allow data to be split into management areas, the resulting sample sizes were small with greater variability for some areas. Because of this, my original preference for a six-area spatial stock assessment was modified to a three-area model. In addition, data limitations meant the spatial assessment model started in 1977 instead of 1960, which was the initial model year for the panmictic model currently used for setting management recommendations. This is likely a common issue not unique to sablefish: simultaneous modeling of temporal and spatial dynamics requires lots of data. Without adequate data, estimating model parameters for critical model components such as selectivity, recruitment, catchability, or natural mortality may not be possible. However, we are continually collecting data, and current data collection methods retain information about location at a much finer scale than in the past. Future spatial stock assessment modeling efforts will benefit from this ever-increasing time series. However, if the goal is to collect finer-scale data for future spatial stock assessments, the optimal data collection scheme may still be different from current methods.

Is it worth it to take a data-rich single-area, panmictic model and make it into a relatively data-poor spatial model? I argue that, yes, it was a valuable research component of

my dissertation. For Alaska sablefish, this spatial model has developed into a useful research model and could be used for continued improvement into a tag-integrated model or as a basis for a three-area model in which the three areas represent coastwide dynamics across the U.S. and Canada.

Simulations developed in Chapters 2 and 3 generated ‘true’ sablefish population dynamics with a spatial operating model (OM) representing six management areas, then used a panmictic, single-area age-based estimation model (EM) to assess the population. These simulation-estimation scenarios revealed that the single-area EM performed well in terms of accurately estimating overall (area-aggregated) biomass and abundance, with a low absolute percentage error between the ‘known’ simulated population and the EM estimated population. Similar results have been reported in other studies; panmictic EMs often have low bias in a panmictic population (Li et al. 2015, Li et al. 2018, Bosley et al. 2021). This result of low bias in overall population abundance estimation was reassuring in the context of Alaska sablefish management and resource sustainability. However, movement rates between the OM areas and assumptions about recruitment were the two largest drivers of simulated sablefish spatial population dynamics under the conditions of the simulations in Chapter 2. This supports further research on spatial recruitment drivers and spawning dynamics because making small gains in knowledge of these issues may help refine assessment model parameterization and structure and lead to continued improvement to the stock assessment used for providing recommendations to fishery managers.

Stakeholder preferences and concerns about the Alaska sablefish management varied within and between user groups who frequently referenced the social and economic consequences to management decisions regarding catch apportionment. Long-term sustainability is not just mandated by the MSA, it was expressed as a core objective of stakeholders. Stakeholders in the pot and longline gear fleets also expressed a desire for interannual stability and optimal yield, and these two objectives can be in direct conflict.

Sablefish recruitment dynamics, movement, and fishing practices are drivers of temporal changes in sablefish spatial distribution. Apportionment of catch to management

areas is a tool by which fisheries managers can direct spatial harvest patterns without a spatial stock assessment model in which modeled areas match management areas. With a spatial model, we could directly examine regional biomass with respect to target biological reference points, and apportionment of allowable catch to management areas would not be necessary. But Chapter 2 revealed some of the data and model challenges to developing a spatial stock assessment model and until those challenges are addressed, there is a continued need to apportion catch to management areas. In the absence of information on stock-recruitment dynamics over space, and lack of data regarding whether some components of the sablefish population or areas contribute more to recruitment success, it is likely best to avoid over-depletion of sablefish in any management area.

Given that fishery stakeholder objectives can at times be in opposition, the logical question becomes: Where is the balance between interannual stability in harvest between regions and avoiding depletion? The final management recommendation from this work was for a harvest apportionment method using a simple five-year moving average of survey abundance across space as a basis for apportionment. It would be valuable to re-evaluate the performance of this apportionment method in the short- and medium-term using the performance metrics of interest to stakeholders, because stock assessment models are continually evolving, the population dynamics resulting from the recent large year classes may not follow historical norms, and the Alaska ecosystem is continually and increasingly changing due to climate changes.

The harvest control rule used by the North Pacific Fishery Management Council specifies target harvest reference points and automatic triggers for reducing fishing mortality when the population estimates fall below specified thresholds (NPFMC 2018a, NPFMC 2018b). One unexpected outcome from the simulation analyses in Chapter 3 was that the harvest control rule (HCR) appeared to be a more influential driver of outcomes than the effect of any of the apportionment scenarios examined. This is valuable validation of the HCR, as the HCR was effective across spatial management alternatives at keeping the population from crashing under the assumptions of these simulations. However, these simulation analyses did not explore alternative stock recruitment assumptions (e.g., a Beverton-Holt stock recruitment

relationship, or spatially divergent recruitment contributions), which may be important drivers of sablefish spatial dynamics and change the outcome of the simulations. Further explorations of alternative recruitment assumptions and how assumptions affect management recommendations could provide more information to fishery managers on how to best ensure the long-term sustainability of the fishery resource.

The unexpected HCR results and caveats about assumptions above highlights another important consideration: the tools we build for research versus tactical management purposes are sometimes different. Research-only models which are more general (i.e., modeling ‘sablefish-like’ population dynamics) provide useful information while allowing flexibility simulation or estimation model parameterization. Many of the modelling choices I made throughout my dissertation were made to condition these models on sablefish with on-the-ground applied management usage at the forefront. I used multiple tools to examine the most pressing management question of how sablefish catch should be apportioned to management areas, while juggling the reality of management deadlines and the need for recommendations. As a result, the single-area EM I used for this research was very similar to the model used for management, but was not robust to all of the spatial OM scenarios I explored. Additionally, recruitment was generally poorly estimated in the panmictic EM, resulting in EM non-convergence. Shifting the modeling tools I developed in this body of research from tactical tools towards ‘research exploration tools’ will help tackle the next set of important sablefish research questions, and the list of questions is long.

Where do we go from here in sablefish and spatial dynamics research? I think the following are the most pressing next topics. First, continued explorations of alternative stock-recruitment relationships for sablefish are a logical next step in the development of these sablefish simulation tools. We might never fully understand sablefish recruitment drivers, but modeling alternative recruitment relationships can help us understand if some management procedures are more robust to incorrect assumptions about recruitment. Second, I recommend the development of a tag-integrated spatial stock assessment model (with further exploration of the choice of spatial areas to represent in the model), using tag data to internally estimate movement rates. There are already models available that do this using a ‘sablefish-like’ premise

for simulation purposes. Lastly, we need to continue making progress on research model projects that explore sablefish population dynamics across their range in the Eastern Pacific Ocean. These models will help managers understand the biases and uncertainties of sablefish management from a broader perspective, and help ensure sustainability across the west coast United States, Alaska, and Canada.

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